MightyConnect™ Security Platform

- Mesh P2P VPN
- Powered by WireGuard & FD.io VPP

MightyLink Lab.
(www.mightyconn.com)
Doc. Revision: 1.01

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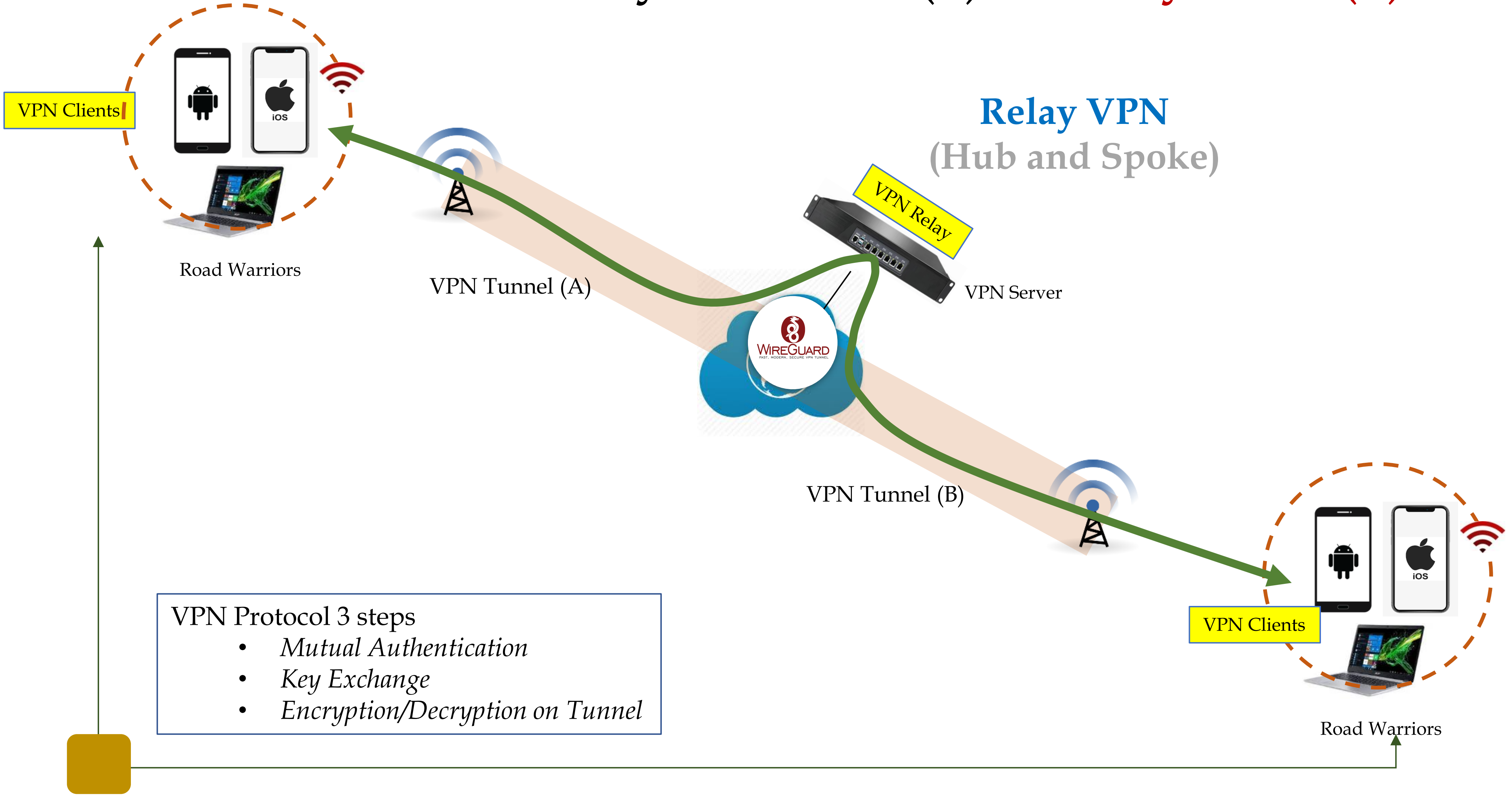
1. Zero Trust P2P Security Platform

Zero Trust P2P Security Platform

- ZTNA/Secure Network
- No modifications for any applications
- Auto Connect, Direct Connect and Relay Connect

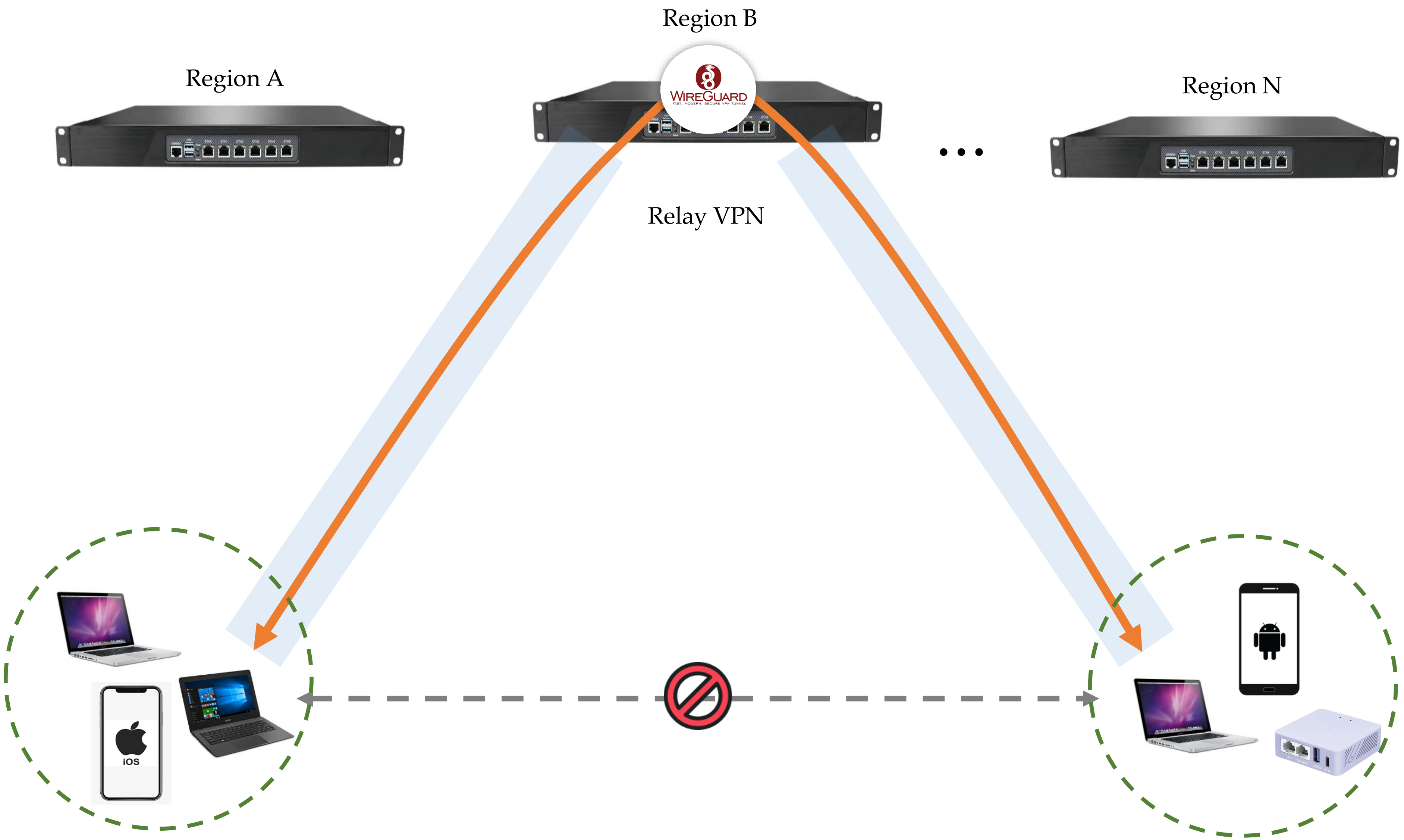
1. Zero Trust P2P Security Platform(1) – Relay VPN(1)



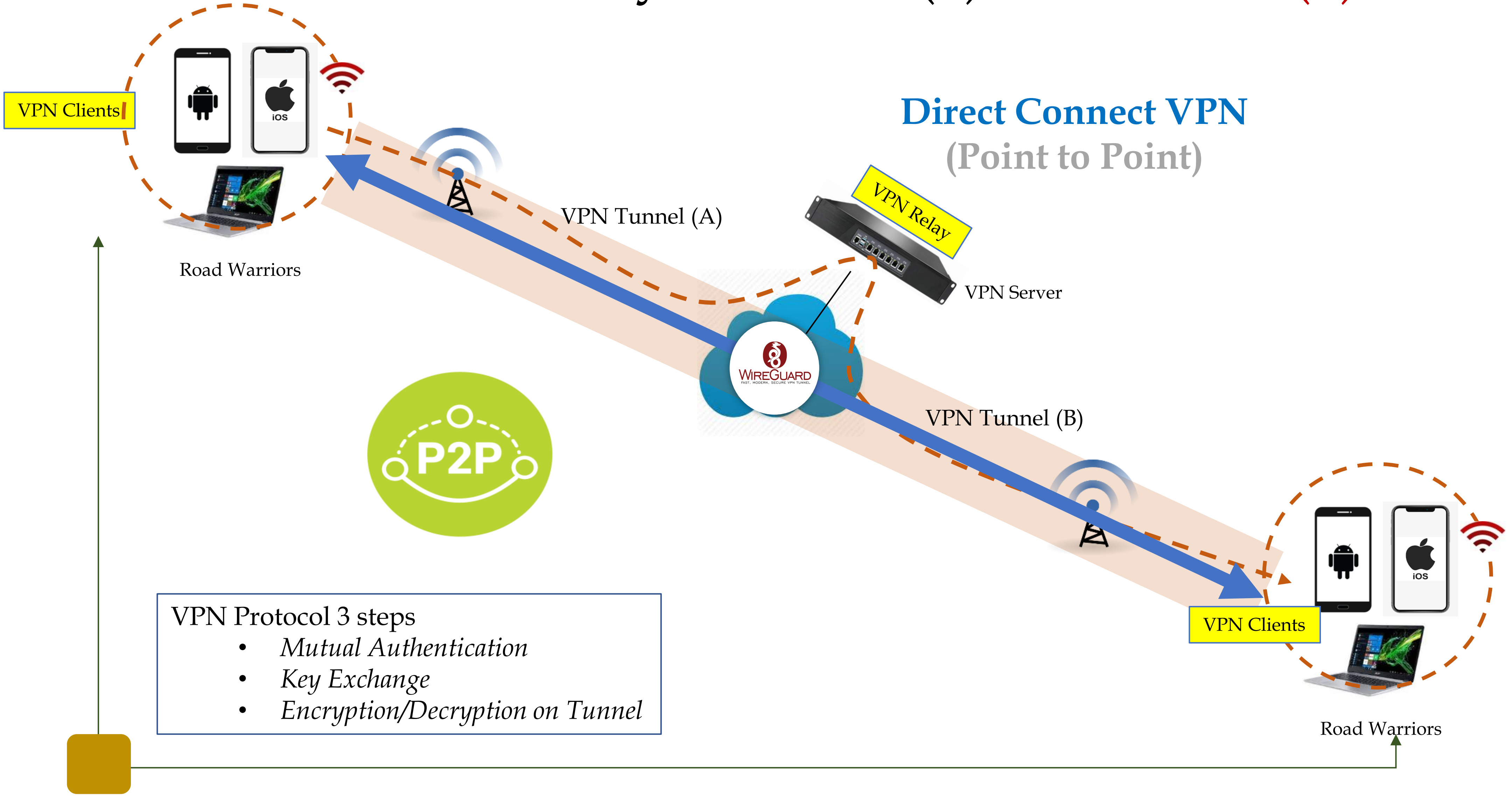
VPN Protocol 3 steps

- *Mutual Authentication*
- *Key Exchange*
- *Encryption/Decryption on Tunnel*

1. Zero Trust P2P Security Platform(1) – Relay VPN(2)



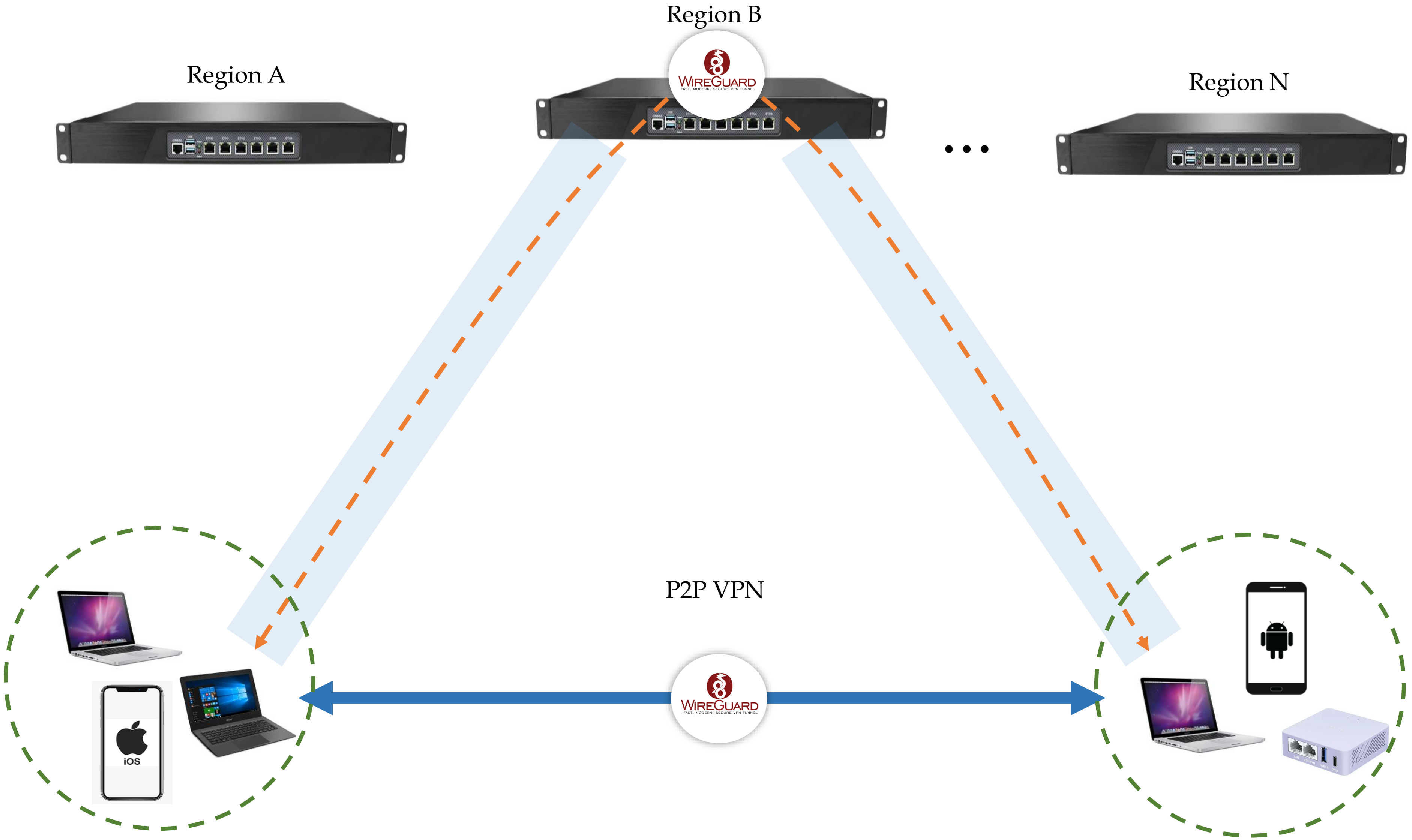
1. Zero Trust P2P Security Platform(2) – P2P VPN(1)



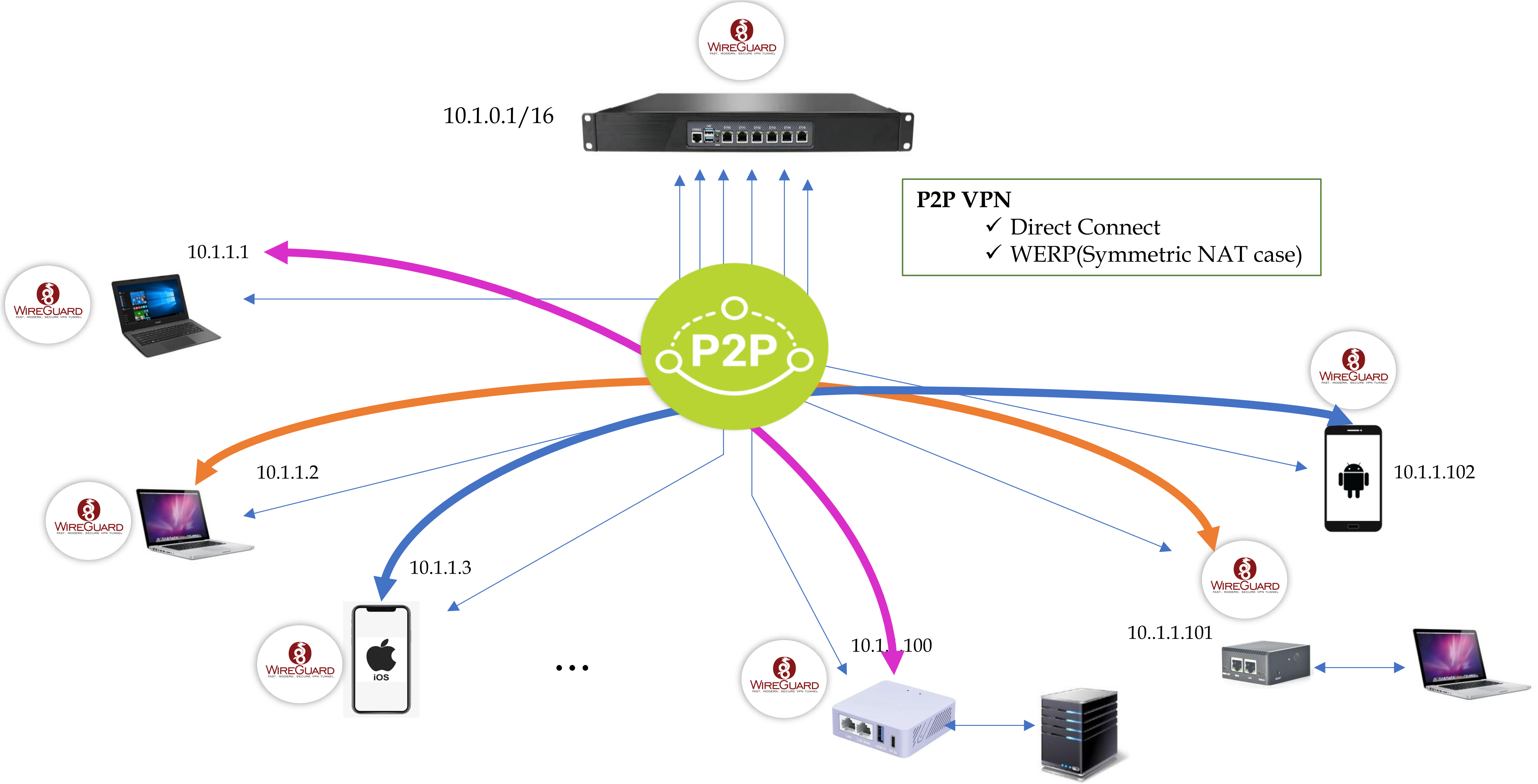
VPN Protocol 3 steps

- *Mutual Authentication*
- *Key Exchange*
- *Encryption/Decryption on Tunnel*

1. Zero Trust P2P Security Platform(2) – P2P VPN(2)



1. Zero Trust P2P Security Platform(2) – P2P VPN(3)

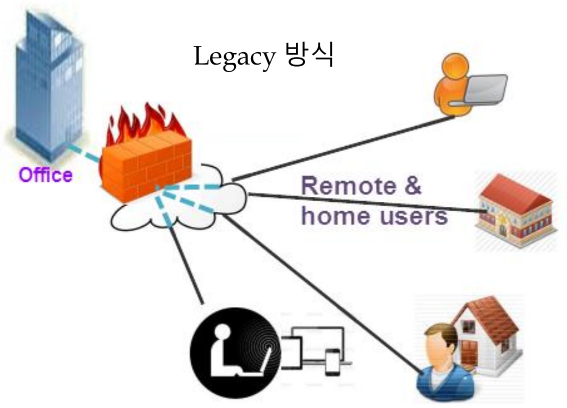


P2P VPN
- ✓ Direct Connect
- ✓ WERP(Symmetric NAT case)

1. Zero Trust P2P Security Platform(3) – ZTNA(1-1)

- **Legacy 방식(Firewall/VPN/IPS/UTM)**
  - Internal network(Trusted Area)과 External network(Untrusted Area) 구분
  - Internal network <=> Perimeter Security 제품 ⬄ External network : 안전한 내부망과 불안전한 외부망 경계(Perimeter)에 보안 제품 설치
  - Static Configuration, IP/Port 기반의 통제

VPN is Dead

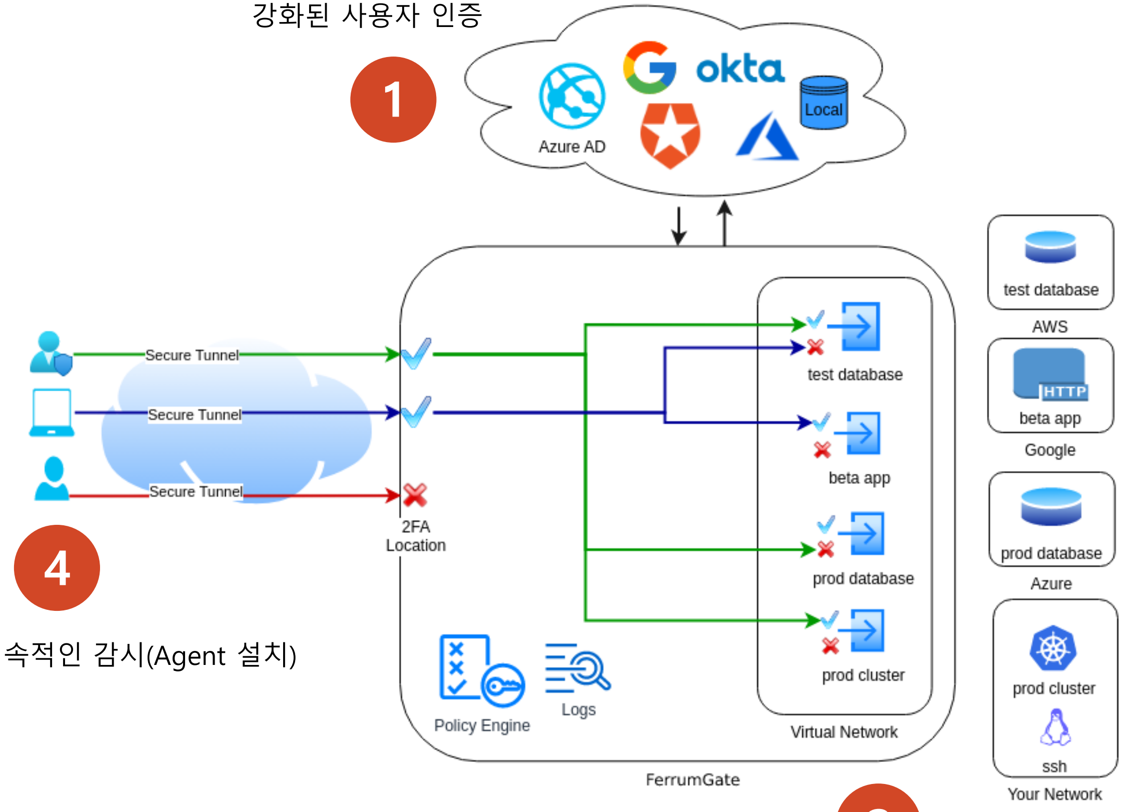


- **ZTNA or SDP(Software Defined Perimeter) 방식**
  - 환경의 변화: **Cloud, Mobile, BYOD**
    - *이제는 중요한 것들이 Perimeter 바깥에 있는 세상이 되었다.*
  - **Zero Trust**: "Internal, External network 모두 안전하지 않다"에서 출발
  - S/W 방식의 **중앙 Controller**에 의한 접근 통제
    - *실제 network에 접근하기 전에, 인증(Authentication) & 접근 통제(Authorization)을 수행하는 개념*
    - *내부 Resource는 Appliance(Gateway)에 의해 모두 Hidden 상태, 인증을 통과한 후에 외부에 Open되는 구조.* **Dark Network(외부에 보이지 않는 network) 개념 출현**
  - Dynamic Control, IP 기반이 아니라, 사용자 기반의 통제

1. Zero Trust P2P Security Platform(3) – ZTNA(1-2)

강화된 사용자 인증

1



4

지속적인 감시(Agent 설치)

2

세분화된 서비스(Authorization)

3

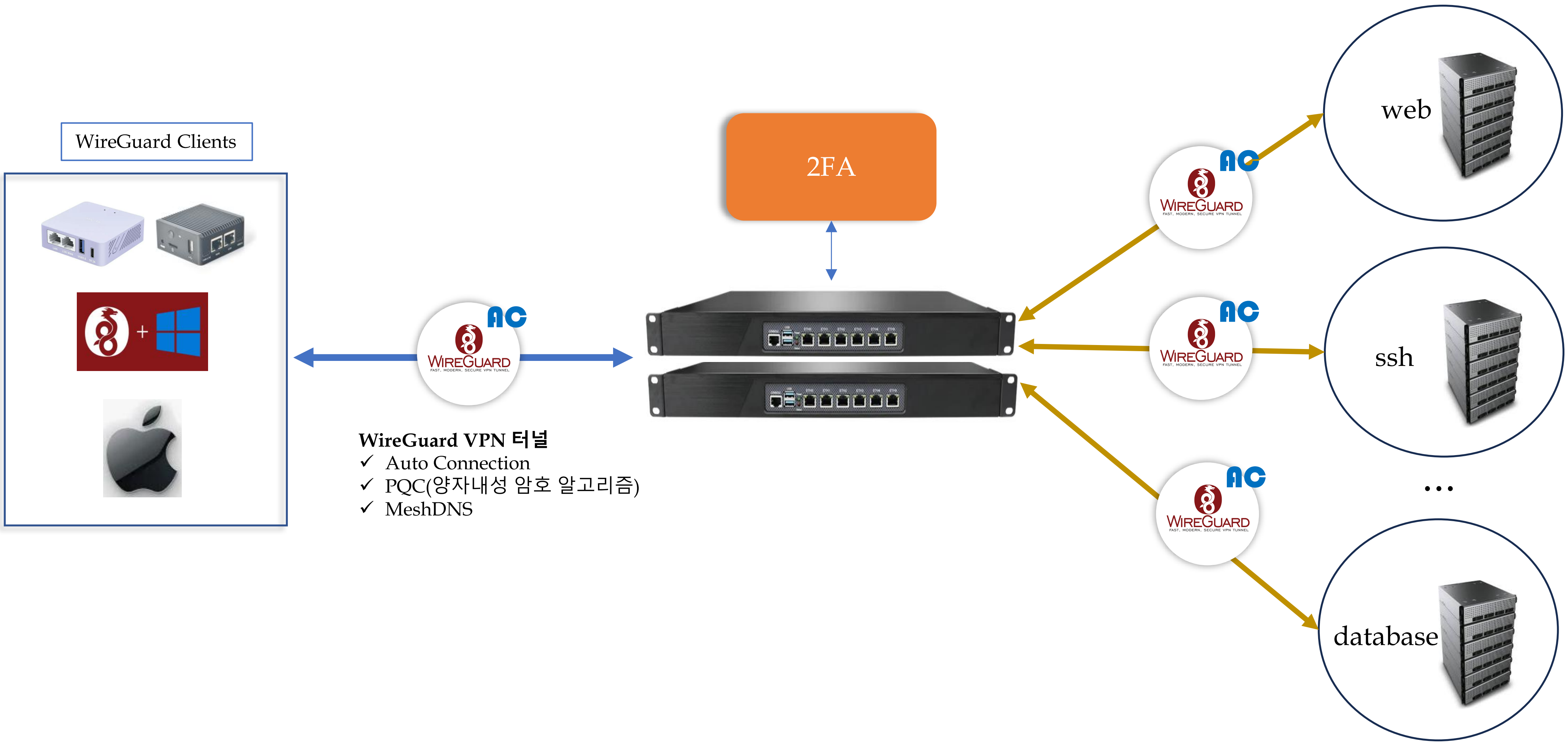
최소한의 권한 부여(Read/Write/Execute 분리)

ZTNA의 요건

- More Authentication Options
  - MFA: SAML, OAUTH2, LDAP, FIDO2
  - Location Check, Time Check, Device Posture Check
- Micro-segmentation
  - Virtual Application Networks
- Least privilege access
  - RBAC(Role Based Access Control)
- Continuous monitoring
  - Every packet, every flow, every connection

[출처] https://ferrumgate.com/

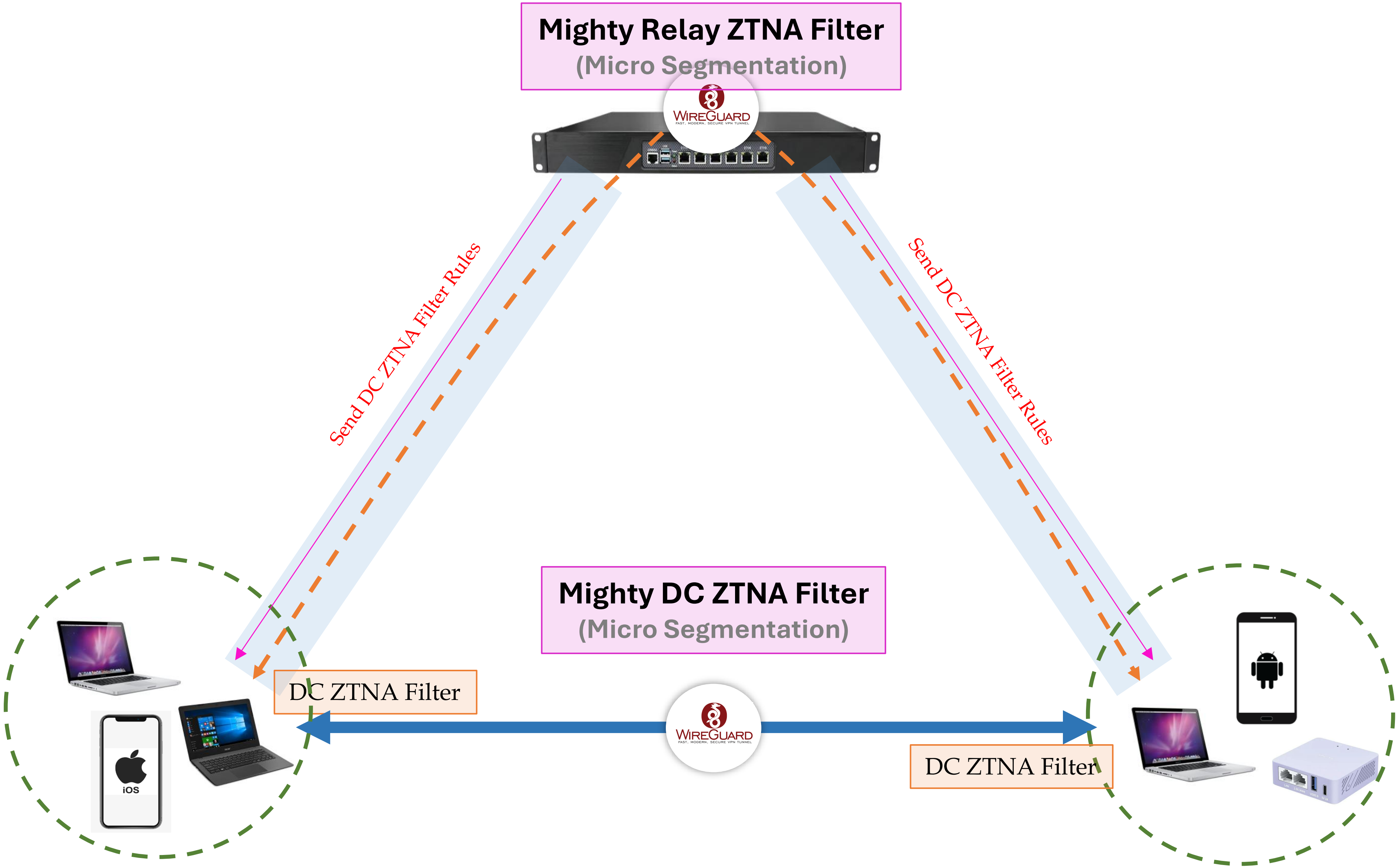
1. Zero Trust P2P Security Platform(3) – ZTNA(2)



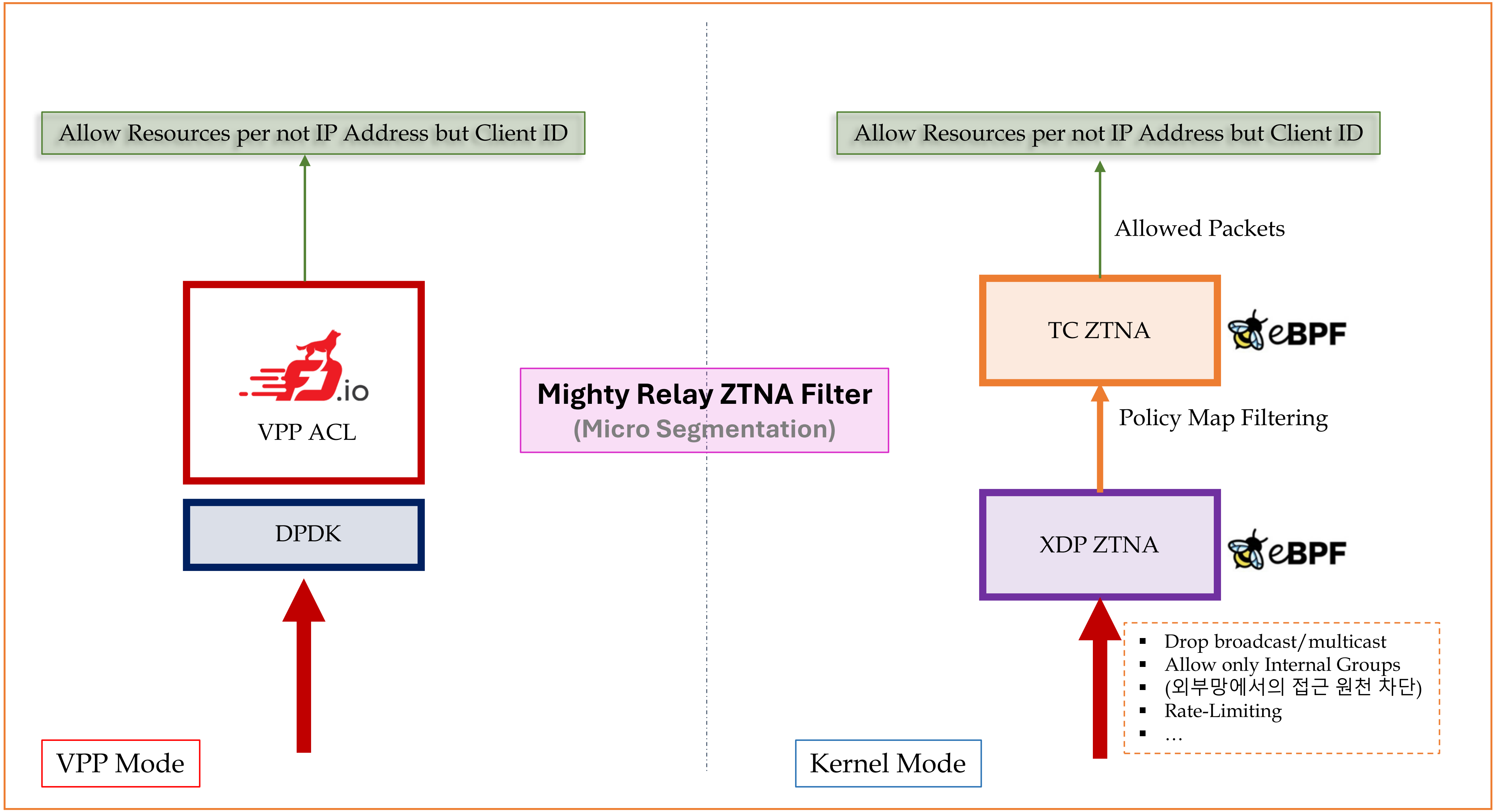
WireGuard VPN 터널

- ✓ Auto Connection
- ✓ PQC(양자내성 암호 알고리즘)
- ✓ MeshDNS

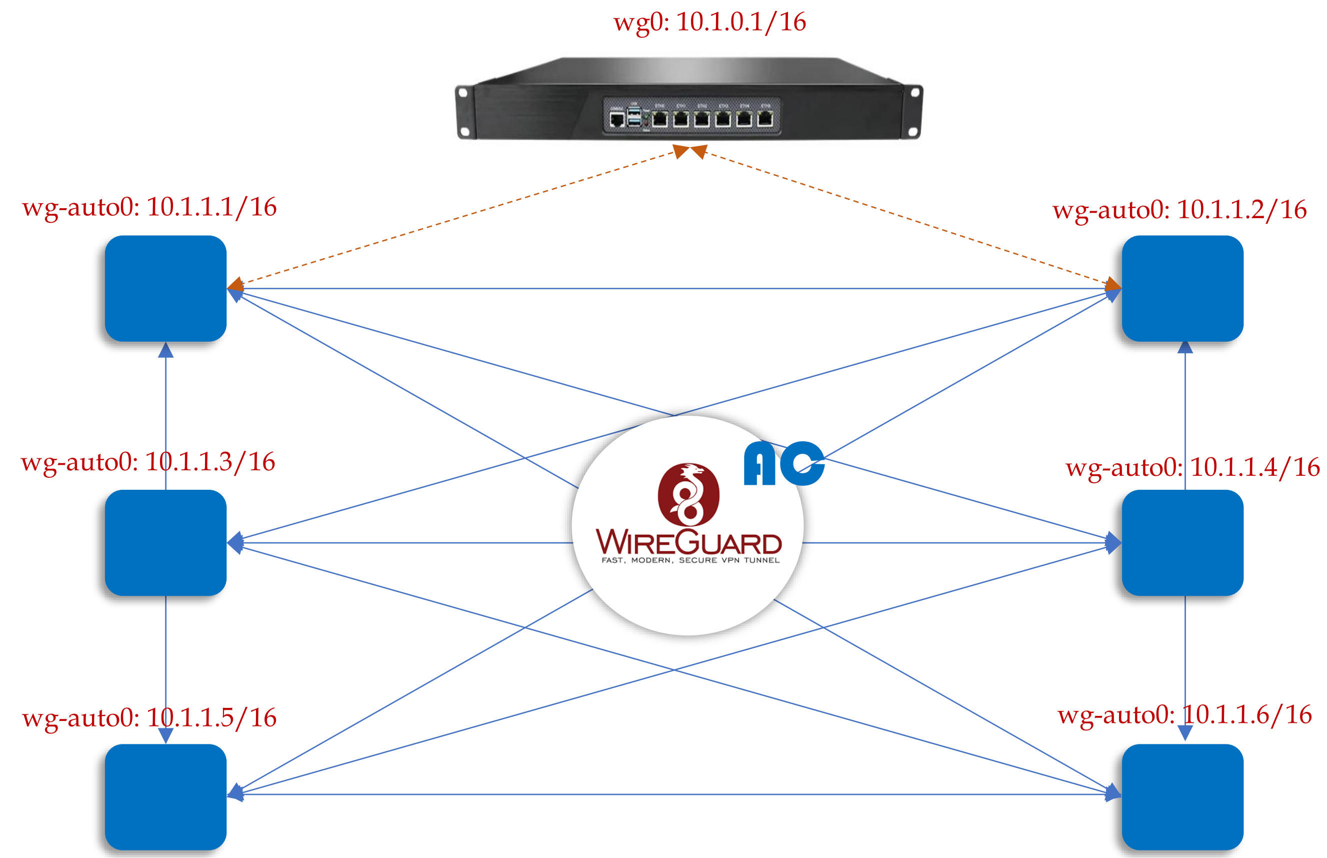
1. Zero Trust P2P Security Platform(3) – ZTNA(3-1)



1. Zero Trust P2P Security Platform(3) – ZTNA(3-2)

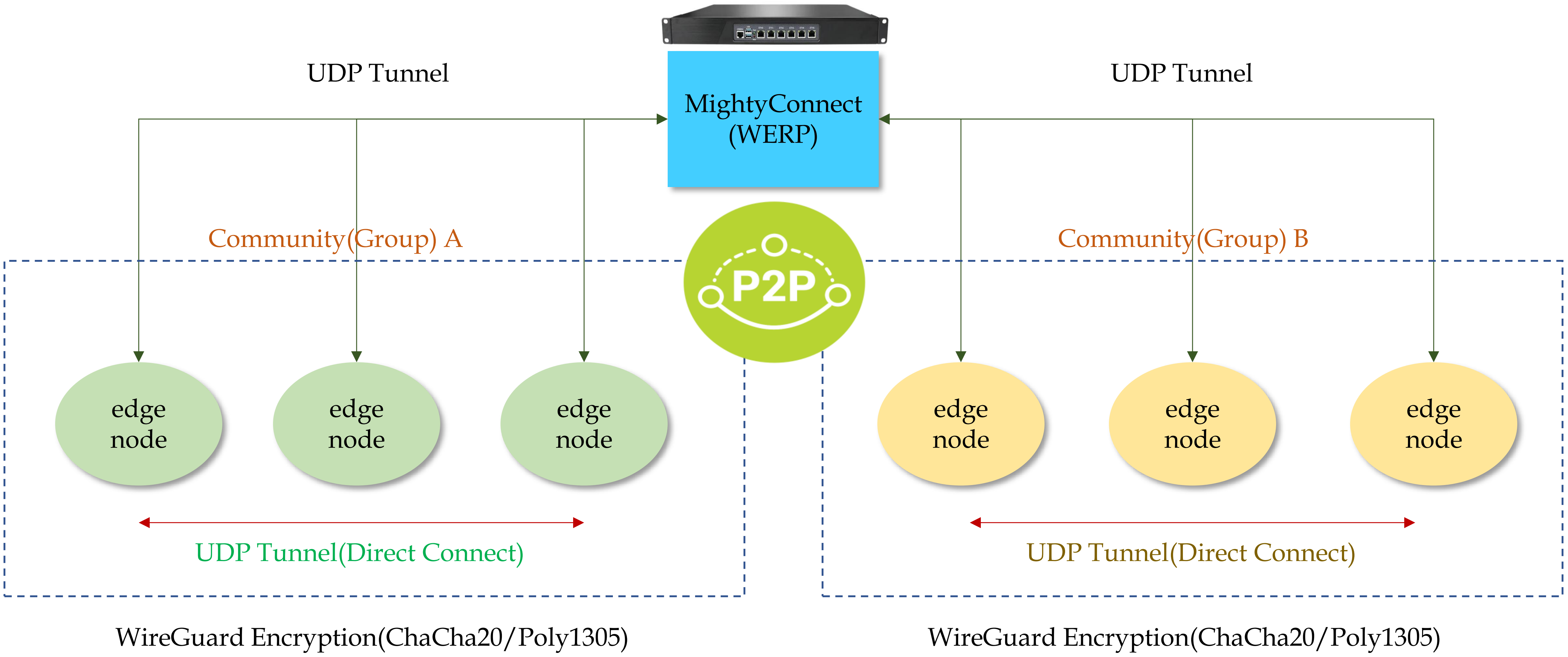


1. Zero Trust P2P Security Platform(4) - Mesh(1)

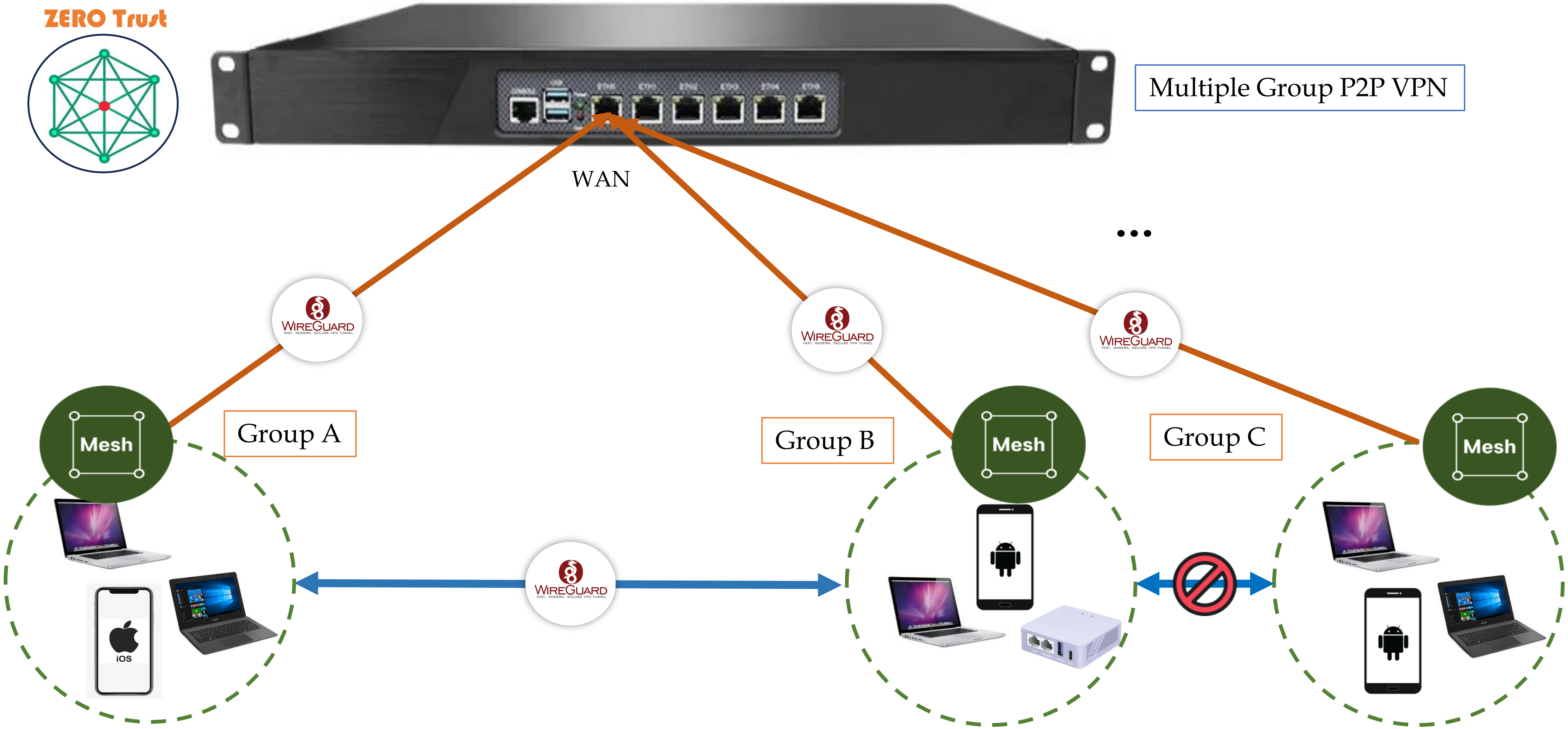


WireGuard Mesh

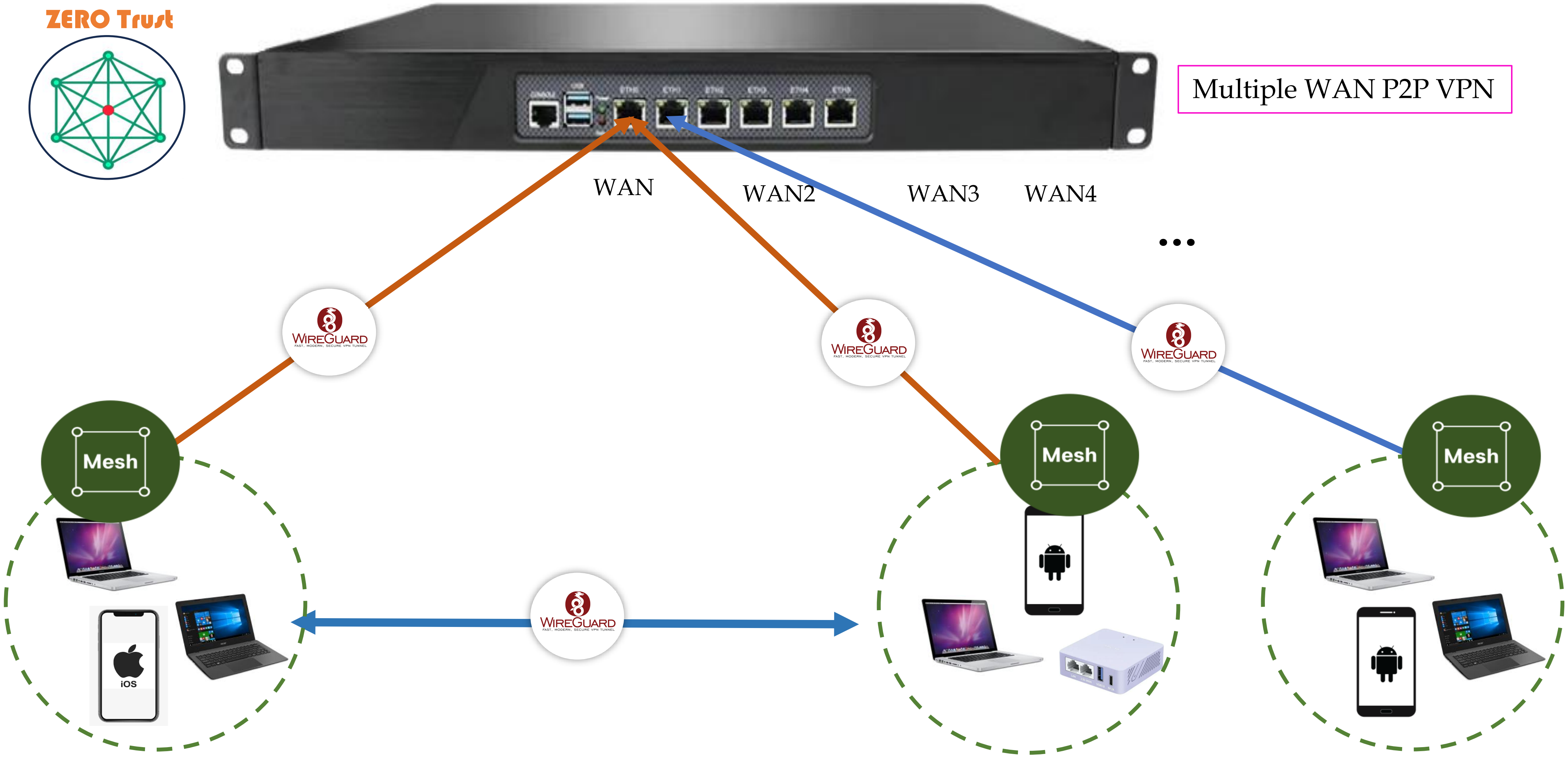
1. Zero Trust P2P Security Platform(4) – Mesh(2)



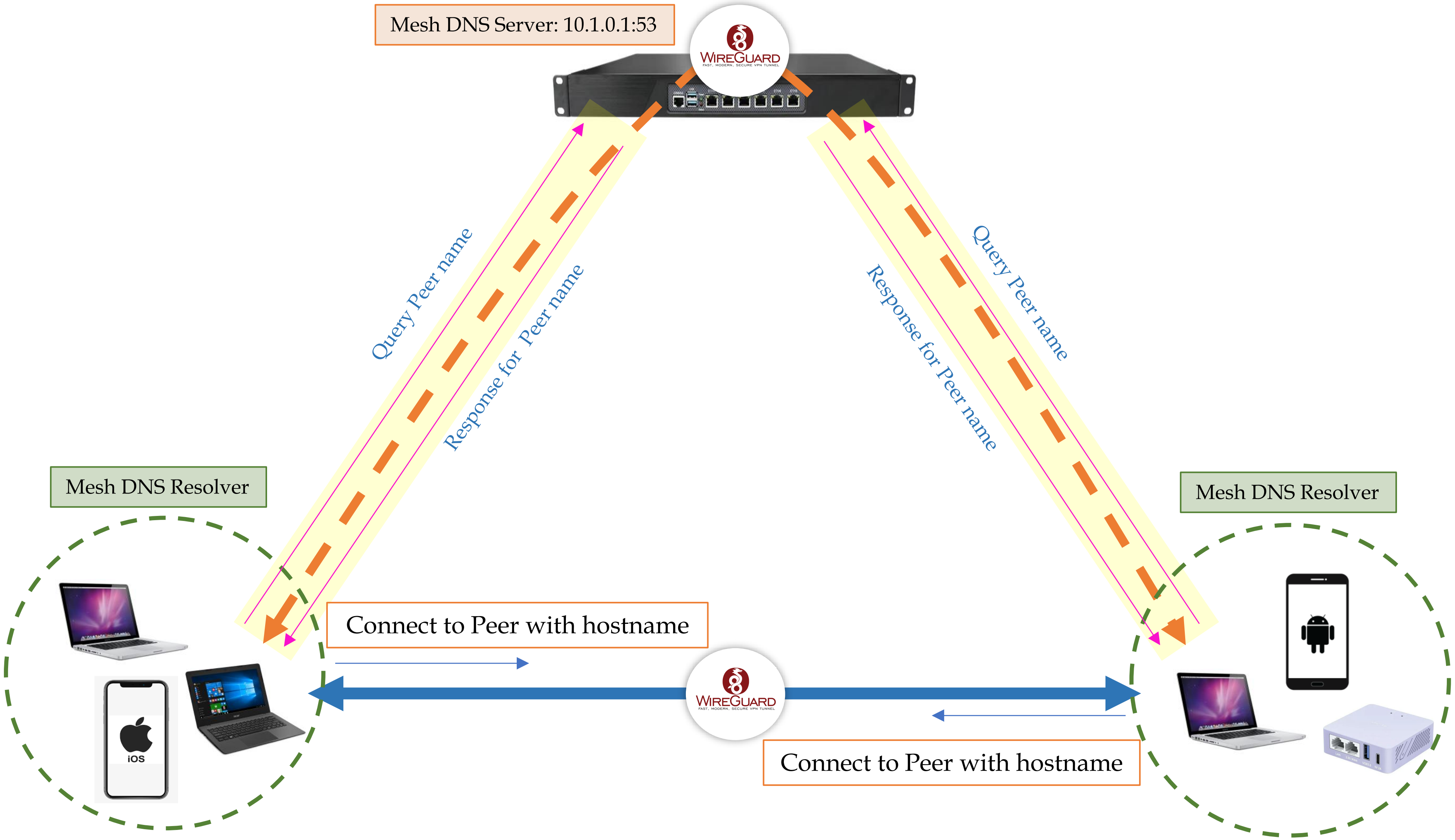
1. Zero Trust P2P Security Platform(4) – Mesh(3)



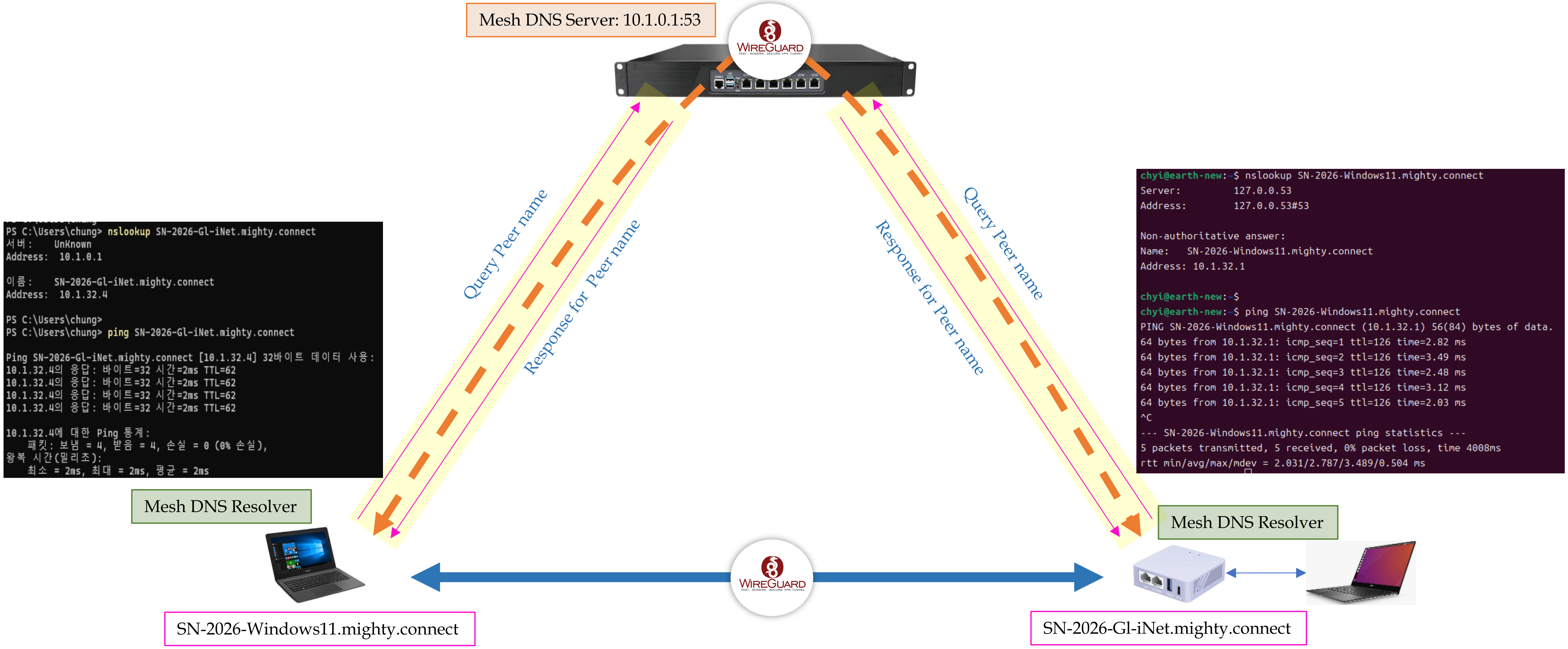
1. Zero Trust P2P Security Platform(4) - Mesh(4)



1. Zero Trust P2P Security Platform(5) – Mesh DNS(1)

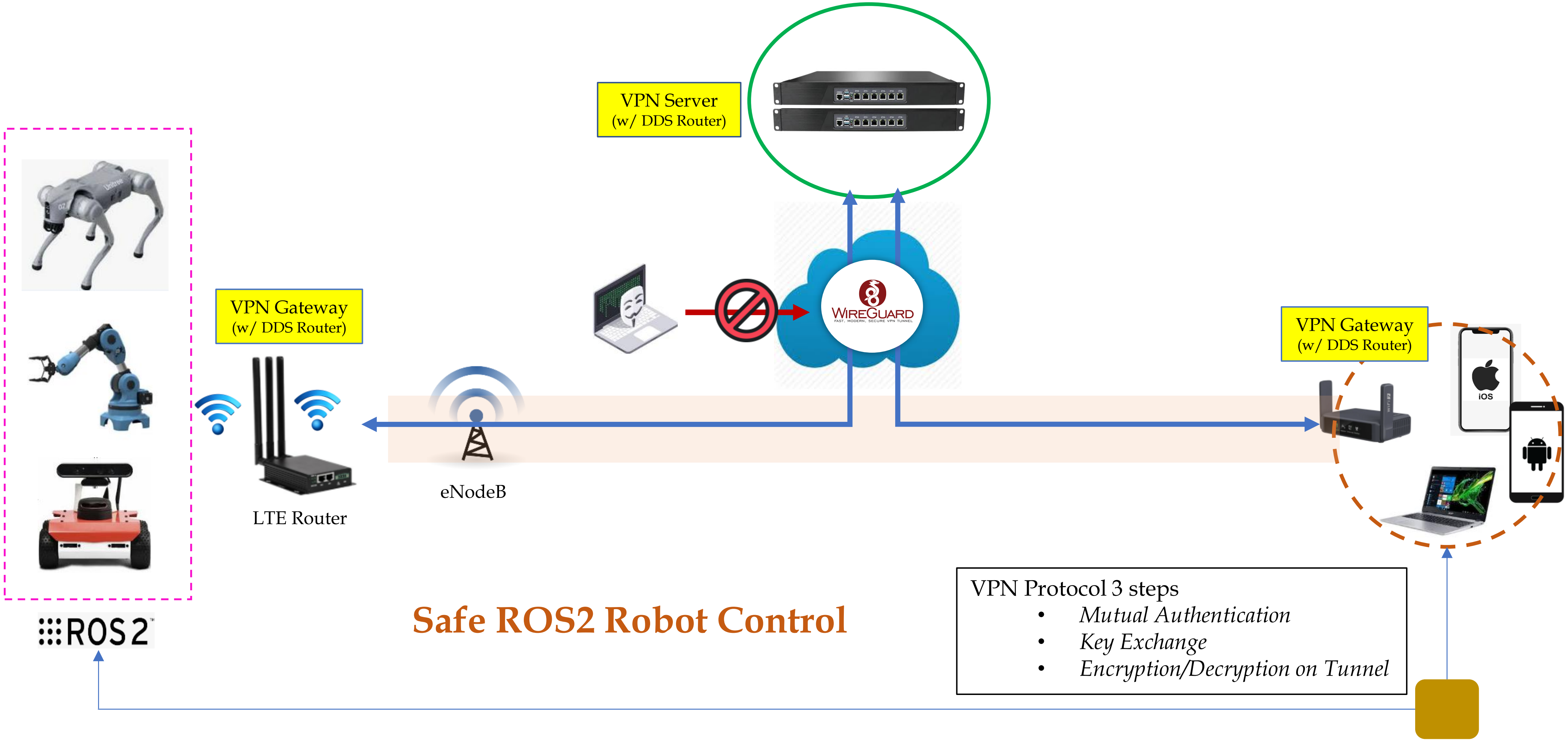


1. Zero Trust P2P Security Platform(5) – Mesh DNS(2)



1. Zero Trust P2P Security Platform(6) – Use Cases(1)

1. Zero Trust P2P Security Platform(6) - Use Cases(2)



VPN Protocol 3 steps

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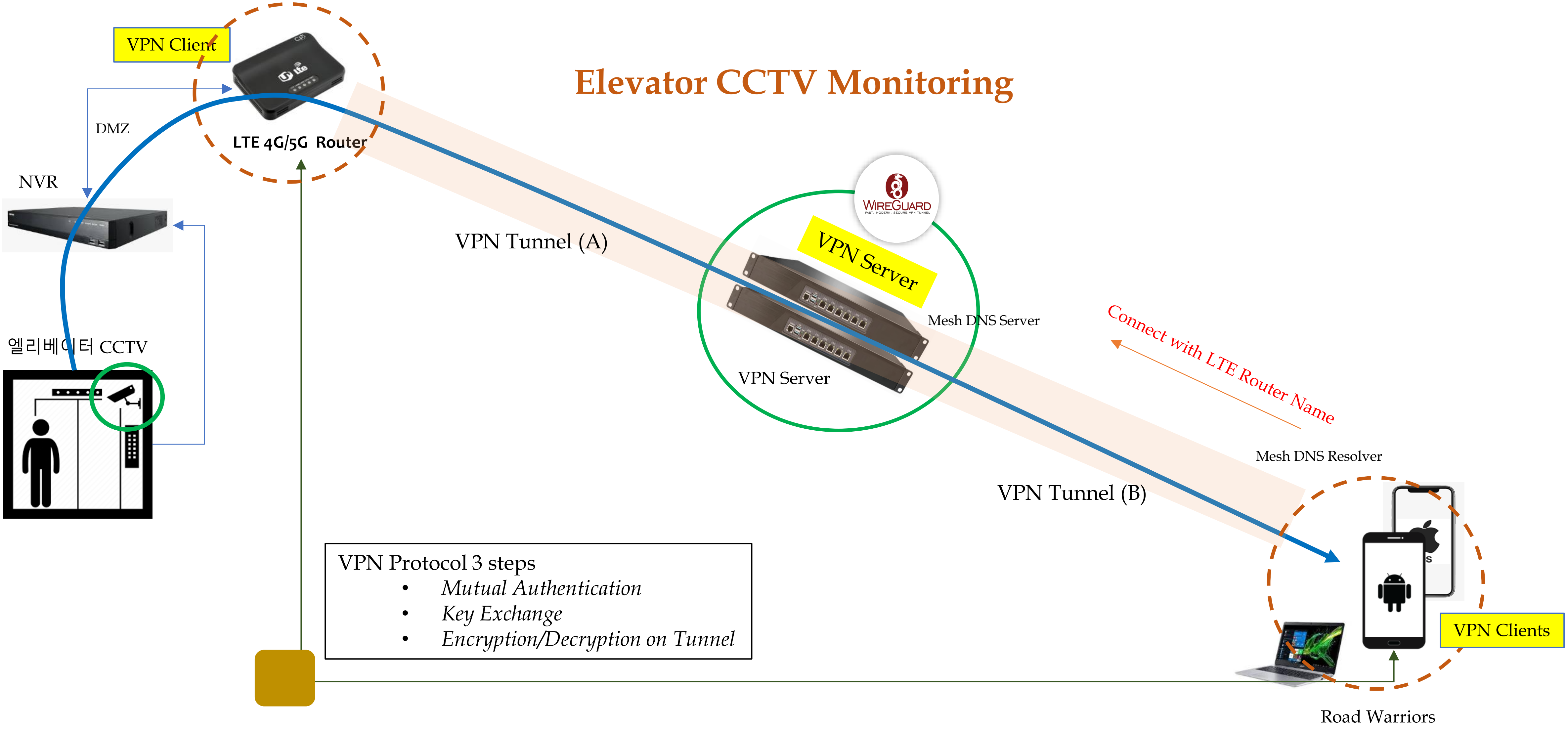
- *Encryption/Decryption on Tunnel*

1. Zero Trust P2P Security Platform(6) - Use Cases(3)



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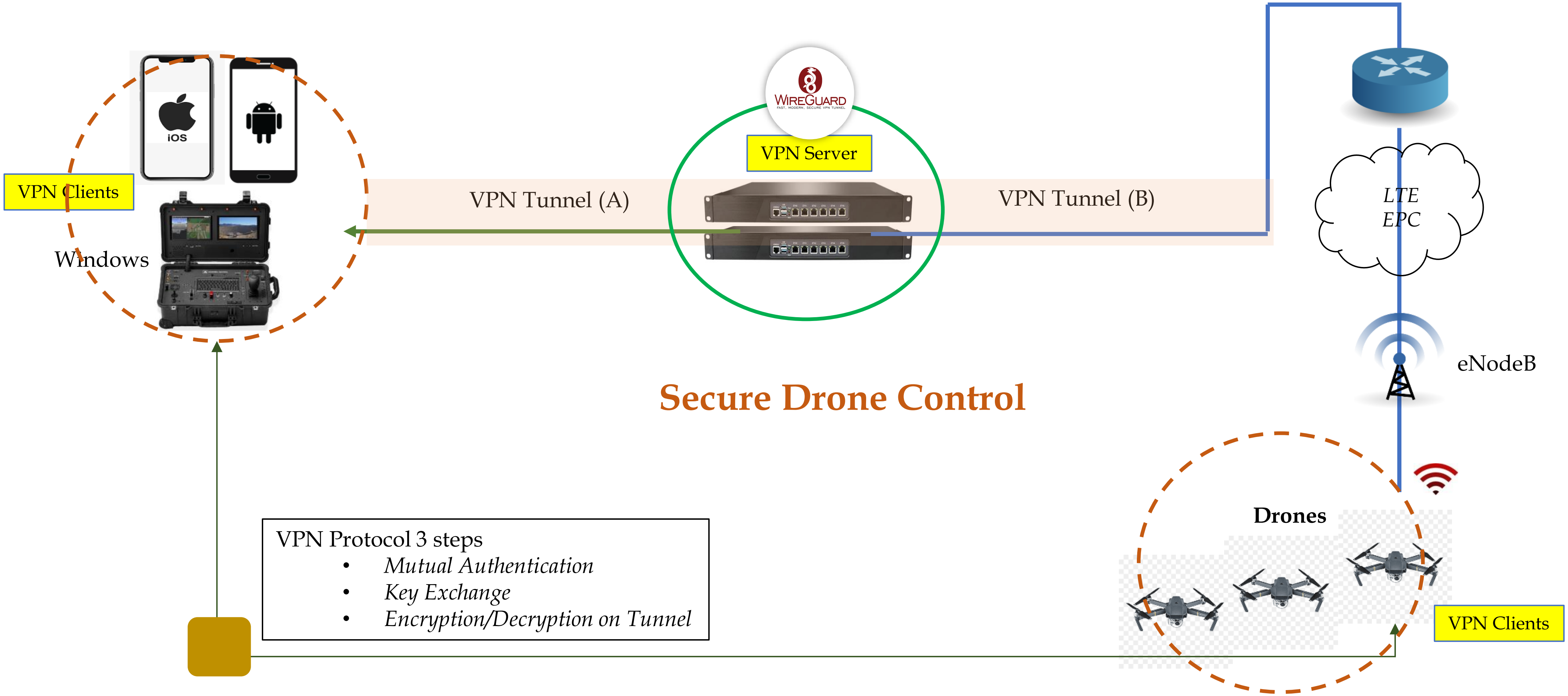

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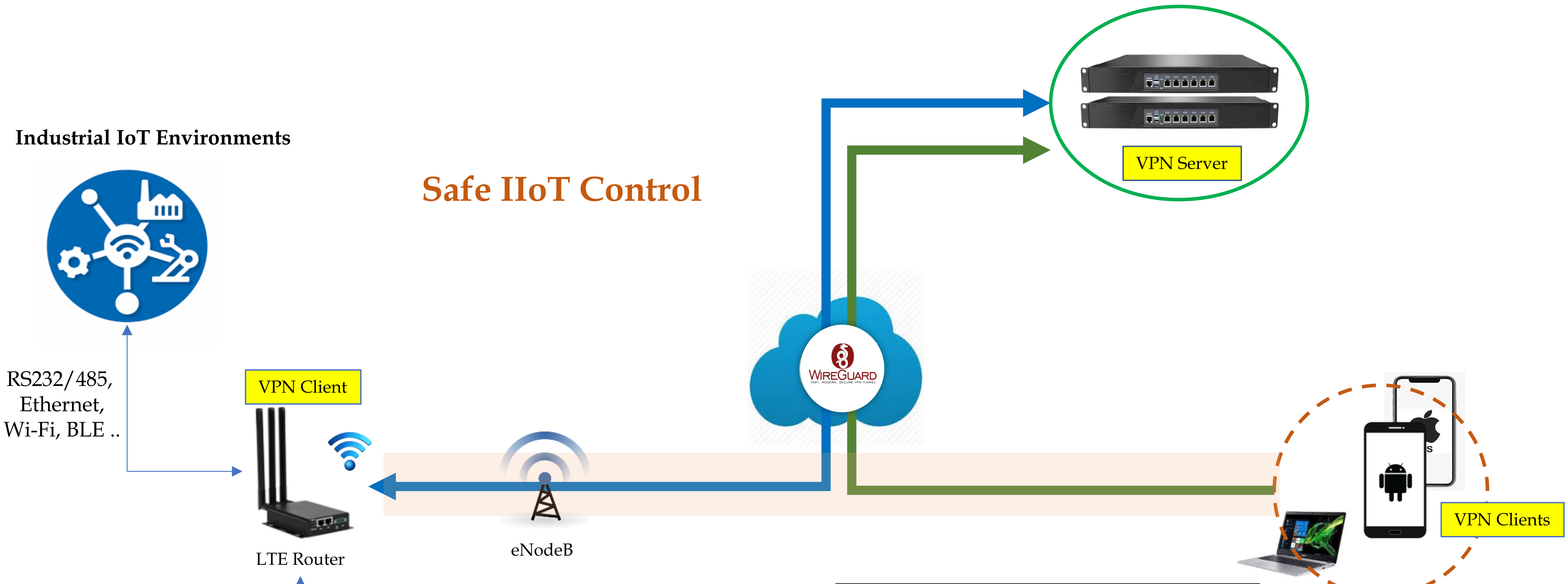
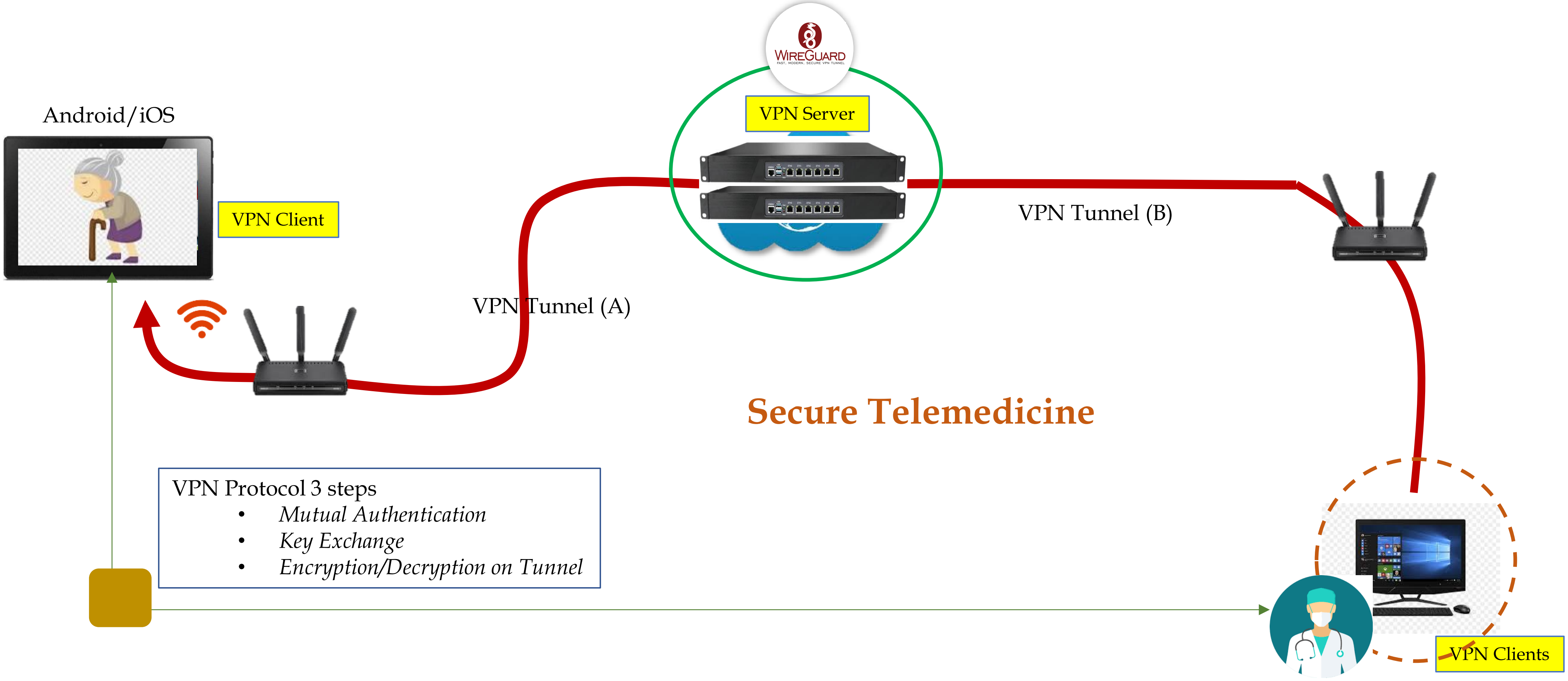


VPN Protocol 3 steps

- *Mutual Authentication*
- *Key Exchange*
- *Encryption/Decryption on Tunnel*

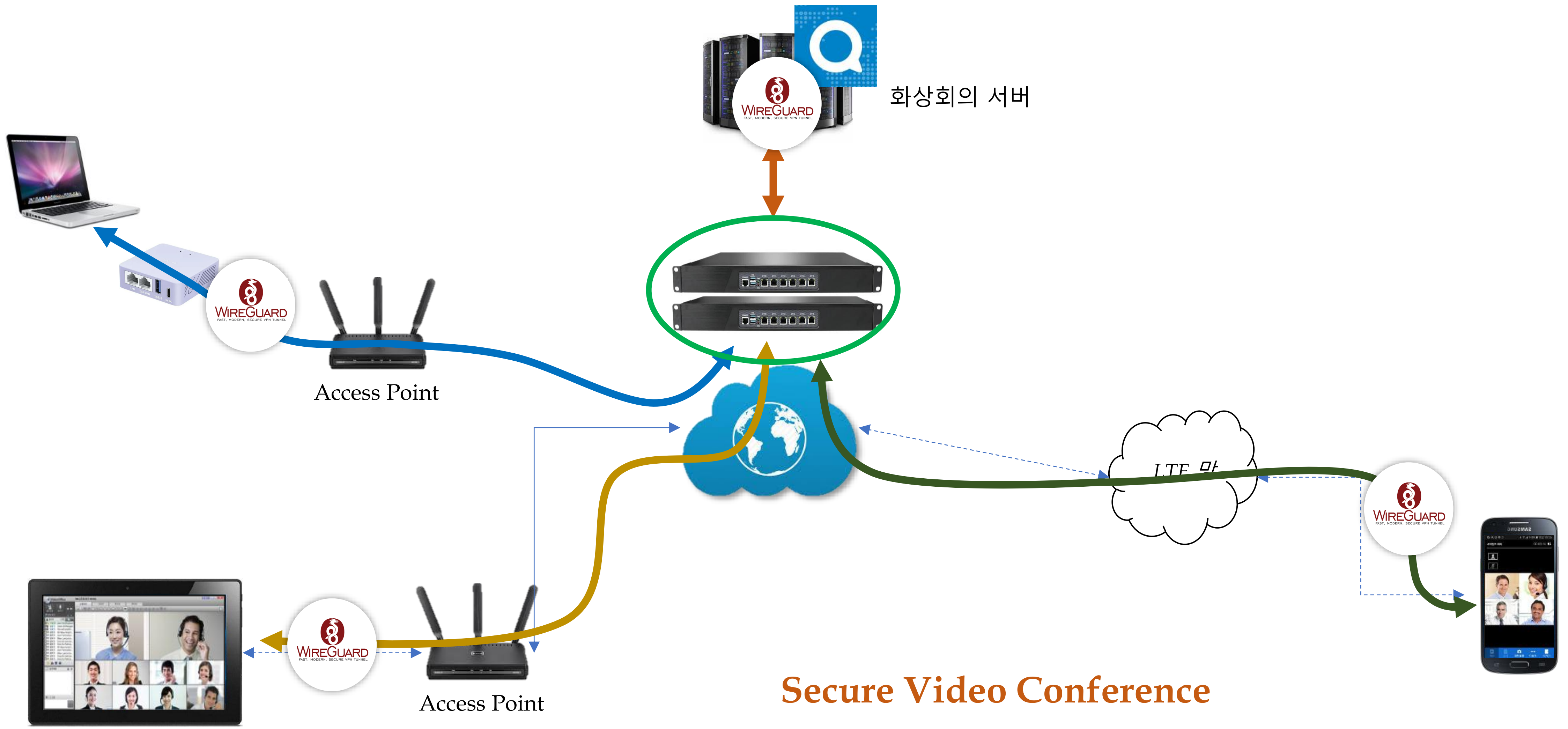
1. Zero Trust P2P Security Platform(6) - Use Cases(6)



VPN Protocol 3 steps

- *Mutual Authentication*
- *Key Exchange*
- *Encryption/Decryption on Tunnel*

1. Zero Trust P2P Security Platform(6) - Use Cases(7)

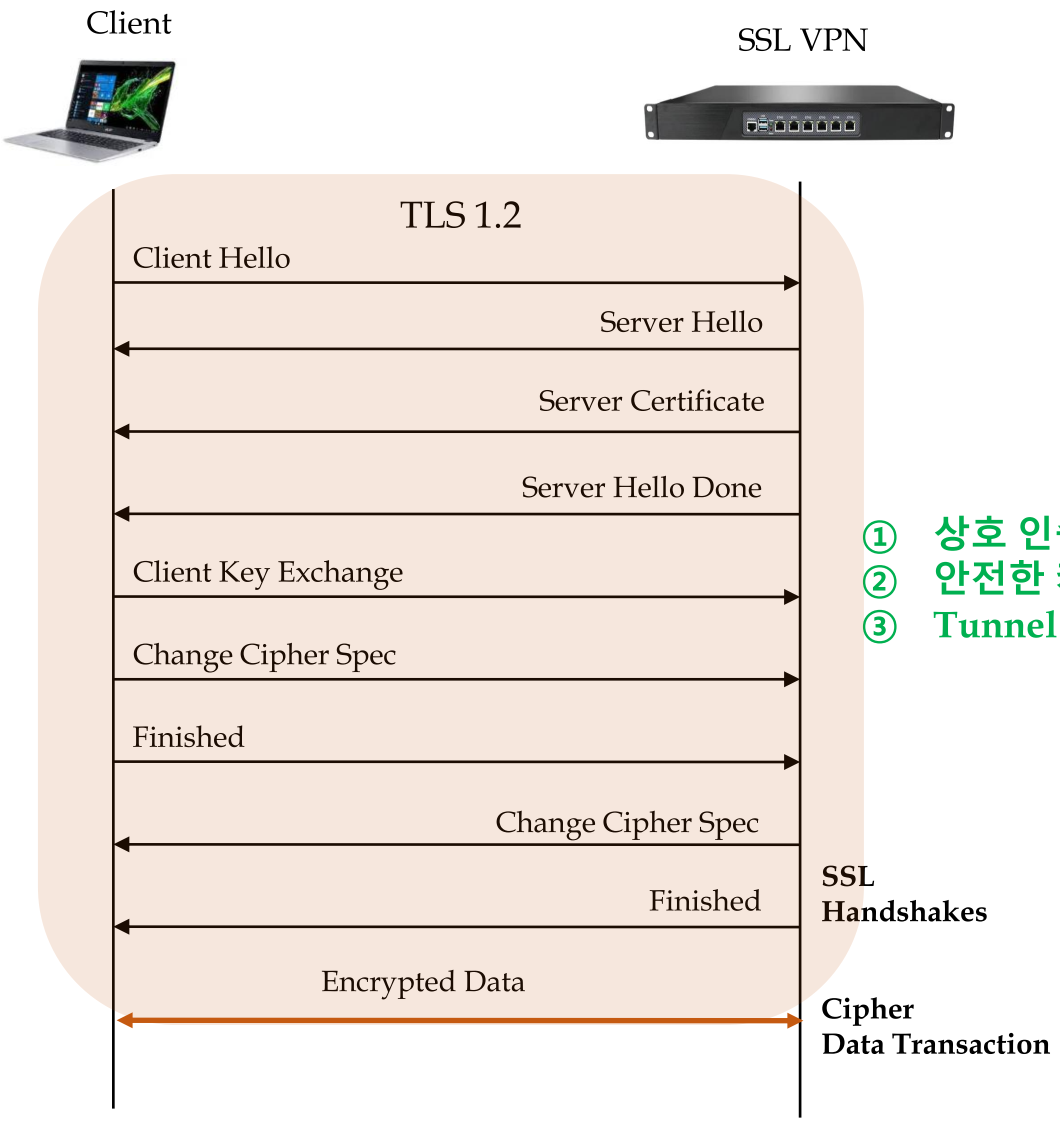


2. WireGuard

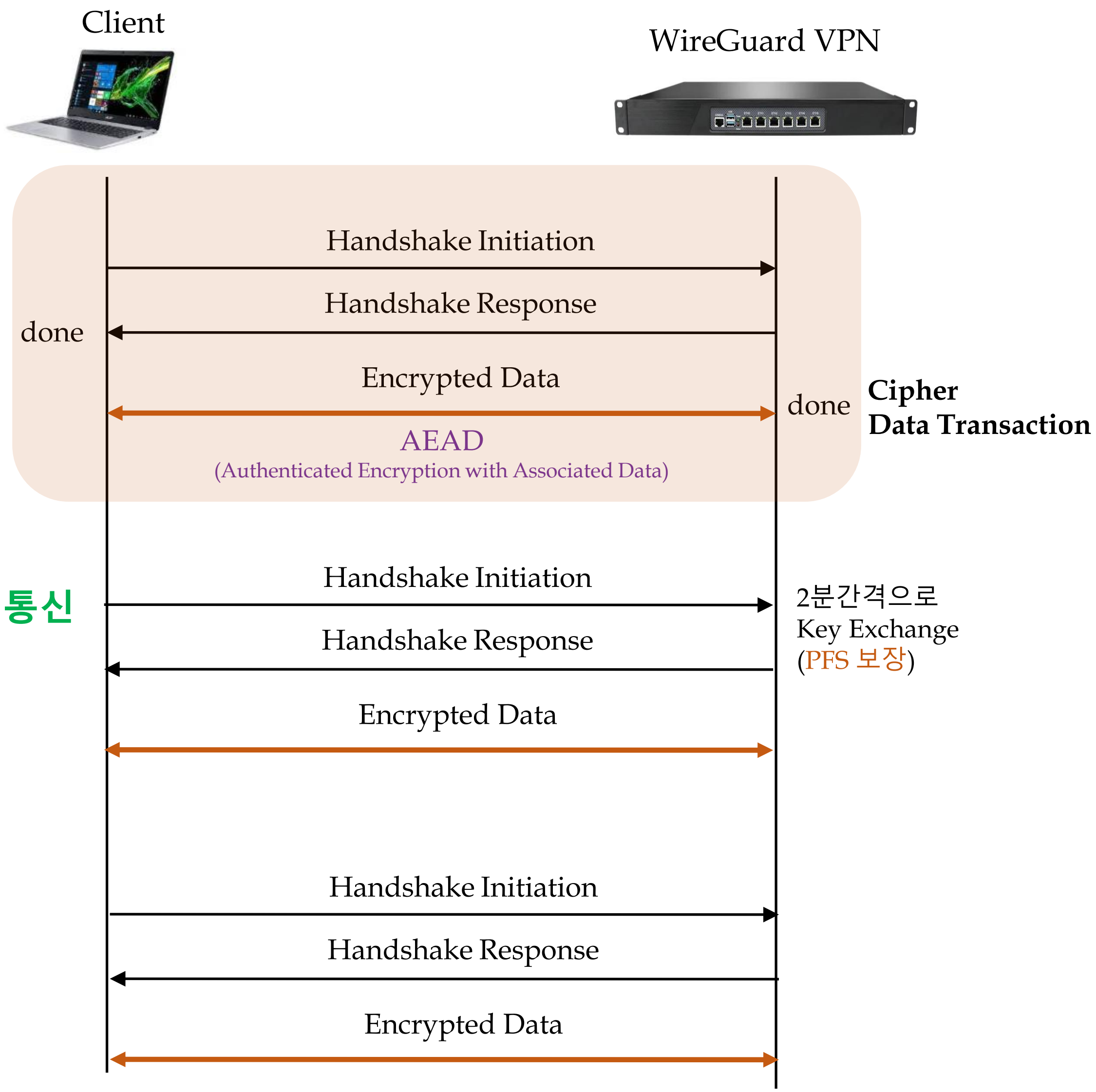
New Generation VPN Protocol

"WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld

2. WireGuard(1) – Architecture(1)

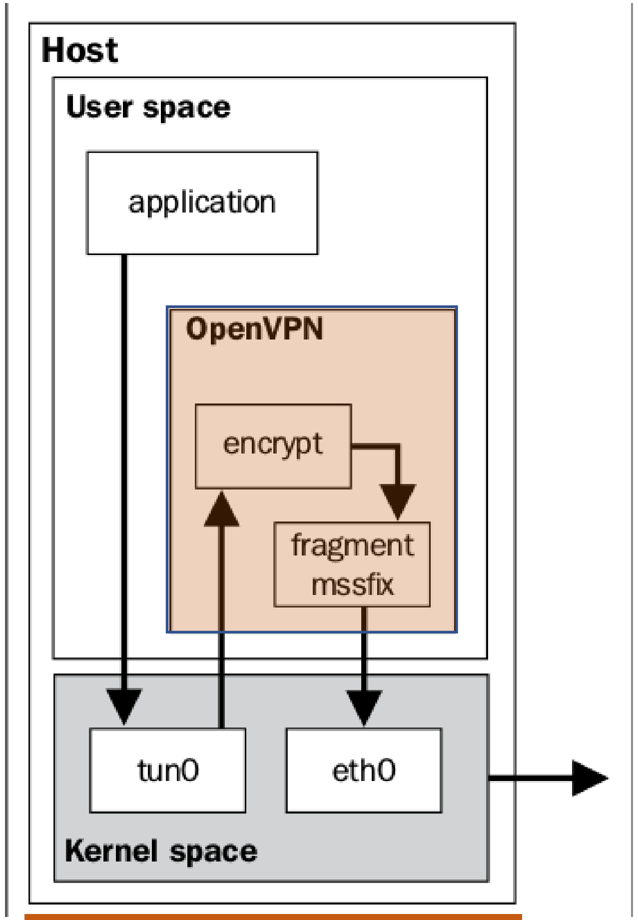


① 상호 인증
② 안전한 키교환
③ Tunnel & 암호 통신



2. WireGuard(1) - Architecture(1)

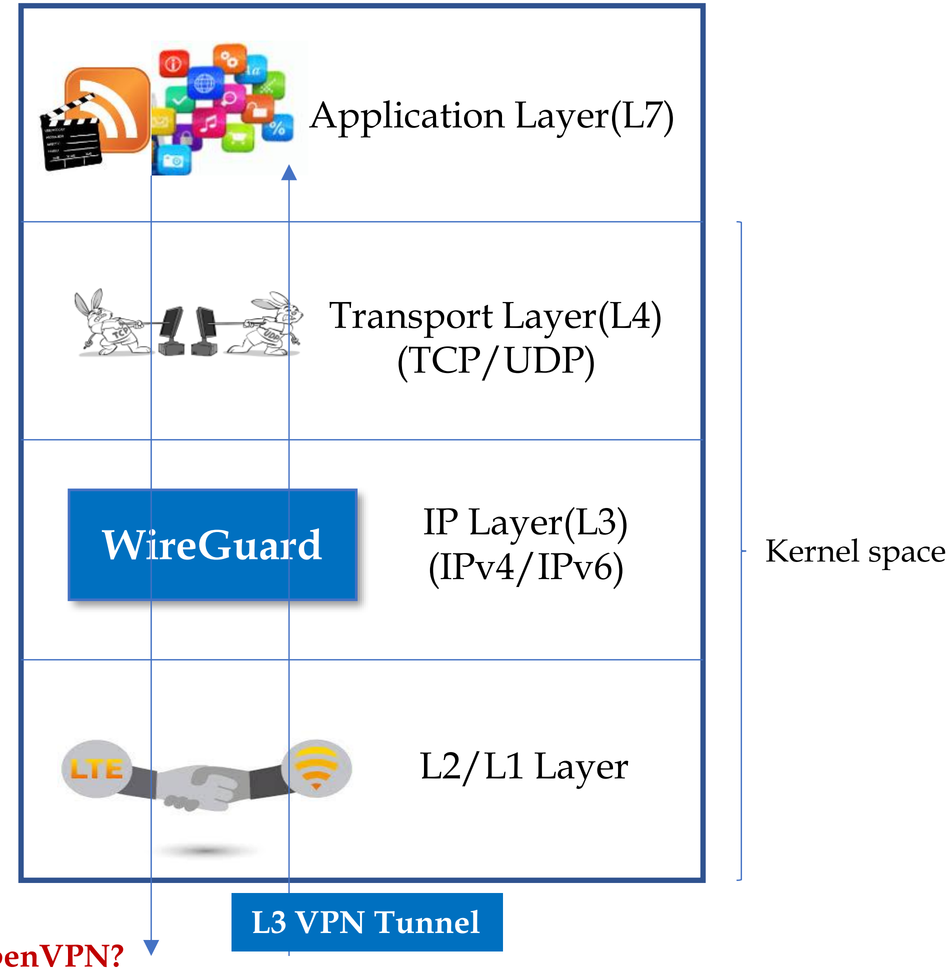
Limitations of OpenVPN



SSL VPN Tunnel(Before DCO)

vs

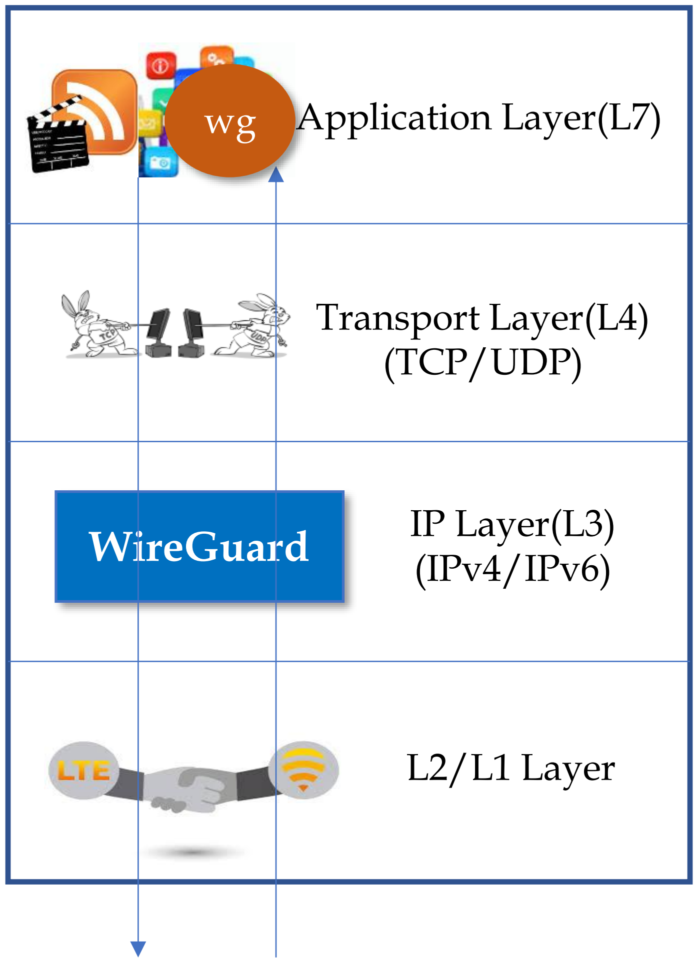
No Limitation. Cryptokey Routing

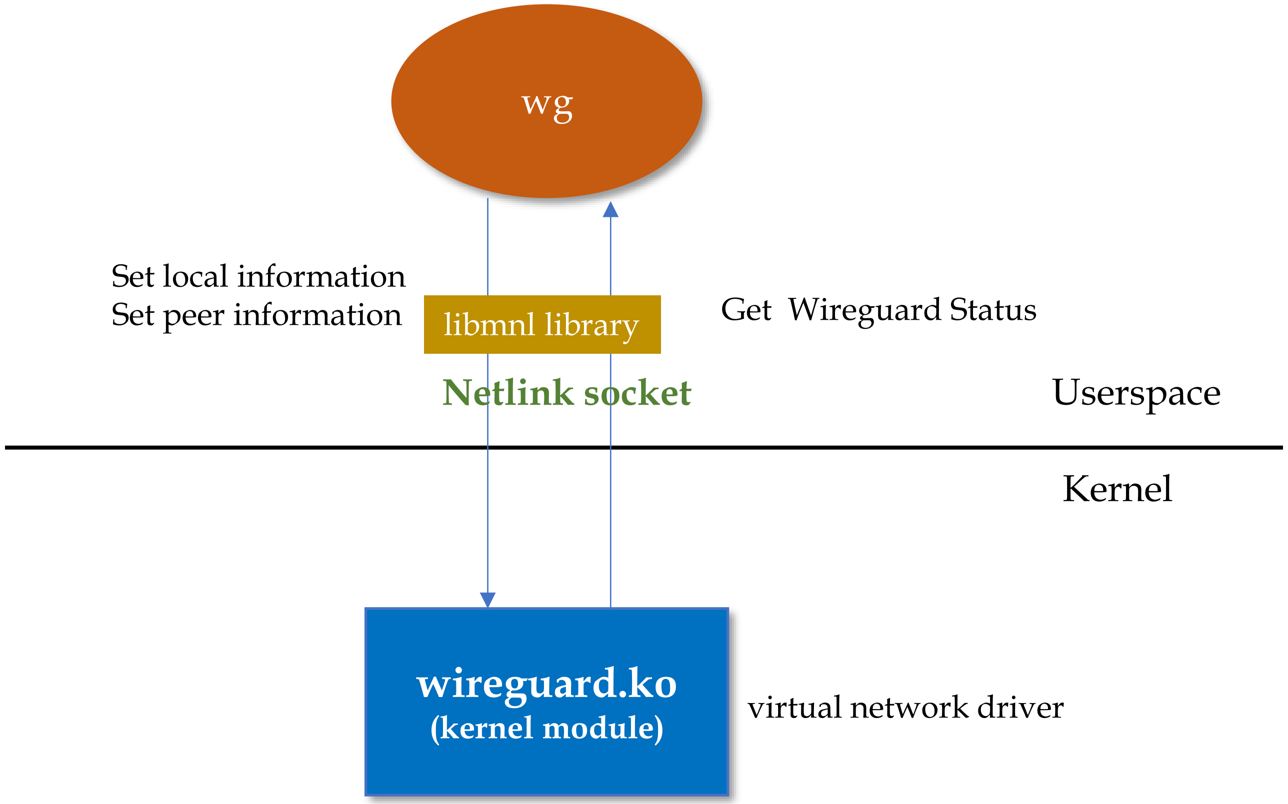


L3 VPN Tunnel

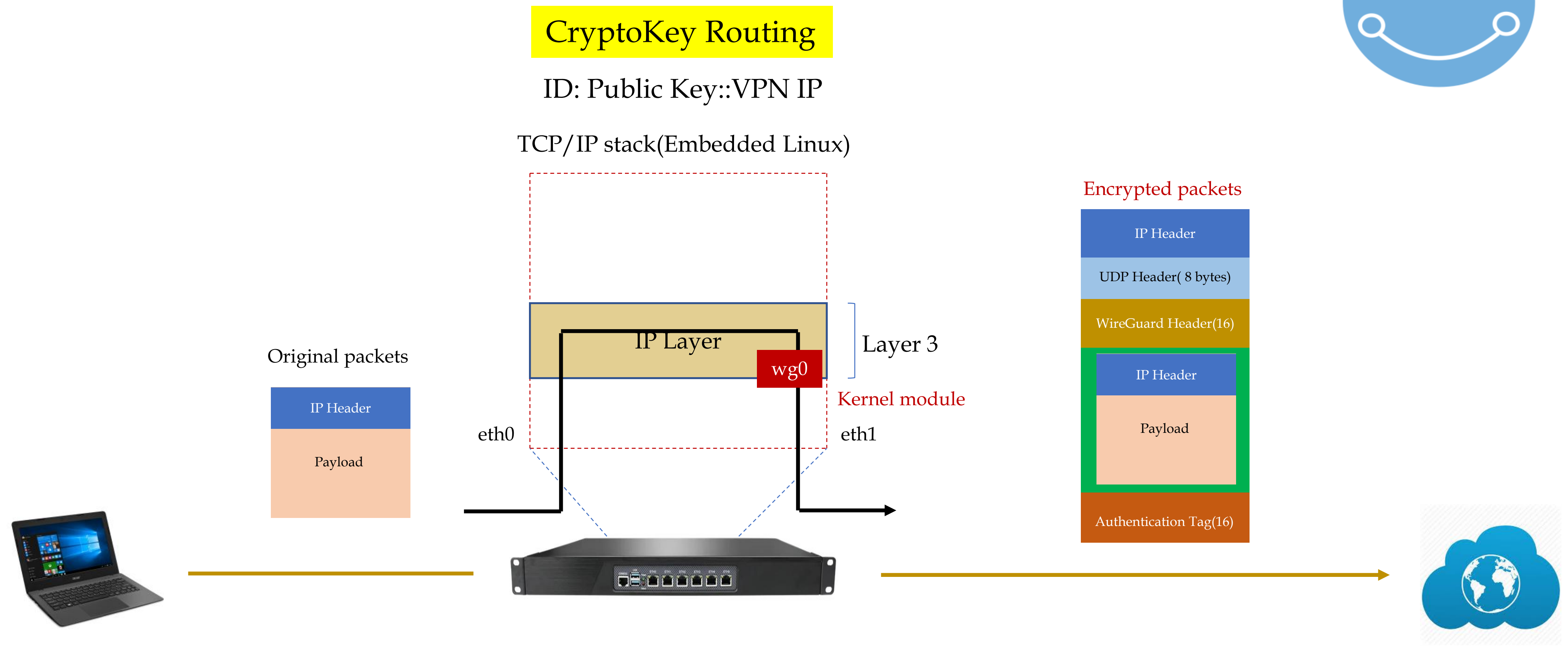
Why is WireGuard faster than OpenVPN?

2. WireGuard(1) - Architecture(2)

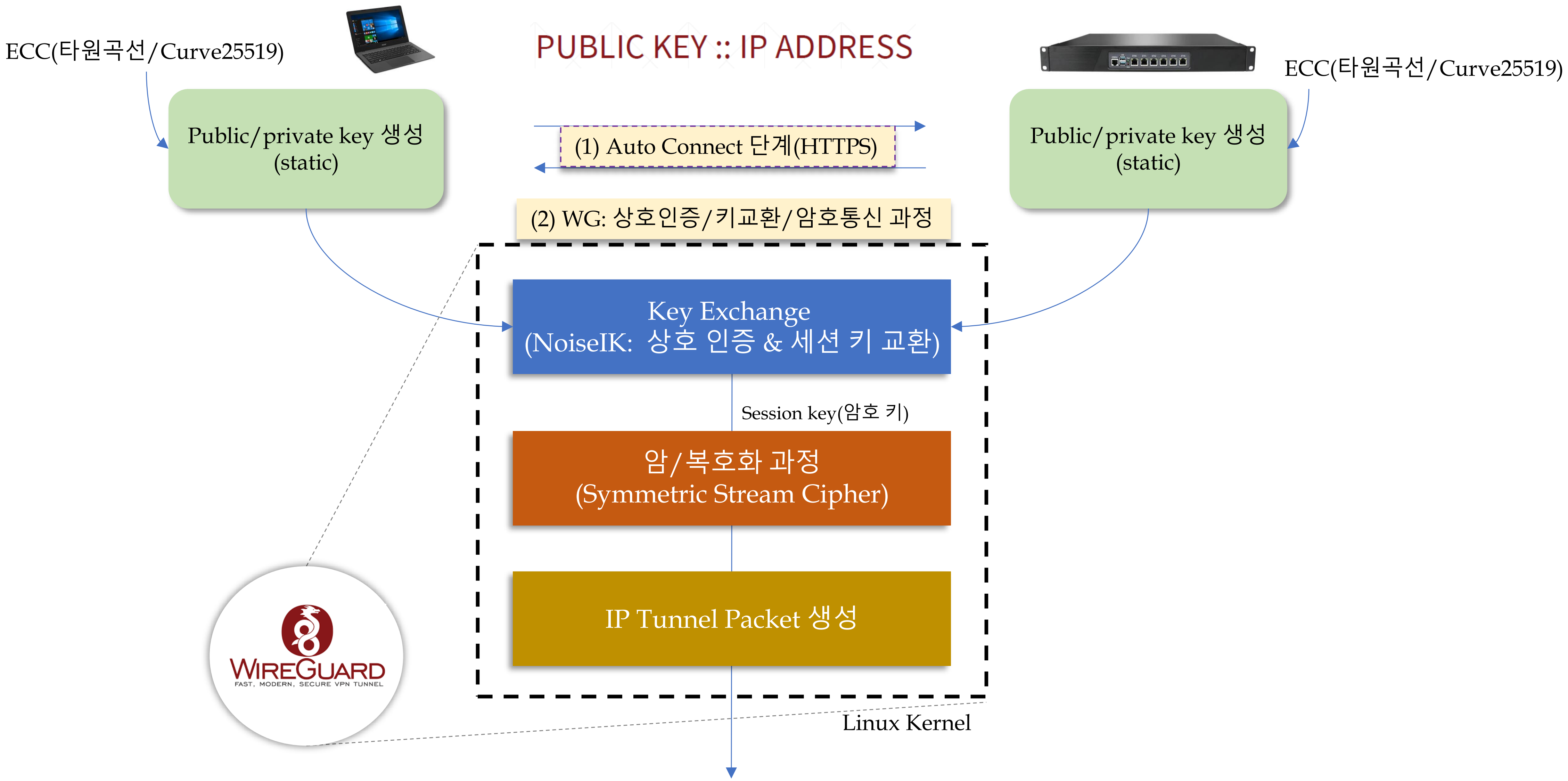




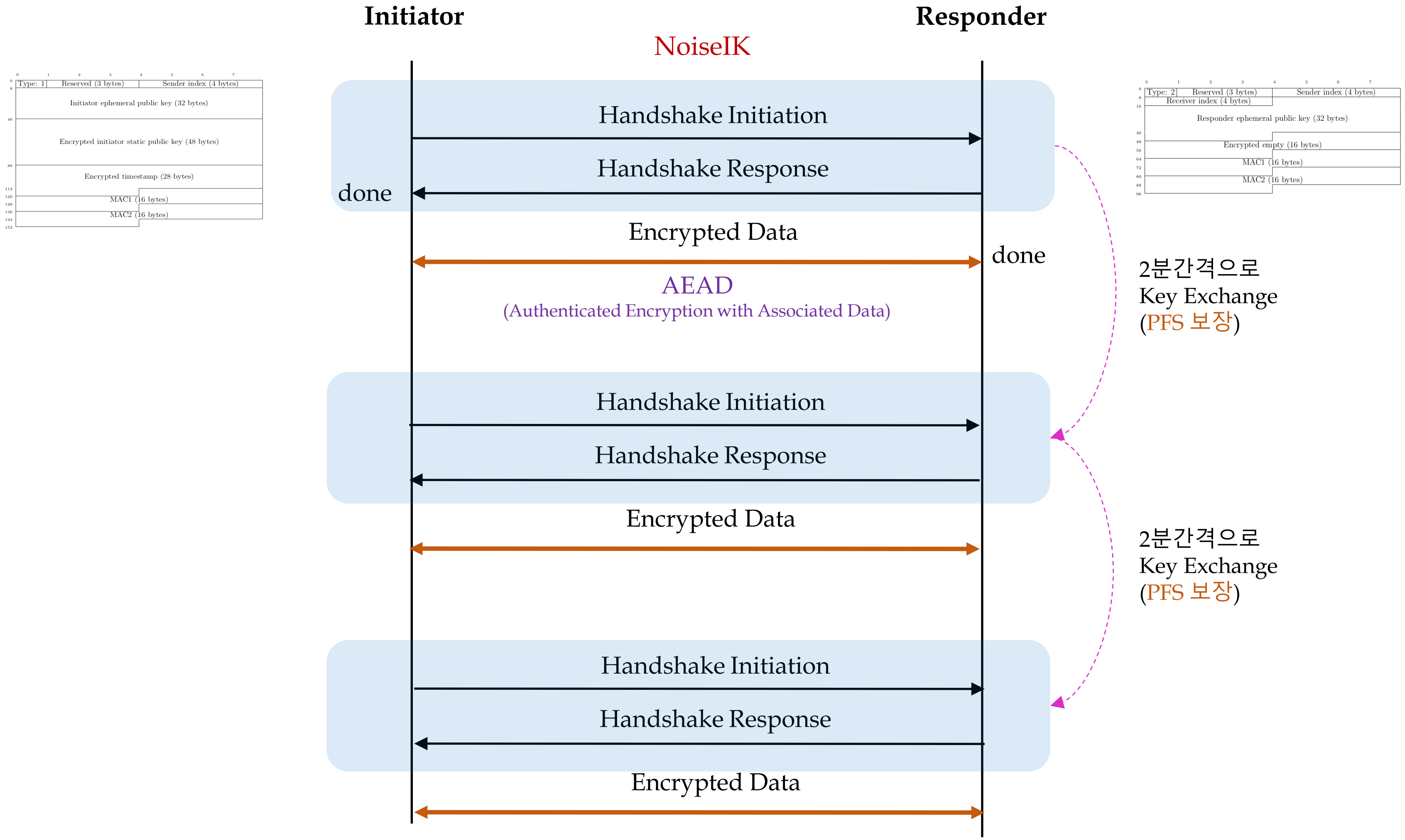
2. WireGuard(1) - Architecture(3)



2. WireGuard(2) - Protocol(1)

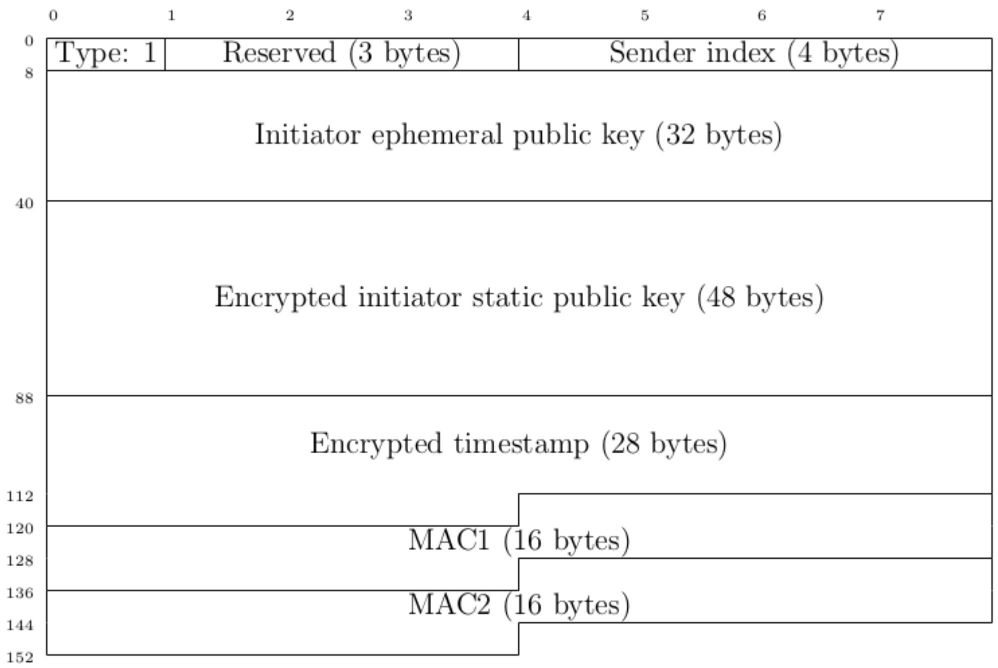


2. WireGuard(2) - Protocol(2)

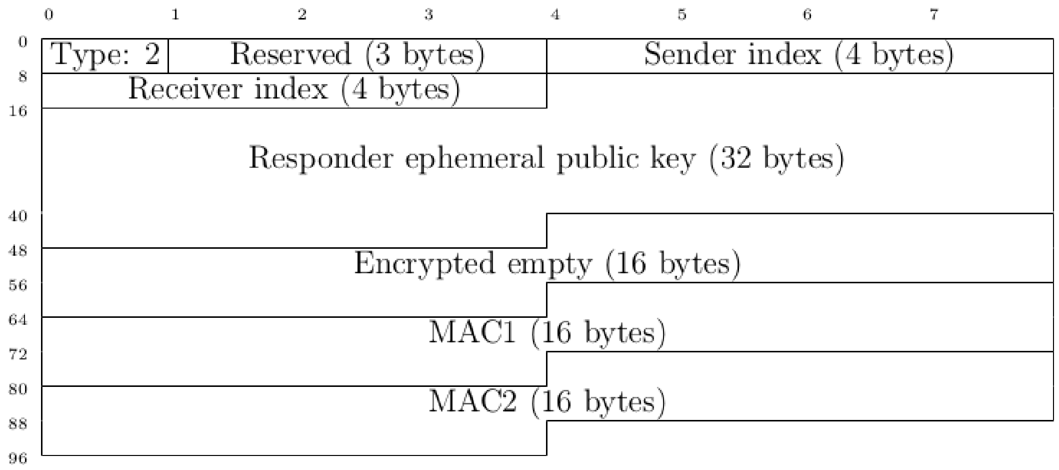


2. WireGuard(2) - Protocol(3)

Initiation Packet Format



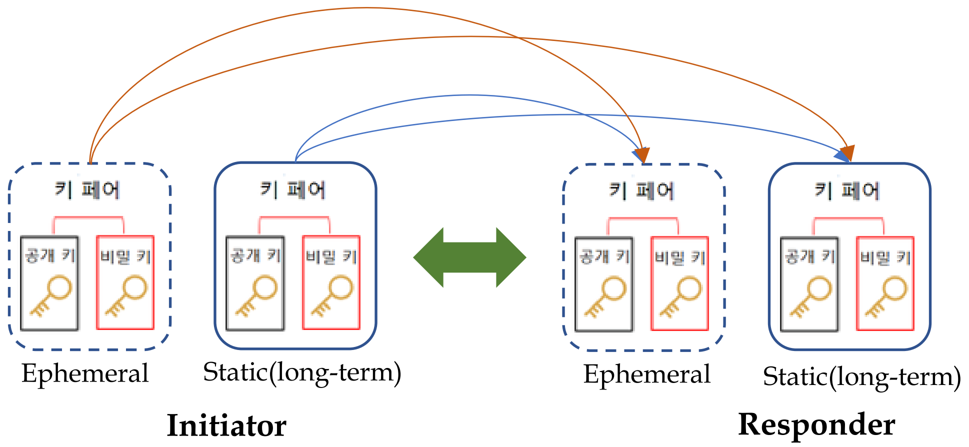
Response Packet Format



Type	Message Name	Length
1	Handshake Initiation	148 bytes
2	Handshake Response	92 bytes
3	Cookie Reply	64 bytes
4	Transport Data	$\geq$ 32 bytes

2. WireGuard(2) - Protocol(4)

Session keys = **Noise**(
ECDH(ephemeral, static),
ECDH(static, ephemeral),
ECDH(ephemeral, ephemeral),
ECDH(static, static)
)



256bits(32bytes)

AEAD

상호 인증
및
키 교환

Initiator has long-term DH key-pair $(\mathbf{ssk}_i, \mathbf{spk}_i)$
Responder has long-term DH key-pair $(\mathbf{ssk}_r, \mathbf{spk}_r)$

Initiator		Responder
$(\mathbf{esk}_i, \mathbf{epk}_i) \leftarrow \mathrm{DH.Gen}()$		
	$\xrightarrow{\mathbf{epk}_i}$	
		$(\mathbf{esk}_r, \mathbf{epk}_r) \leftarrow \mathrm{DH.Gen}()$
	$\xleftarrow{\mathbf{epk}_r}$	
$k_1 \leftarrow \mathrm{DH.Shared}(\mathbf{esk}_i, \mathbf{spk}_r)$		$k_2 \leftarrow \mathrm{DH.Shared}(\mathbf{ssk}_r, \mathbf{epk}_i)$
$k_2 \leftarrow \mathrm{DH.Shared}(\mathbf{ssk}_i, \mathbf{epk}_r)$		$k_3 \leftarrow \mathrm{DH.Shared}(\mathbf{esk}_r, \mathbf{spk}_i)$
$k_3 \leftarrow \mathrm{DH.Shared}(\mathbf{esk}_i, \mathbf{epk}_r)$		$k_4 \leftarrow \mathrm{DH.Shared}(\mathbf{esk}_r, \mathbf{epk}_i)$
$k_4 \leftarrow \mathrm{DH.Shared}(\mathbf{ssk}_i, \mathbf{spk}_r)$		$k_1 \leftarrow \mathrm{DH.Shared}(\mathbf{ssk}_r, \mathbf{spk}_i)$

Derive session key from k_1, k_2, k_3, and k_4

2. WireGuard(3) – Algorithms(1)

Layer 3(IPv4, IPv6 지원) VPN Protocol

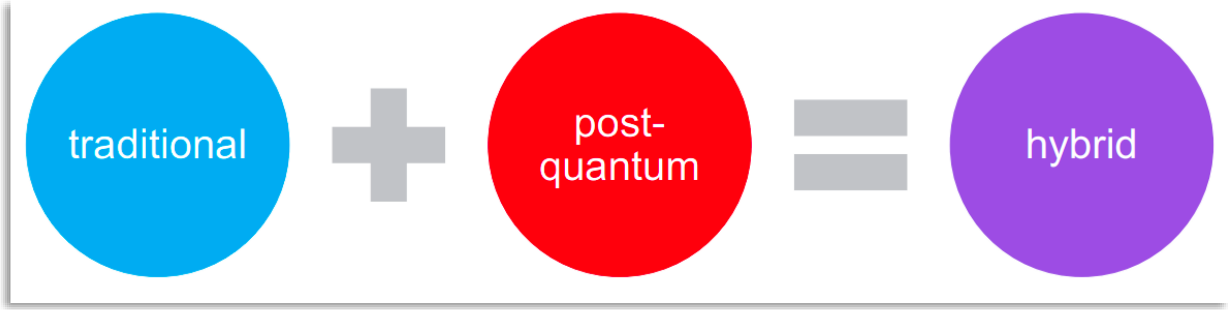
handshake pattern	IKpsk2	(Noise Protocol Framework 기반)
DH function	X25519	
cipher function	ChaCha20-Poly1305	
hash function	BLAKE2s	
prologue	`WireGuard v1 zx2c4 Jason@zx2c4.com` (34 bytes)	

2. WireGuard(3) - Algorithms(2)

보안 알고리즘	상세 내용
NoiseIK **Key 교환 방식** **및** **상호 인증**	**NoiseIK handshake 방식(Noise IKpsk2)** ▪ ECDH(Diffie-Hellman) 기반 ▪ **Curve25519** Public key(32 byte)를 교환 후, 이를 통해 안전하게 shared secret 생성 ▪ Static/Ephemeral private/public key(2 쌍) 이용 ▪ Key 교환 시 아래 기능 보장 ▪ *Perfect forward secrecy 보장, replay attack 방지 기능* ▪ *Identity 감춤 기능 제공, DoS 공격 완화 기능(Cookie),* ▪ *PSK 사용 시, post-quantum security 보장* **Hash 알고리즘** ▪ **BLAKE2** – fast secure hashing 알고리즘 ▪ SHA series 보다 빠름. 즉 MD5 수준임. SHA-3 보안 강도. **PRF & MAC 및 Key Derivation 함수** ▪ **SipHash4** (Hash table lookup 용으로 사용) ▪ **HKDF** (HMAC-based Key Derivation Function) 사용
암호 알고리즘 **Authenticator 알고리즘** **(ChaCha20-Poly1305 AEAD)**	**ChaCha20** – 256 bits stream cipher(20 round cipher Salsa20 기반) ▪ Stream cipher는 일반적인 block cipher(예: AES-256-CBC)에 비해 속도가 빠름 ▪ key(32 bytes)는 대칭키를 사용(즉, 암호화 용 키와 복호화 용 키 동일) ▪ **Video/Audio 등 stream 암호화에 적합**
	Poly1305 - message authentication code 알고리즘(16 bytes output 생성)

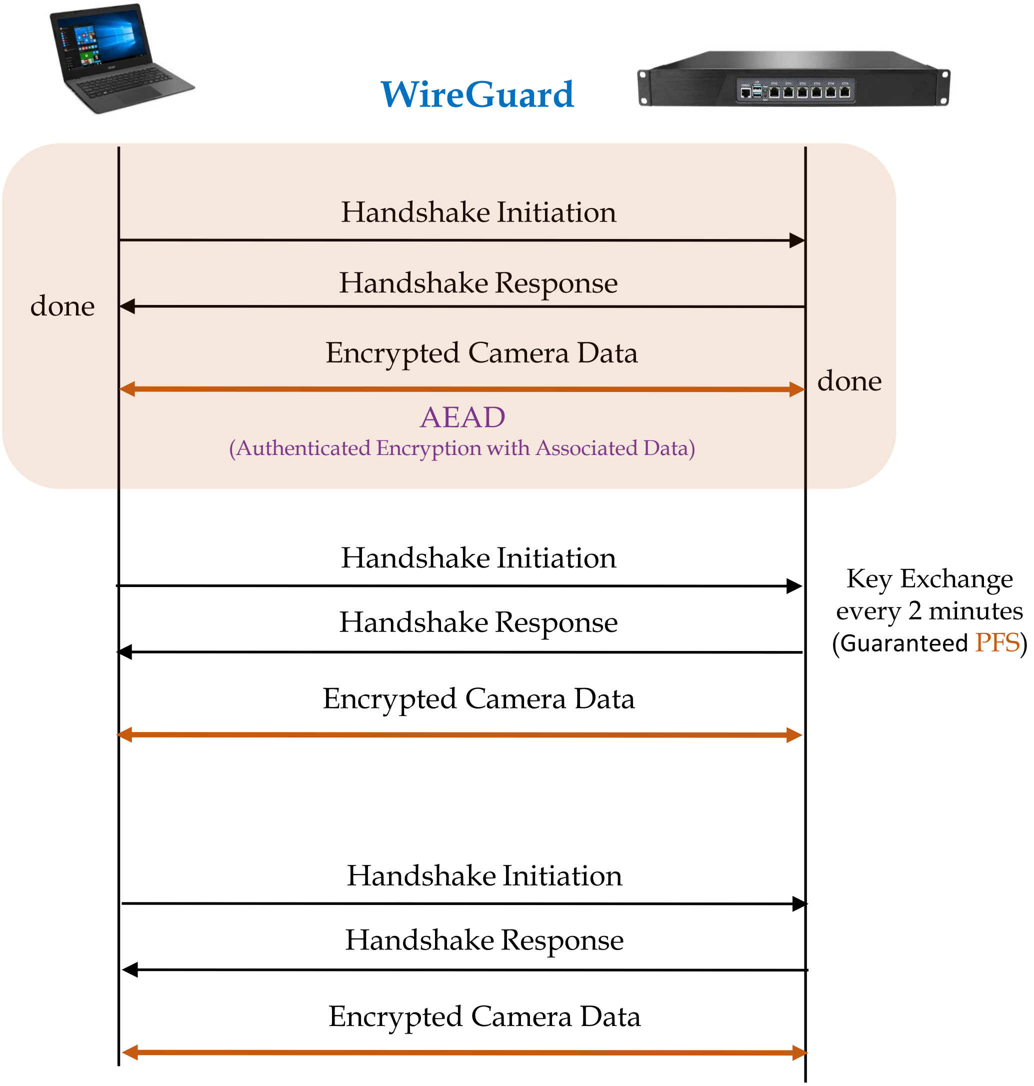
Noise_IKpsk2_25519_ChaChaPoly_BLAKE2s

2. WireGuard(4) – PQC(1)

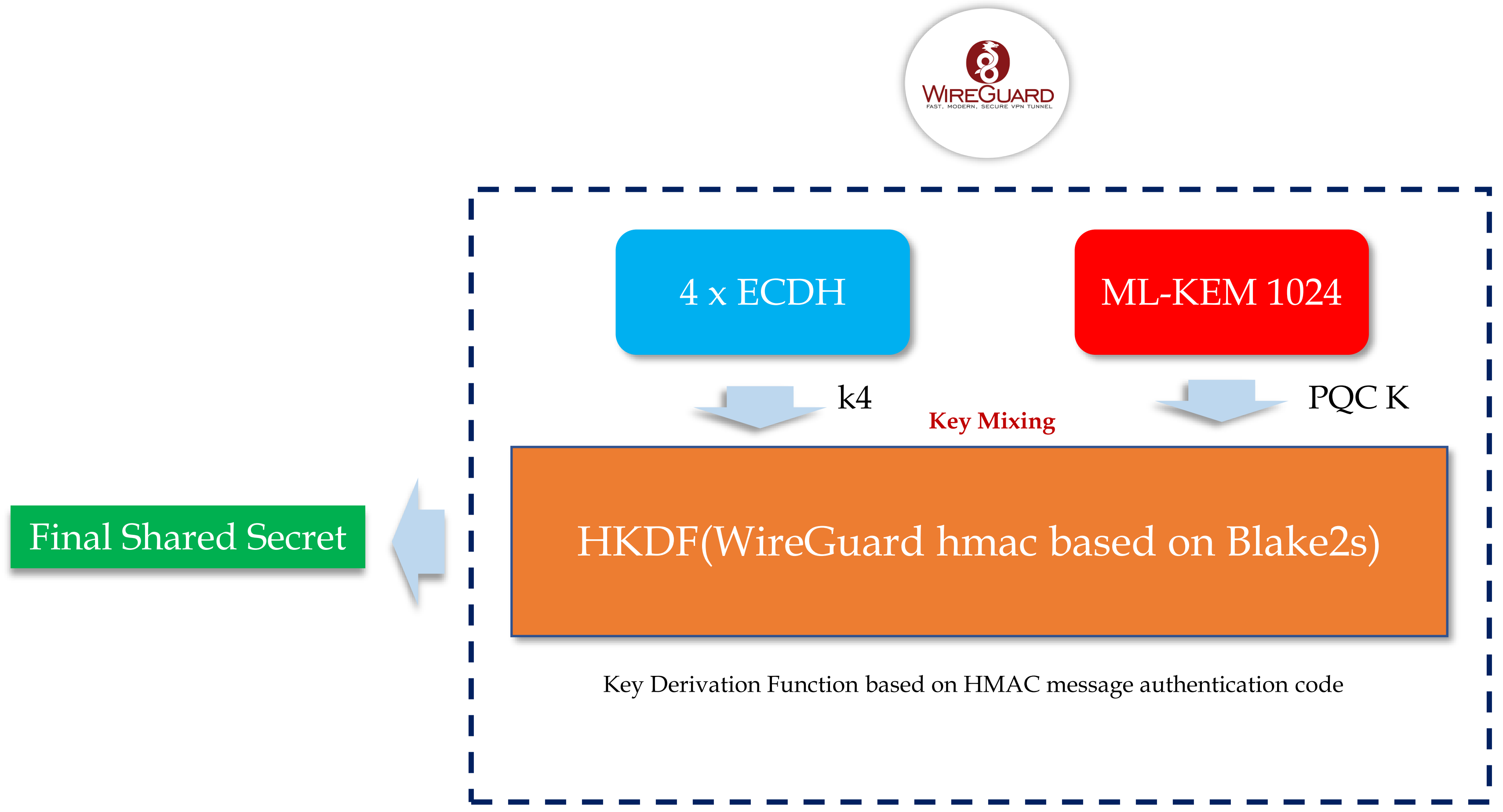


WireGuard **ML-KEM 1024** **Quantum Safe WireGuard**

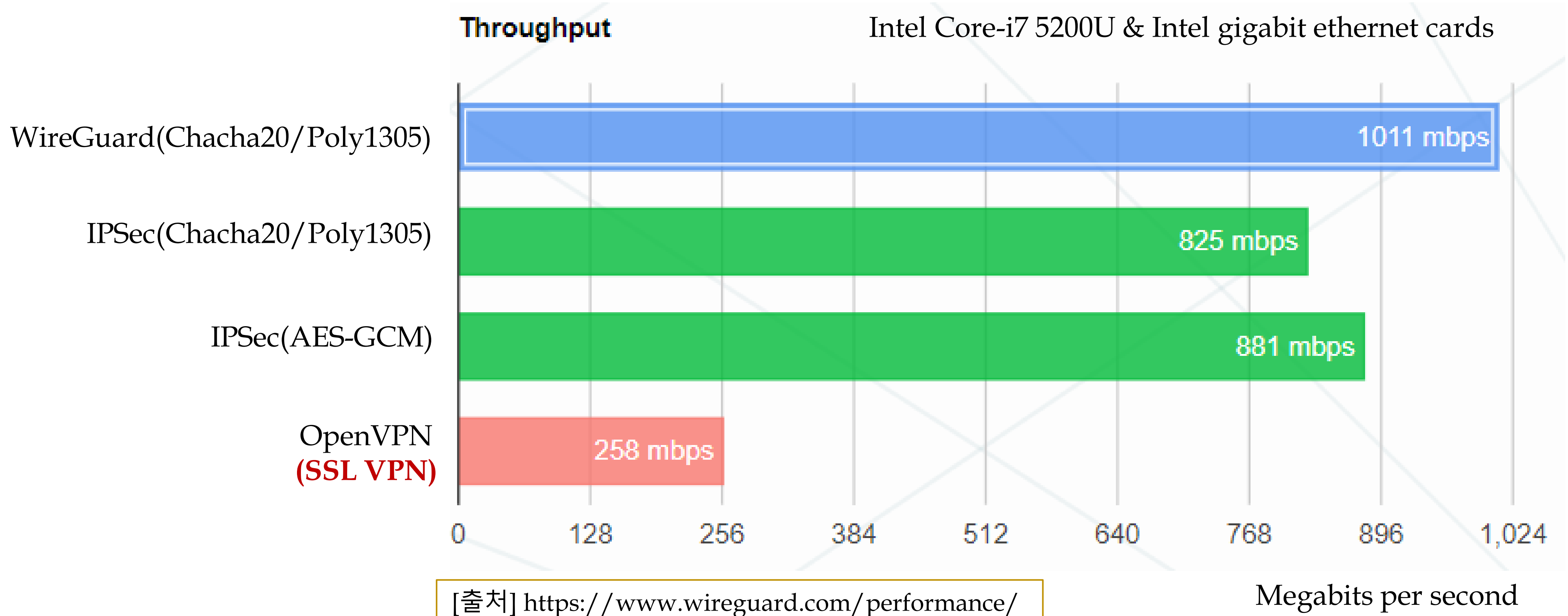
PQC 알고리즘이 통합된 WireGuard Protocol



2. WireGuard(4) – PQC(2)



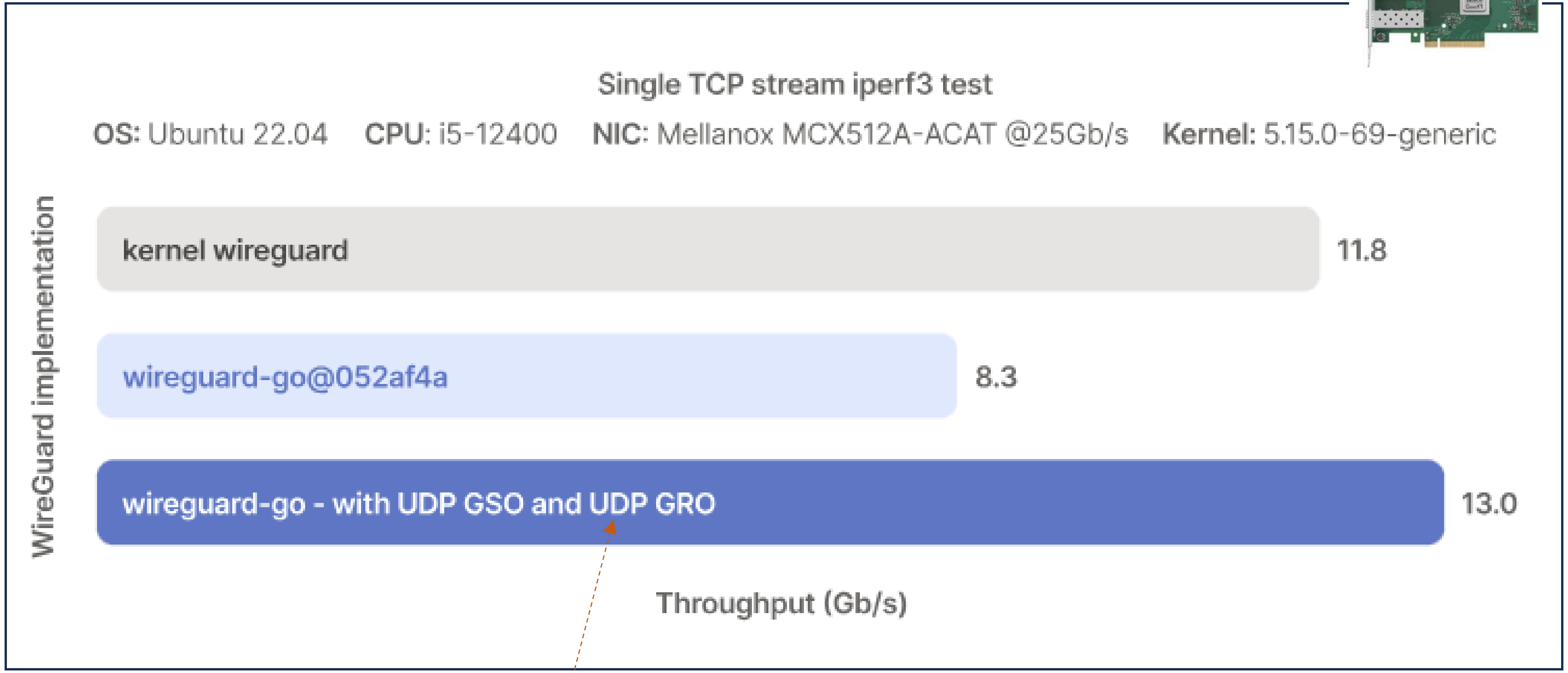
2. WireGuard(5) – Performance(1)



[출처] https://www.wireguard.com/performance/

[주의] OpenVPN DCO(Data Channel Offload)는 위의 그림보다는 훨씬 빠르다.

2. WireGuard(5) - Performance(2)



Packet Vector Processing
UDP GRO/GSO
Checksum Offload

그림 출처: https://tailscale.com/blog/more-throughput

2. WireGuard(5) - Performance(3)

- **WireGuard VPN의 우수한 성능 (Multi-Core 환경)**

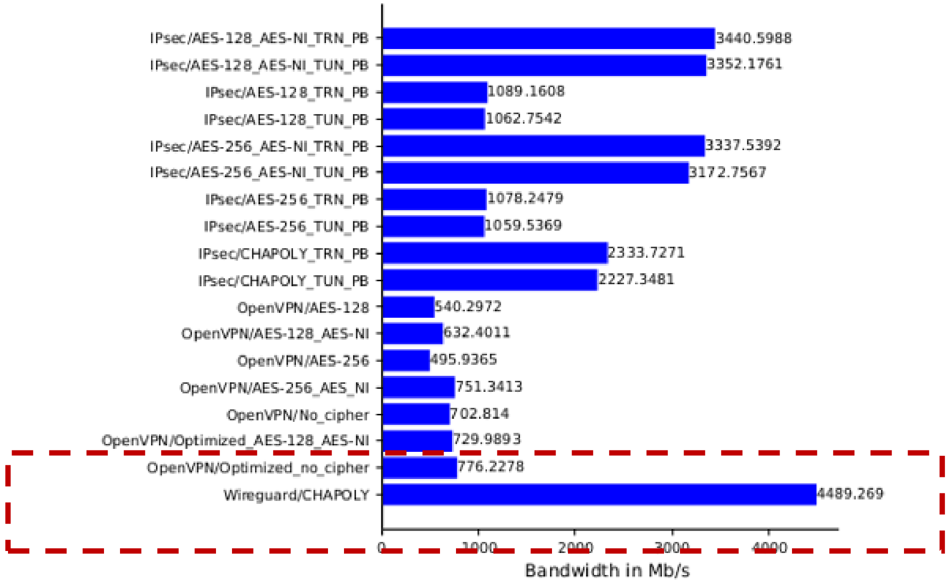
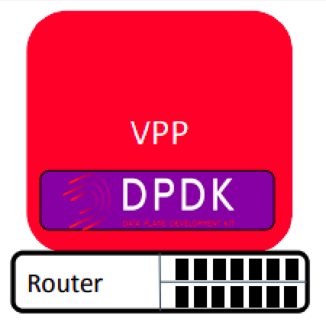


그림 출처: Performance Comparison of VPN Solutions, Lukas Osswald

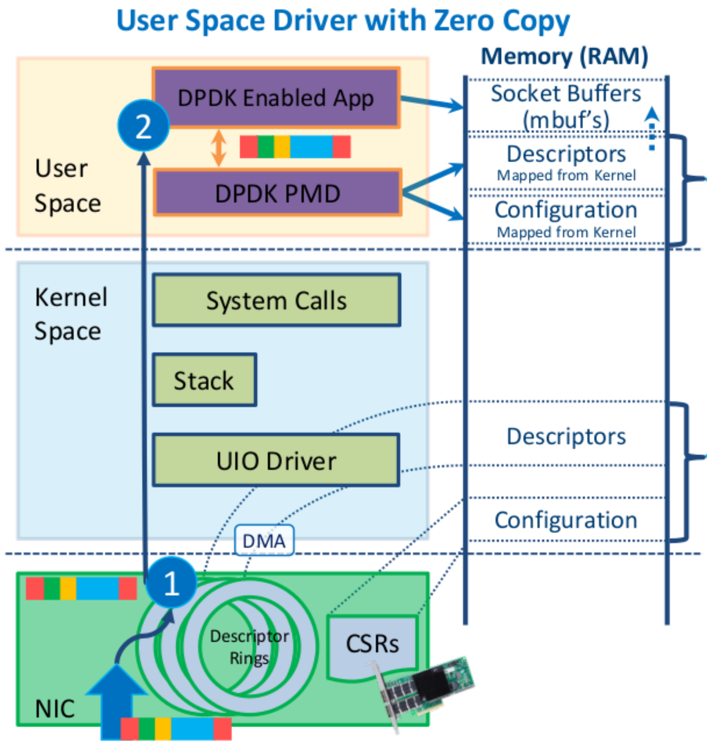
- **Signature 없이 인증 & 키 교환하는 AKE 기법 도입**
- **최신 암호 알고리즘 및 암호 가속 기능으로 무장**
  - Curve25519, Blake2s, ChaCha20, Poly1305, SipHash4
  - X86_64 AVX2, AVX512
- **Perfect Forward Secrecy 보장(2분 간격 Handshaking)**
  - 1-RTT만에 Handshaking 완료
- **Replay Attack 방지 기능(Counter 필드 유지/관리)**
- **Identity Hiding 기능**
- **DoS 공격 완화 기능(Cookie 활용)**

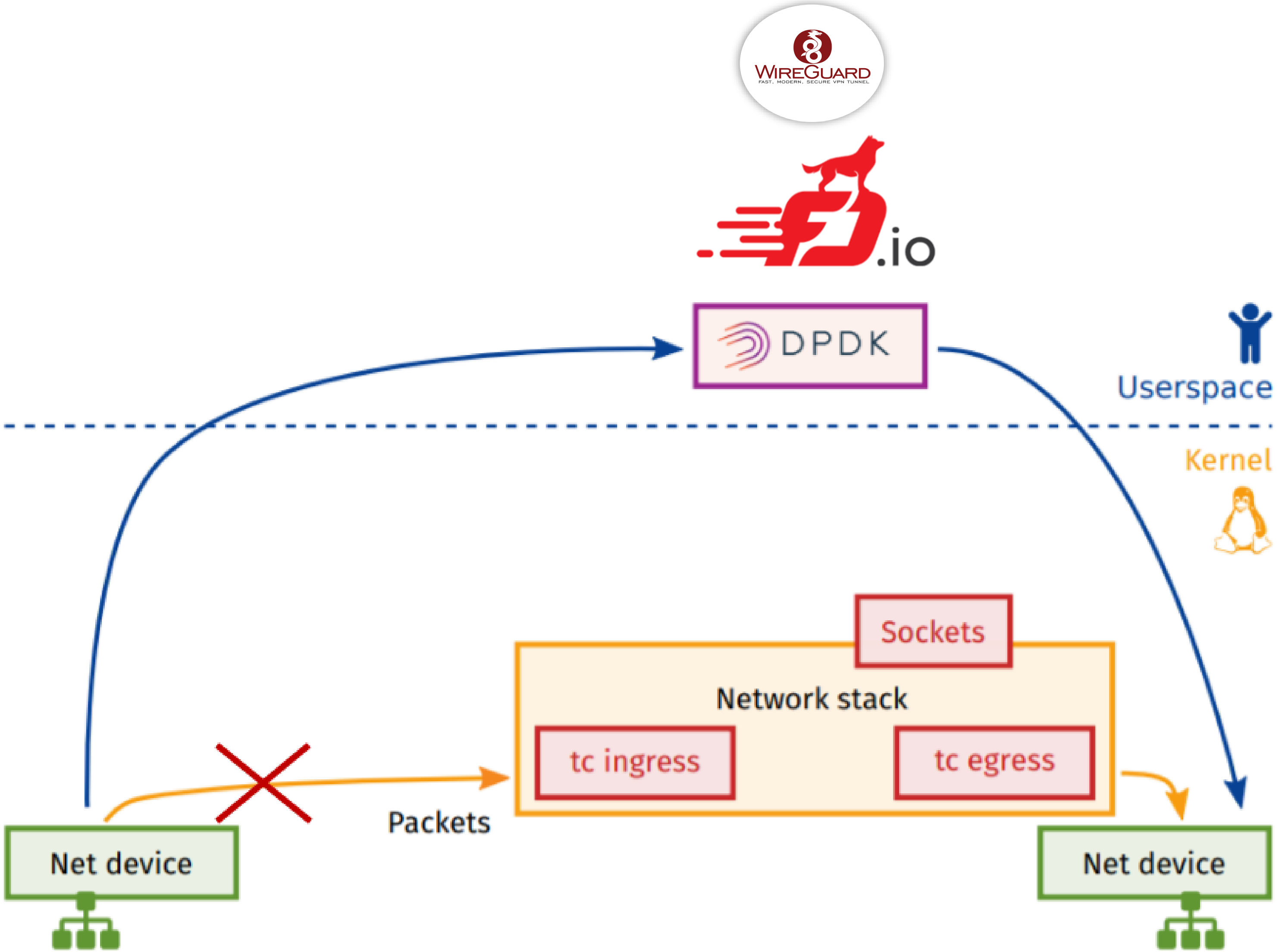
3. FD.io VPP



Software ASIC

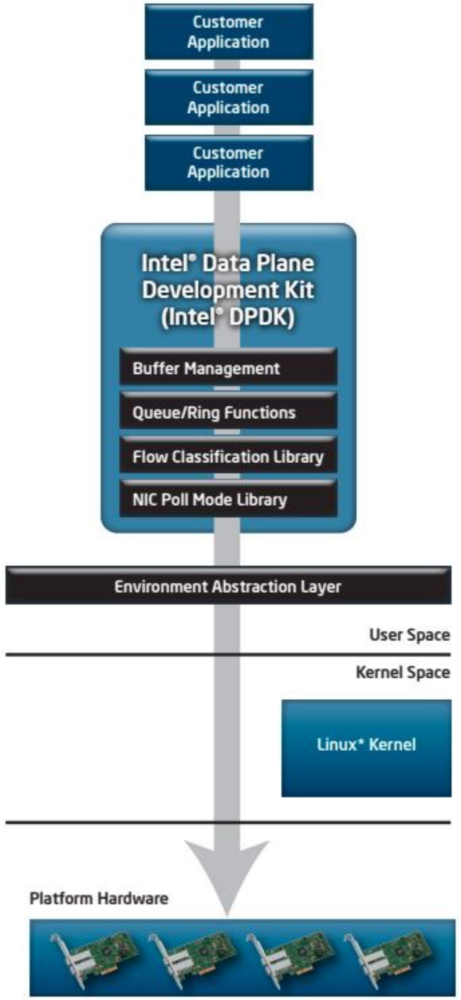
3. FD.io VPP(1)

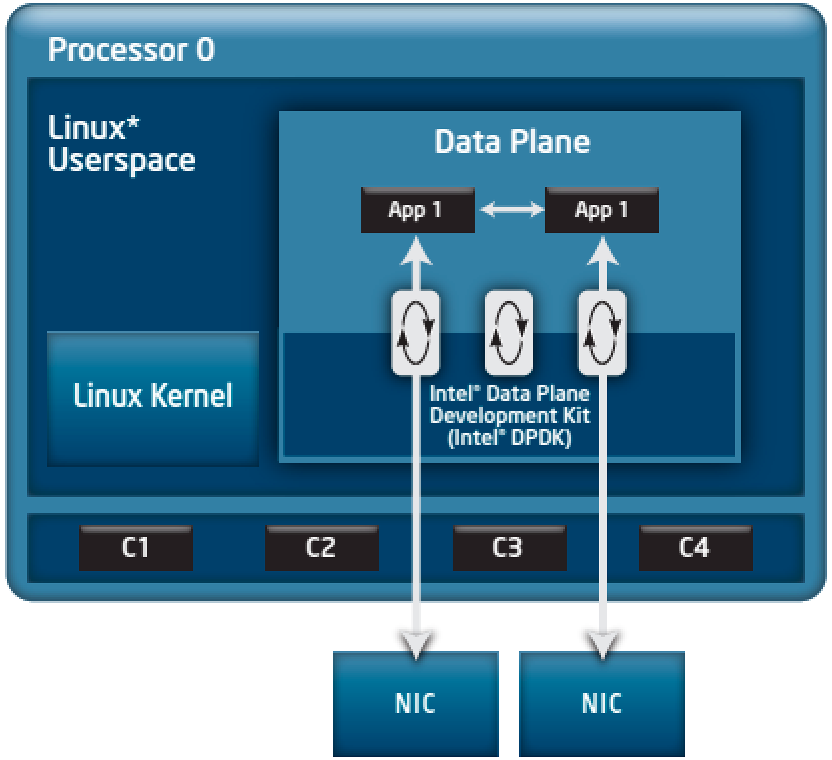




Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

3. FD.io VPP(2)

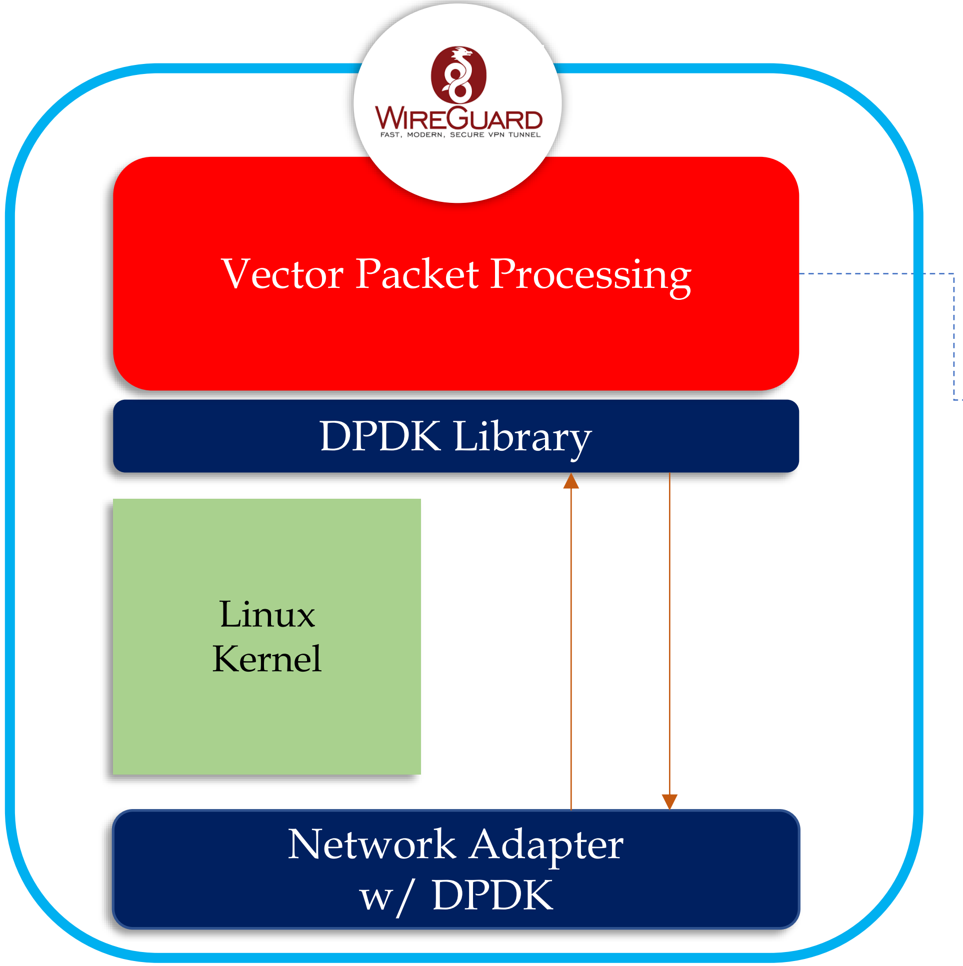


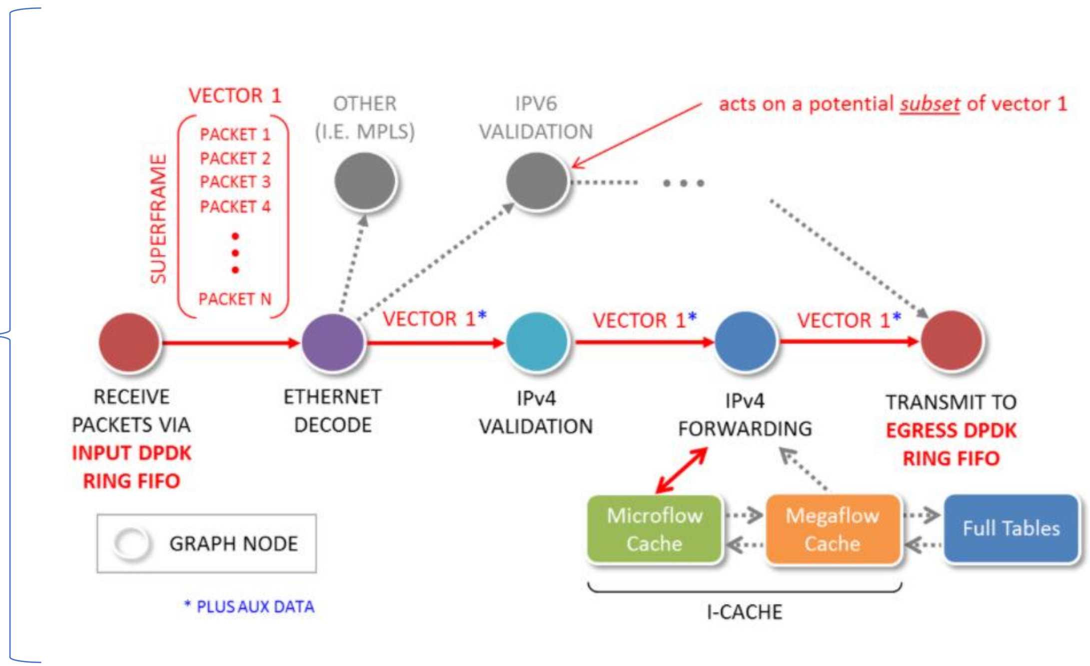


Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation

3. FD.io VPP(3)

Ref) https://fd.io/technology/

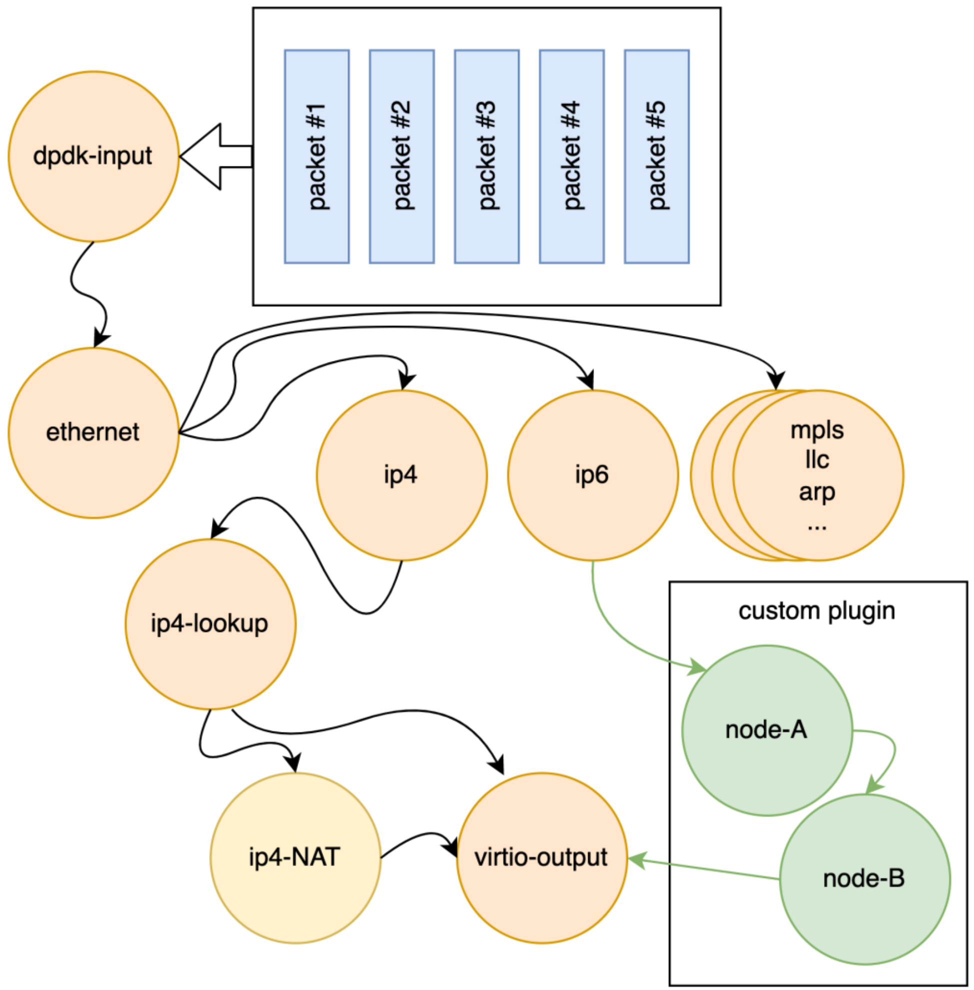


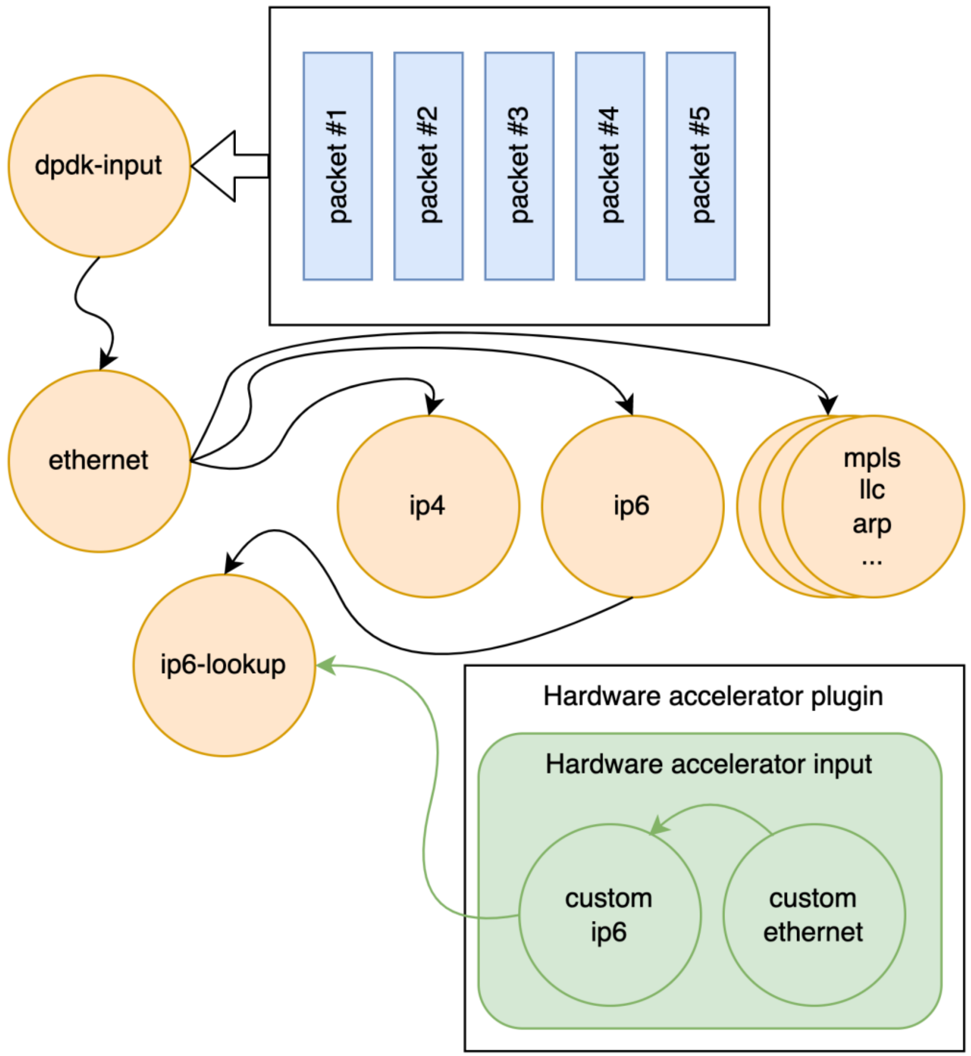


VPP

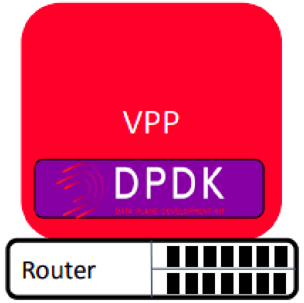
3. FD.io VPP(4)

Ref) https://fd.io/technology/

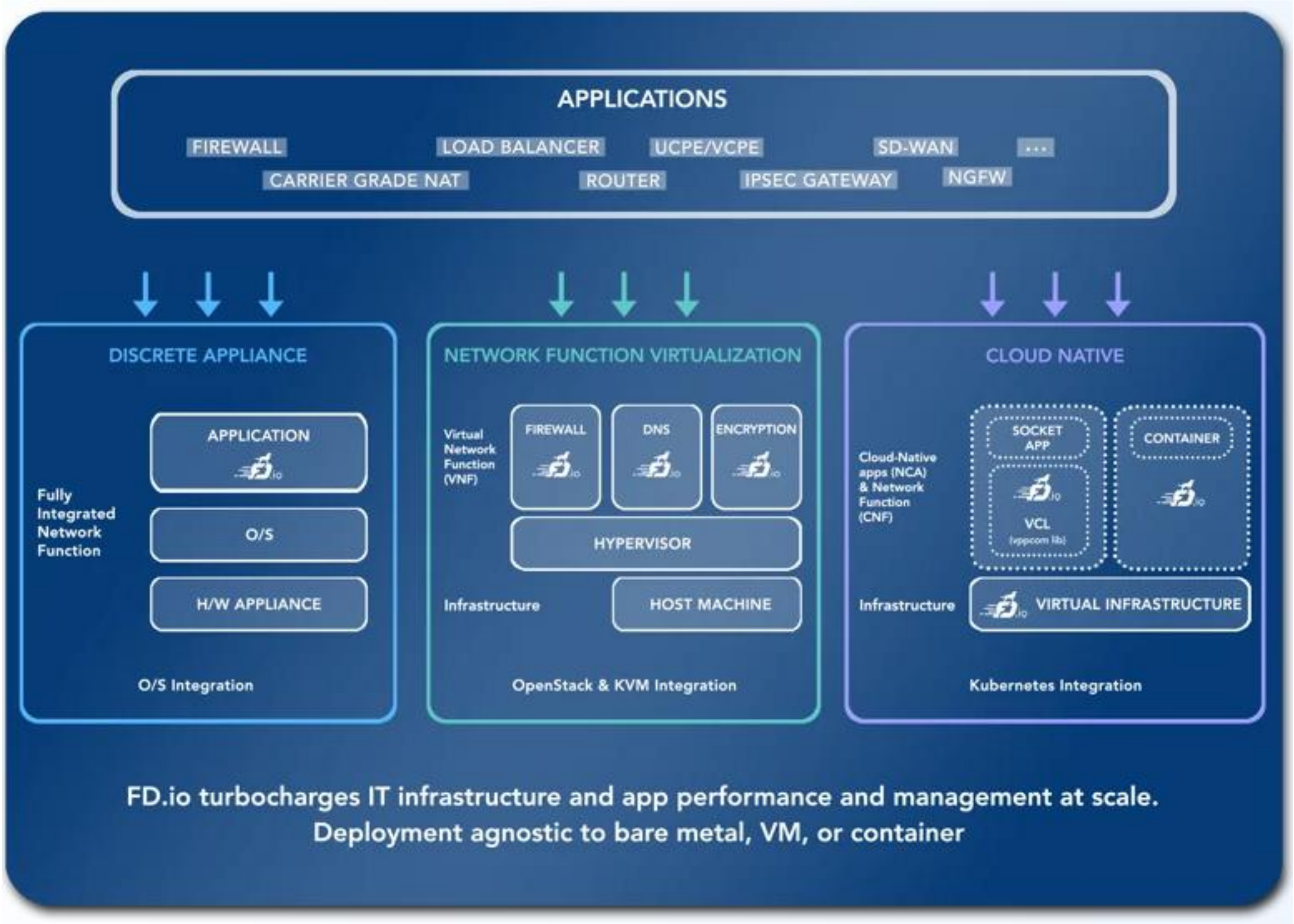




3. FD.io VPP(5)

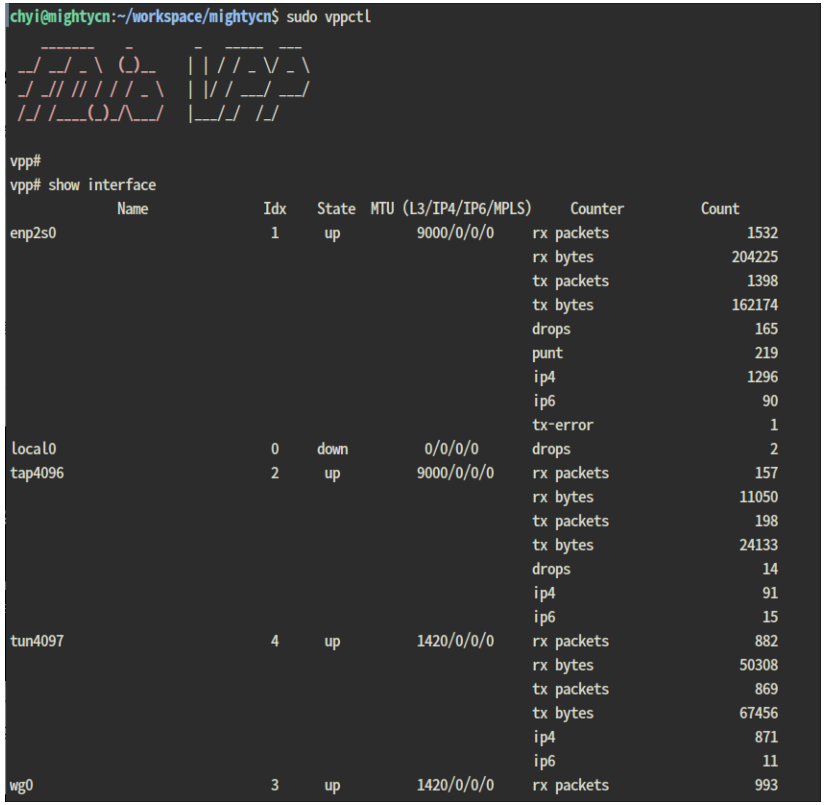


- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- …

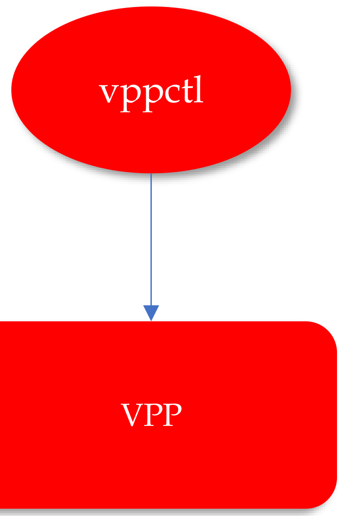


Ref) https://www.netgate.com/resources/articles-vector-packet-processing

3. FD.io VPP(6)

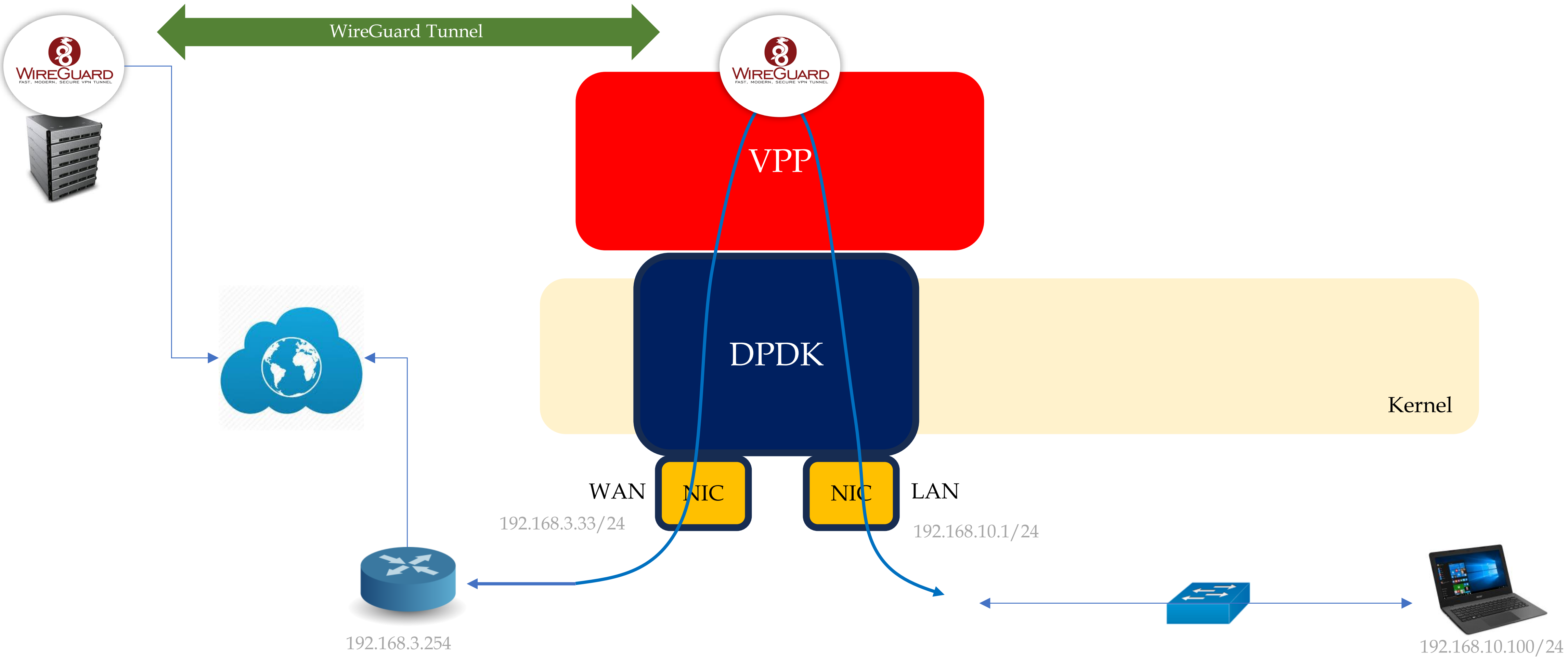


```
chyi@mightycn:~/workspace/mightycn$ sudo vppctl

vpp#
vpp# show interface
              Name               Idx    State  MTU (L3/IP4/IP6/MPLS)     Counter          Count
enp2s0                            1      up          9000/0/0/0     rx packets                  1532
                                                                    rx bytes                  204225
                                                                    tx packets                  1398
                                                                    tx bytes                  162174
                                                                    drops                        165
                                                                    punt                         219
                                                                    ip4                         1296
                                                                    ip6                           90
                                                                    tx-error                       1
local0                            0     down           0/0/0/0      drops                          2
tap4096                           2      up          9000/0/0/0     rx packets                   157
                                                                    rx bytes                   11050
                                                                    tx packets                   198
                                                                    tx bytes                   24133
                                                                    drops                         14
                                                                    ip4                           91
                                                                    ip6                           15
tun4097                           4      up          1420/0/0/0     rx packets                   882
                                                                    rx bytes                   50308
                                                                    tx packets                   869
                                                                    tx bytes                   67456
                                                                    ip4                          871
                                                                    ip6                           11
wg0                               3      up          1420/0/0/0     rx packets                   993
```

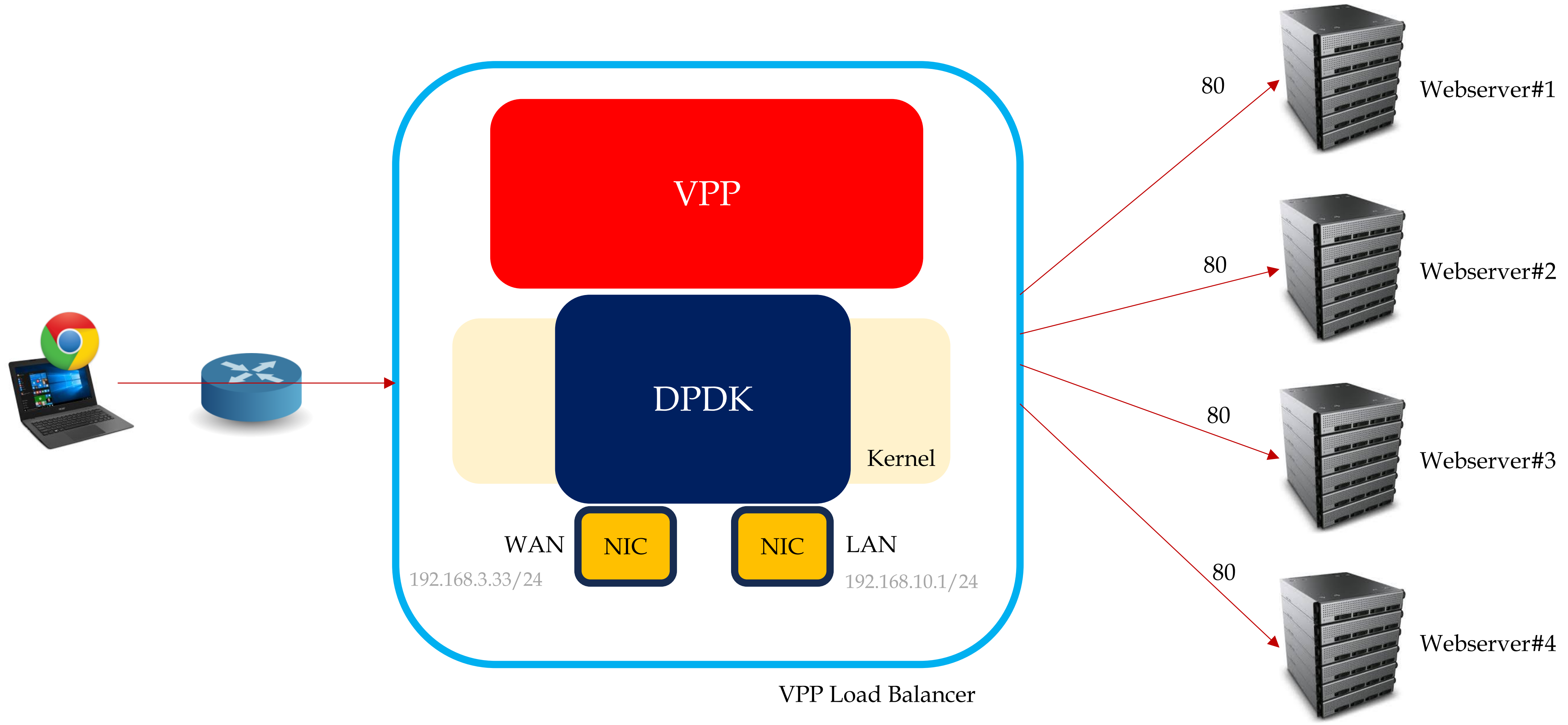


/usr/bin/vpp –c /etc/vpp/startup.conf

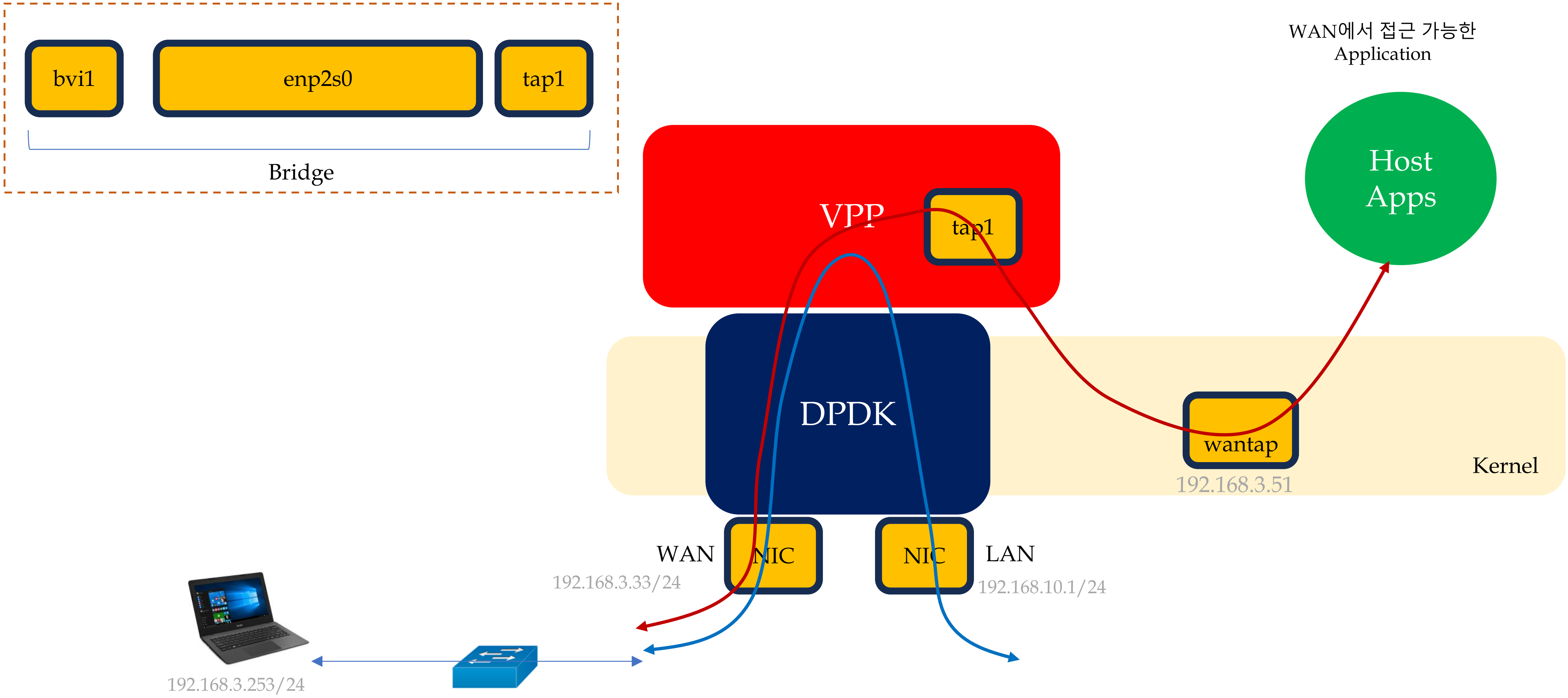
3. FD.io VPP(7) - WireGuard



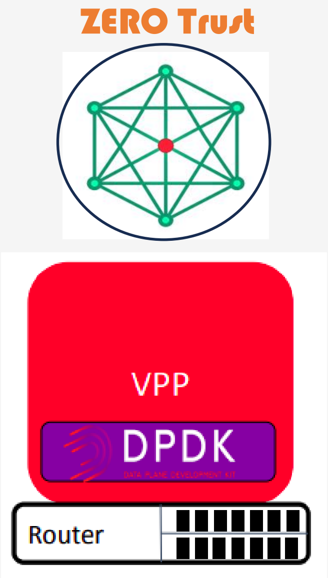
3. FD.io VPP(8) – NAT & Load Balancing



3. FD.io VPP(9) – Legacy App Access



4. MightyConnect Security Box

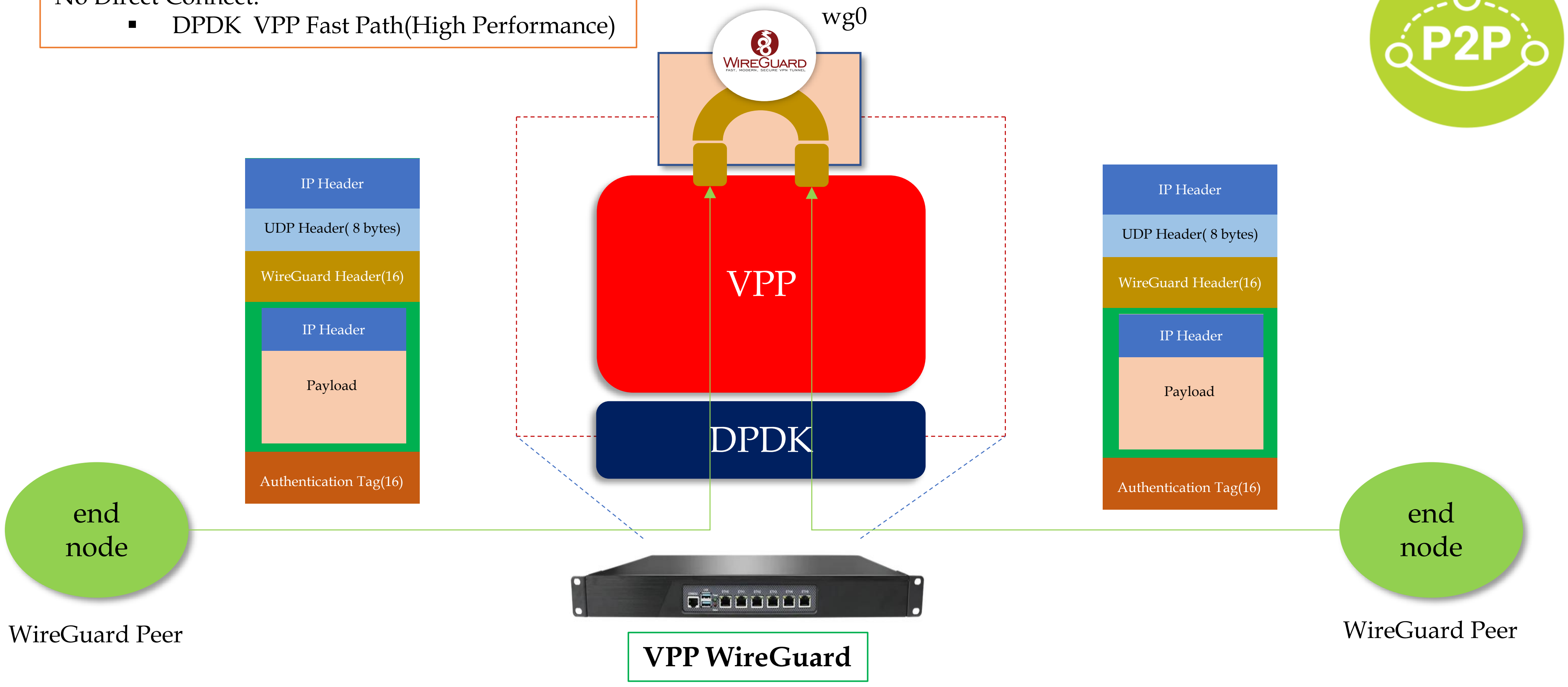


Software ASIC

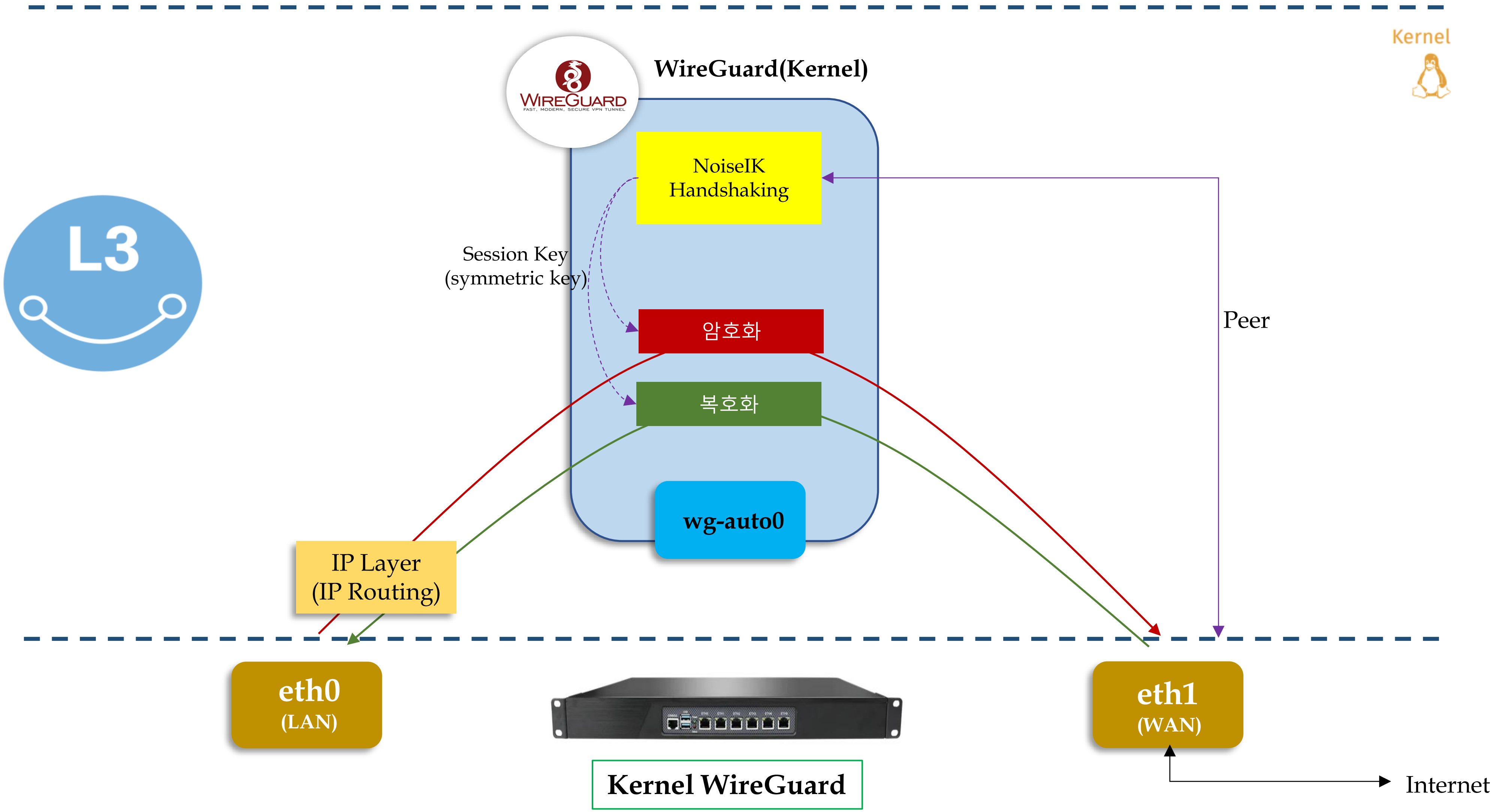
4. MightyConnect Security Box(1) – WireGuard(1)

No Direct Connect:

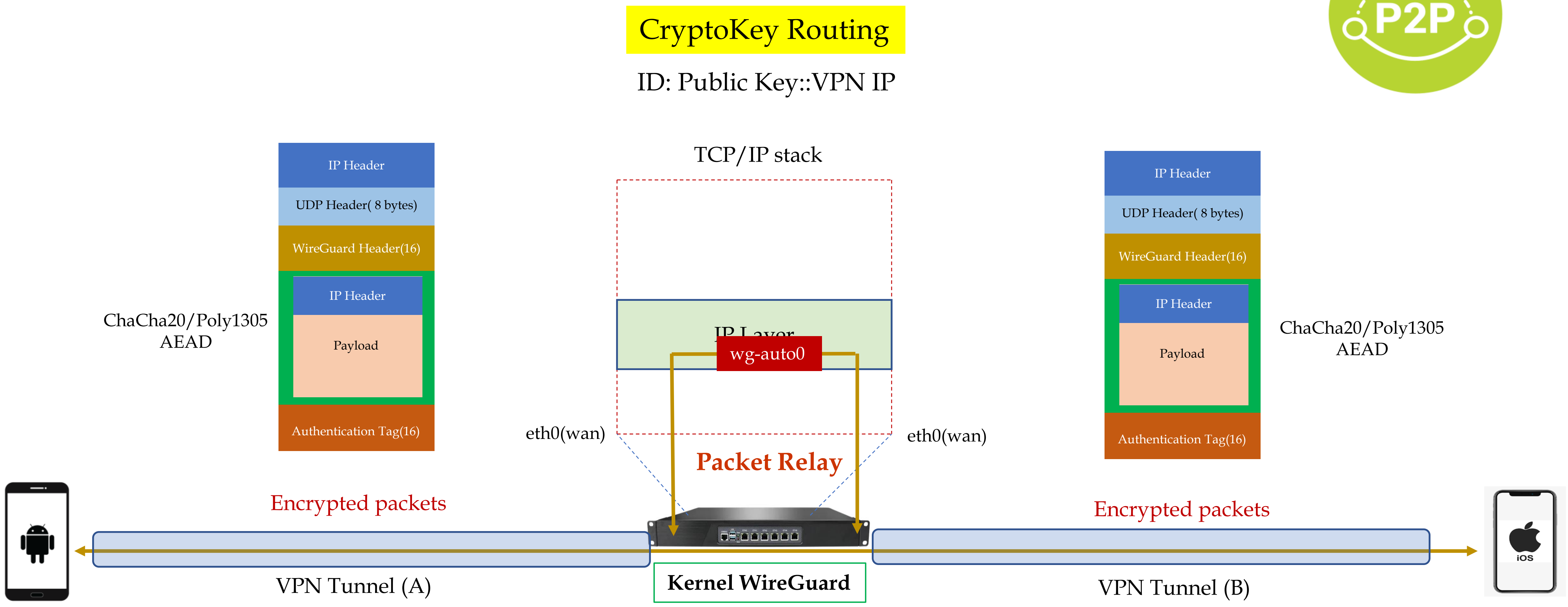
- DPDK VPP Fast Path(High Performance)



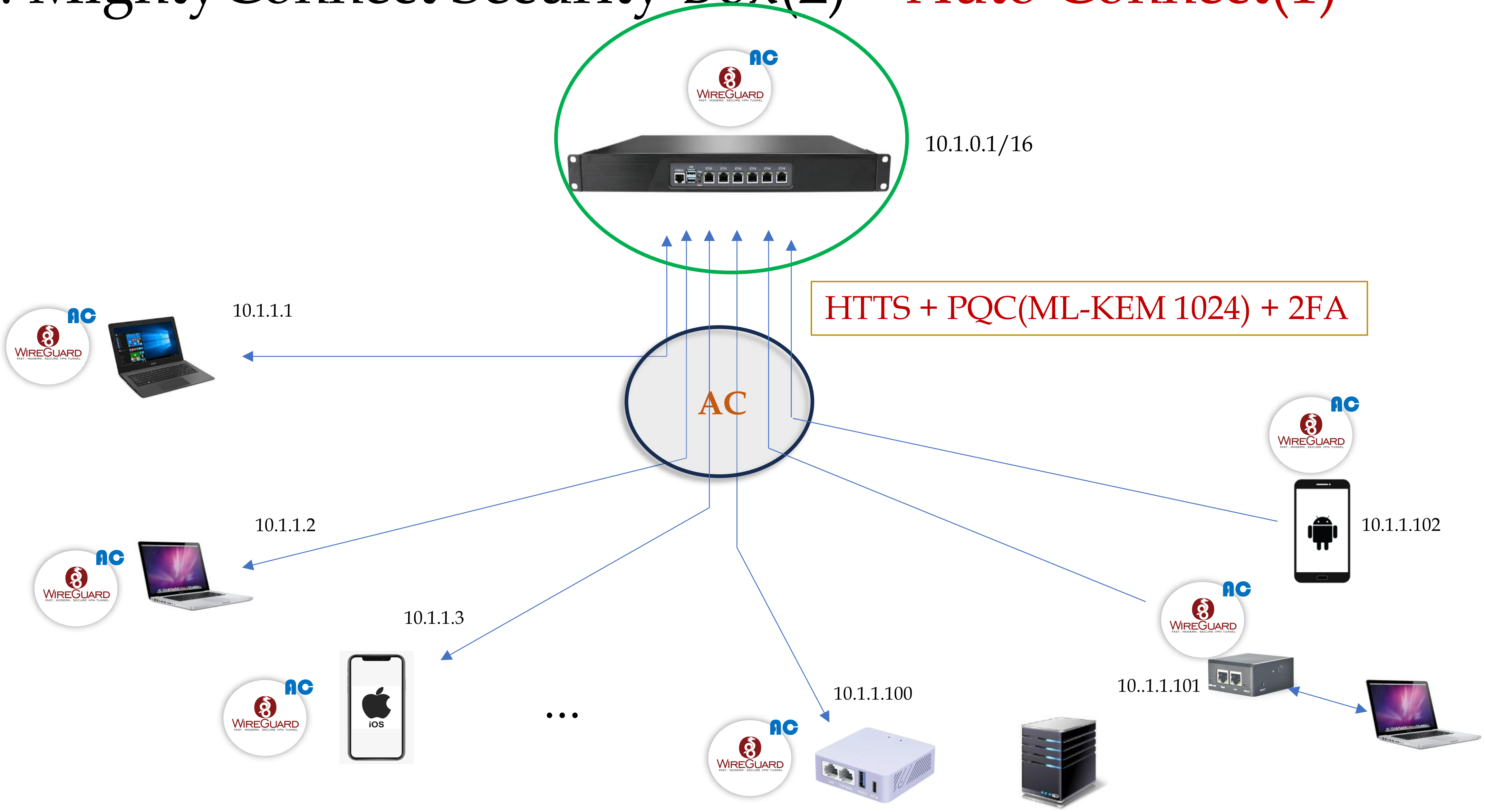
4. MightyConnect Security Box(1) – WireGuard(2-1)



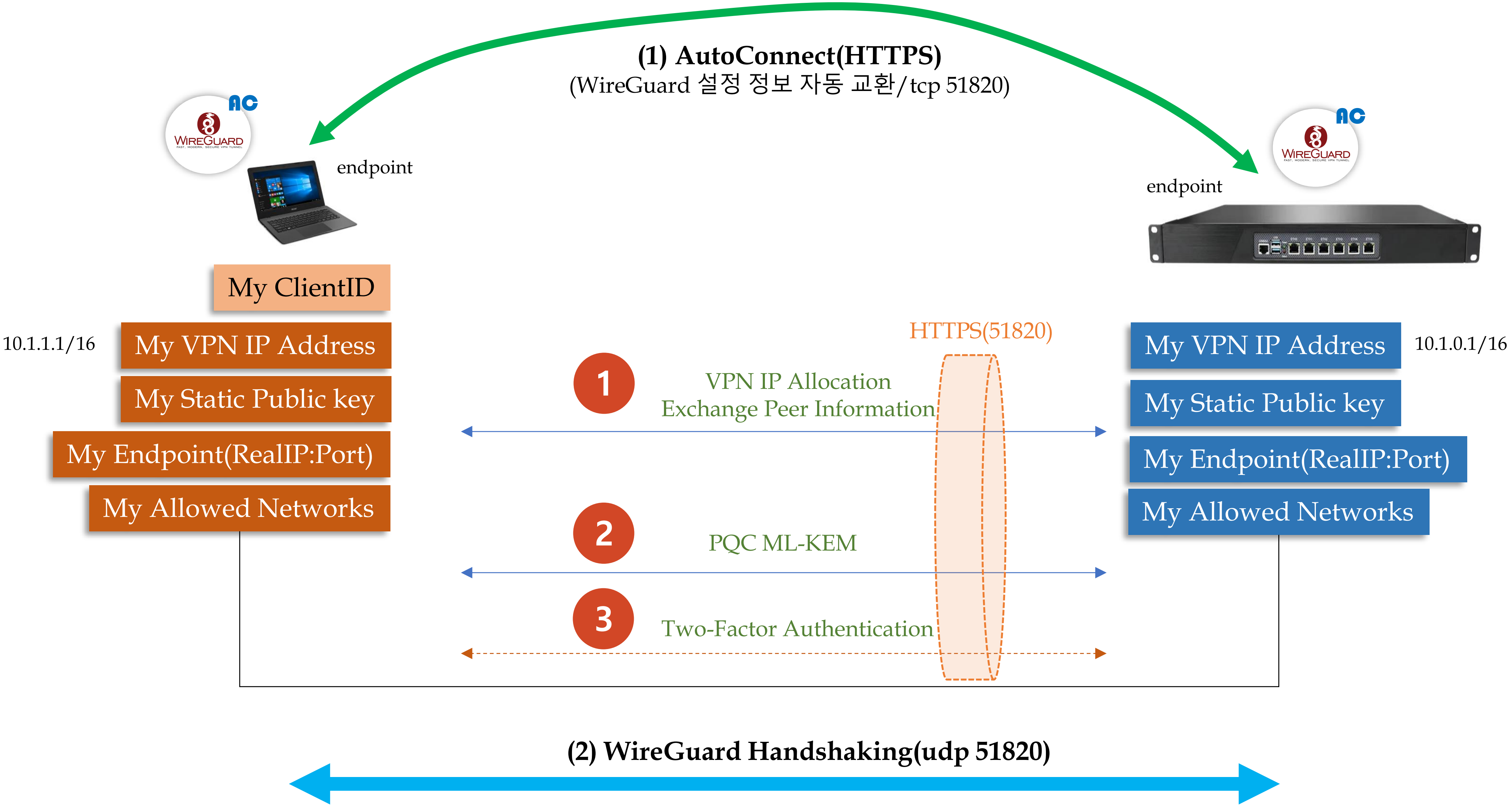
4. MightyConnect Security Box(1) – WireGuard(2-2)



4. MightyConnect Security Box(2) – Auto Connect(1)



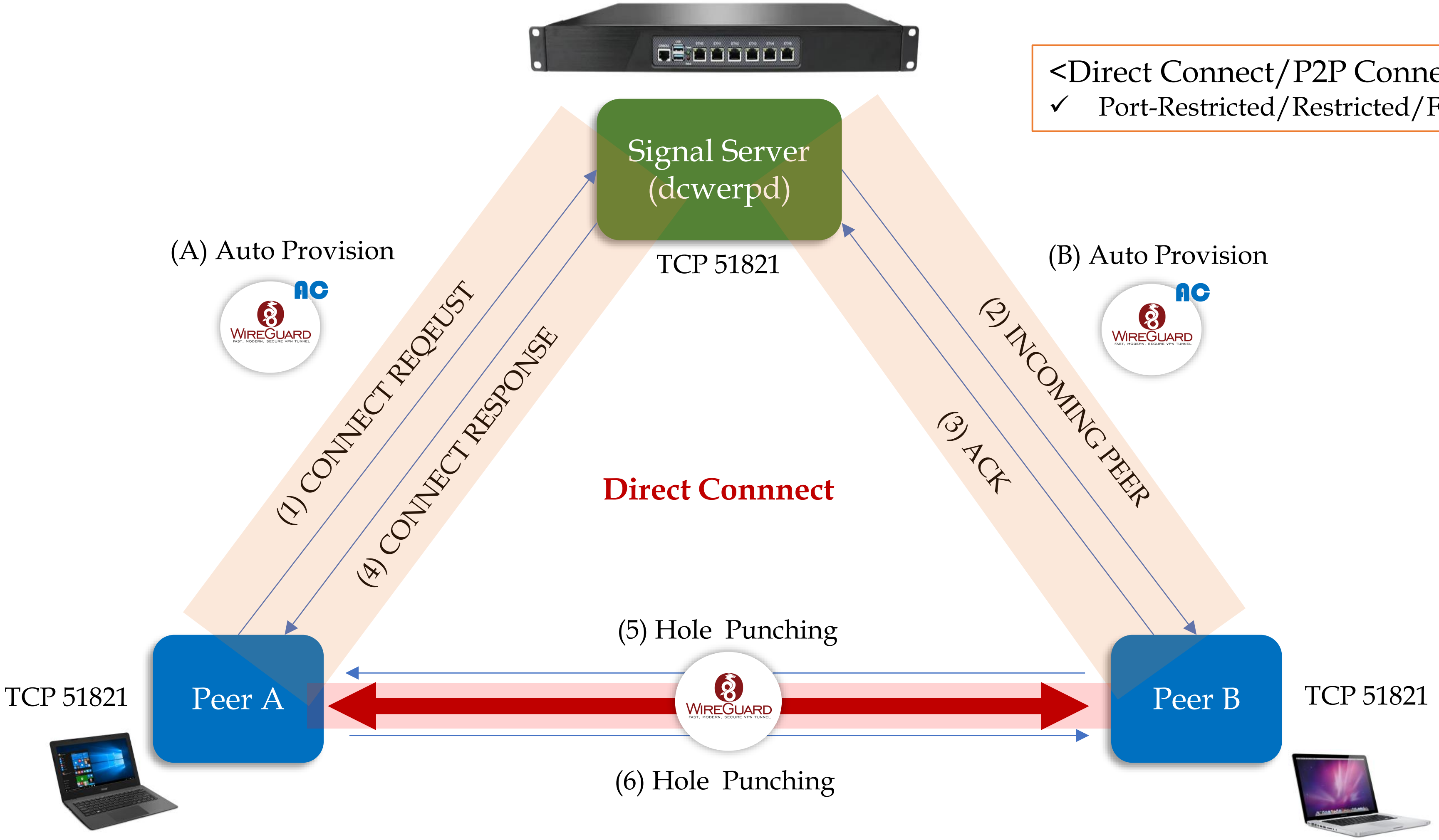
4. MightyConnect Security Box(2) – Auto Connect(2)



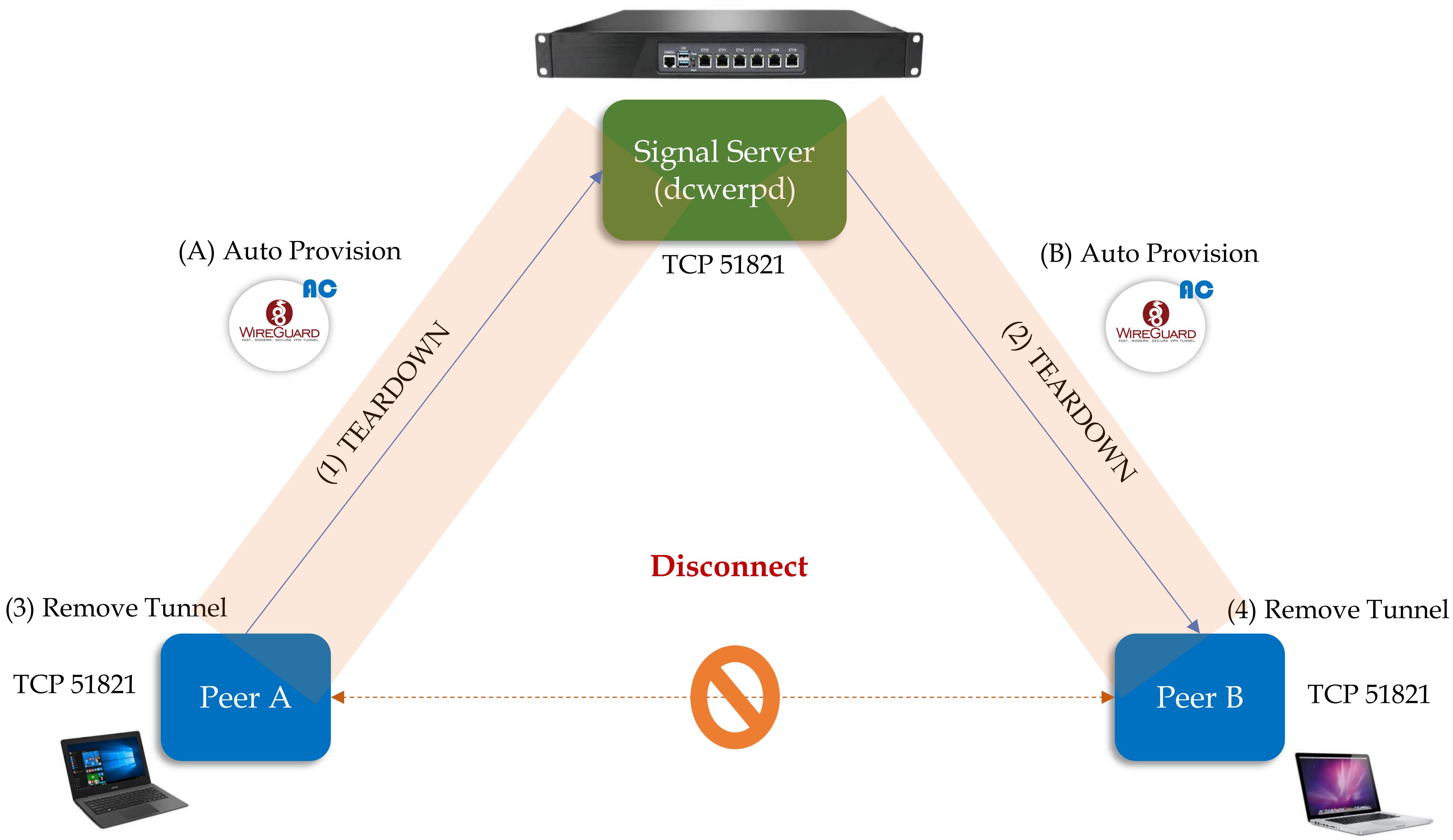
4. MightyConnect Security Box(3) – Direct Connect(1)

<Direct Connect/P2P Connect>

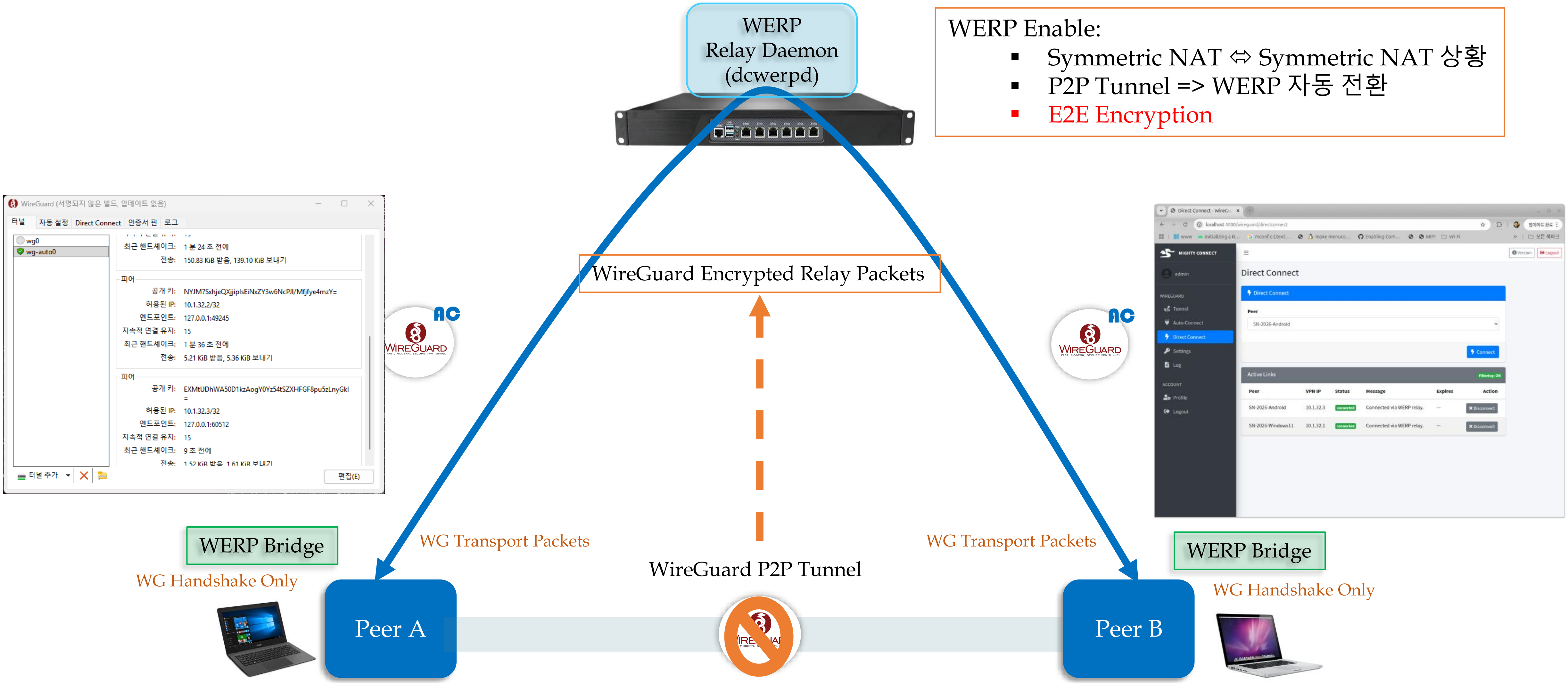
✓ Port-Restricted/Restricted/Full Cone NAT 지원



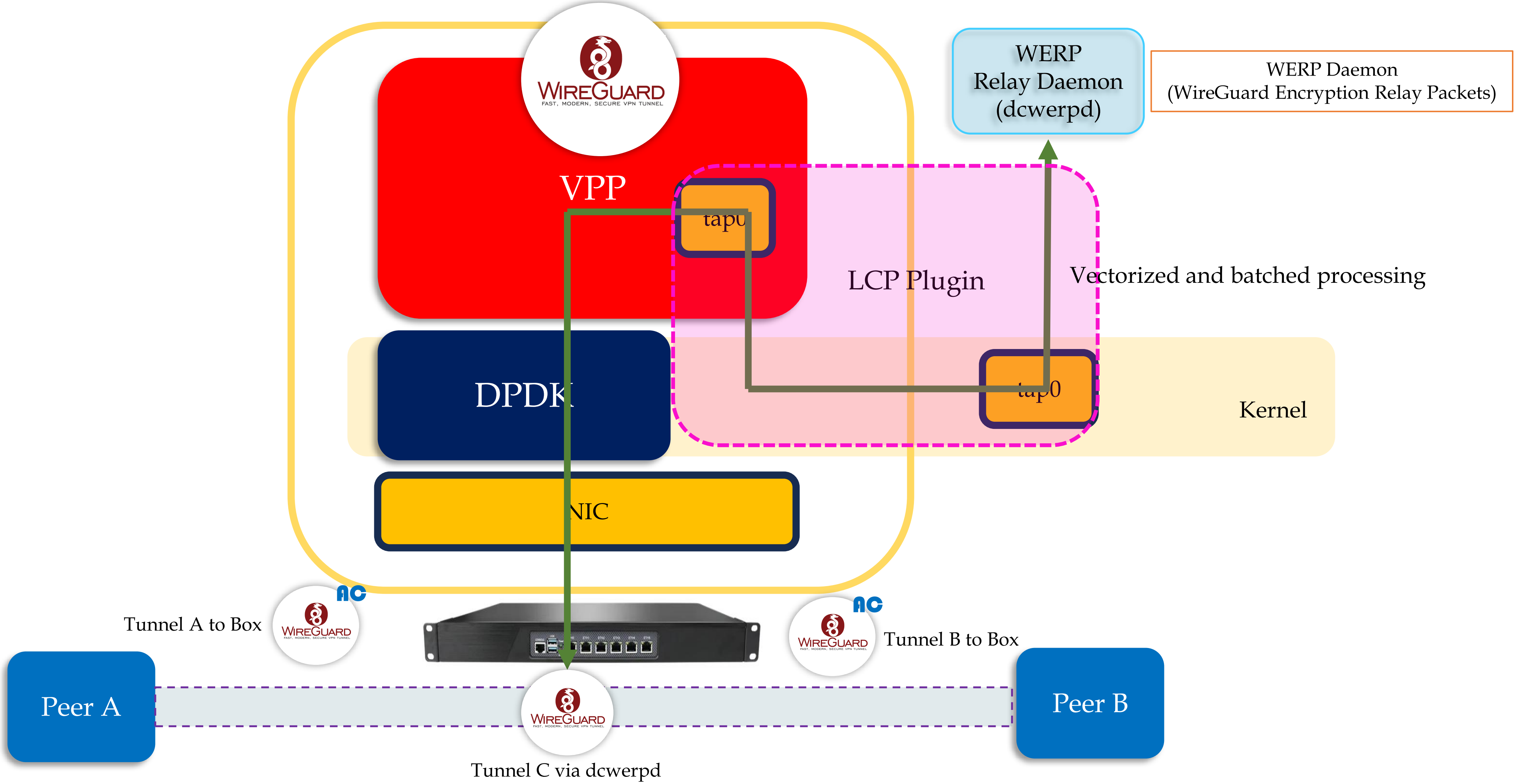
4. MightyConnect Security Box(3) – Direct Connect(2)



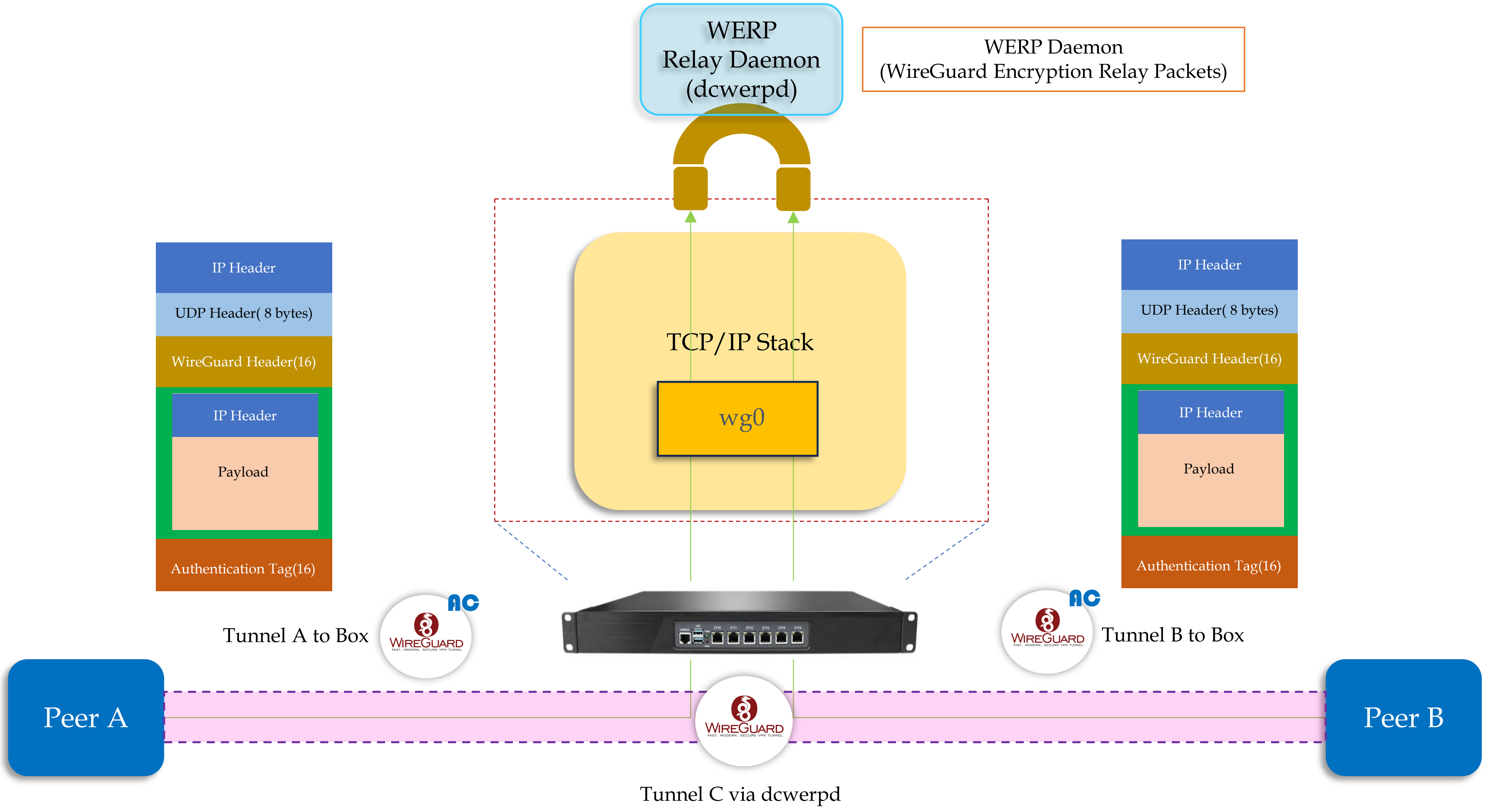
4. MightyConnect Security Box(3) – Direct Connect(3-1)



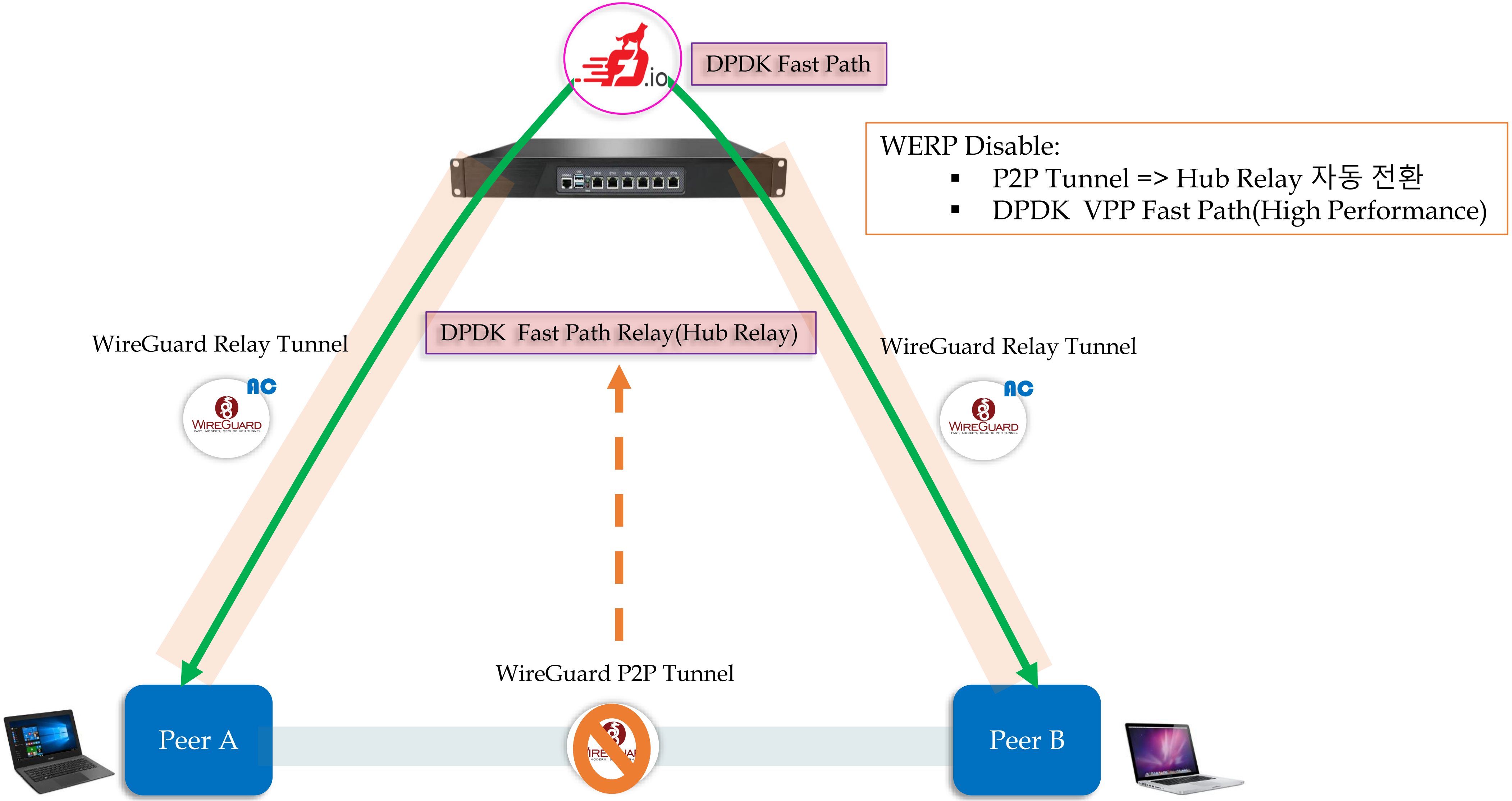
4. MightyConnect Security Box(3) – Direct Connect(3-2)



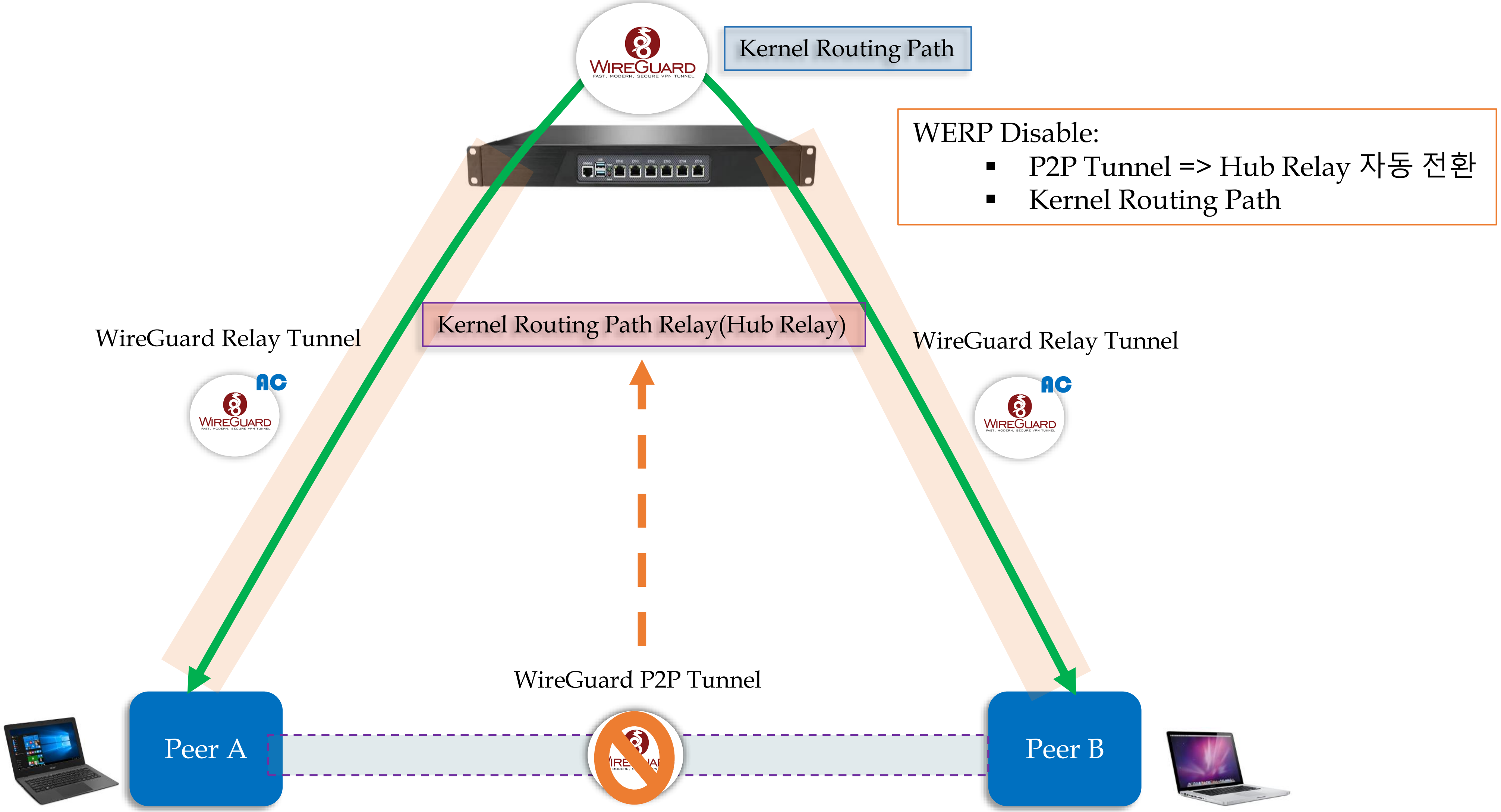
4. MightyConnect Security Box(3) – Direct Connect(3-3)



4. MightyConnect Security Box(4) – Fast Hub Relay(1)



4. MightyConnect Security Box(4) – Fast Hub Relay(2)



4. MightyConnect Security Box(5) – VPP ZTNA(1)

Allow Resources per not IP Address but Client ID

.io

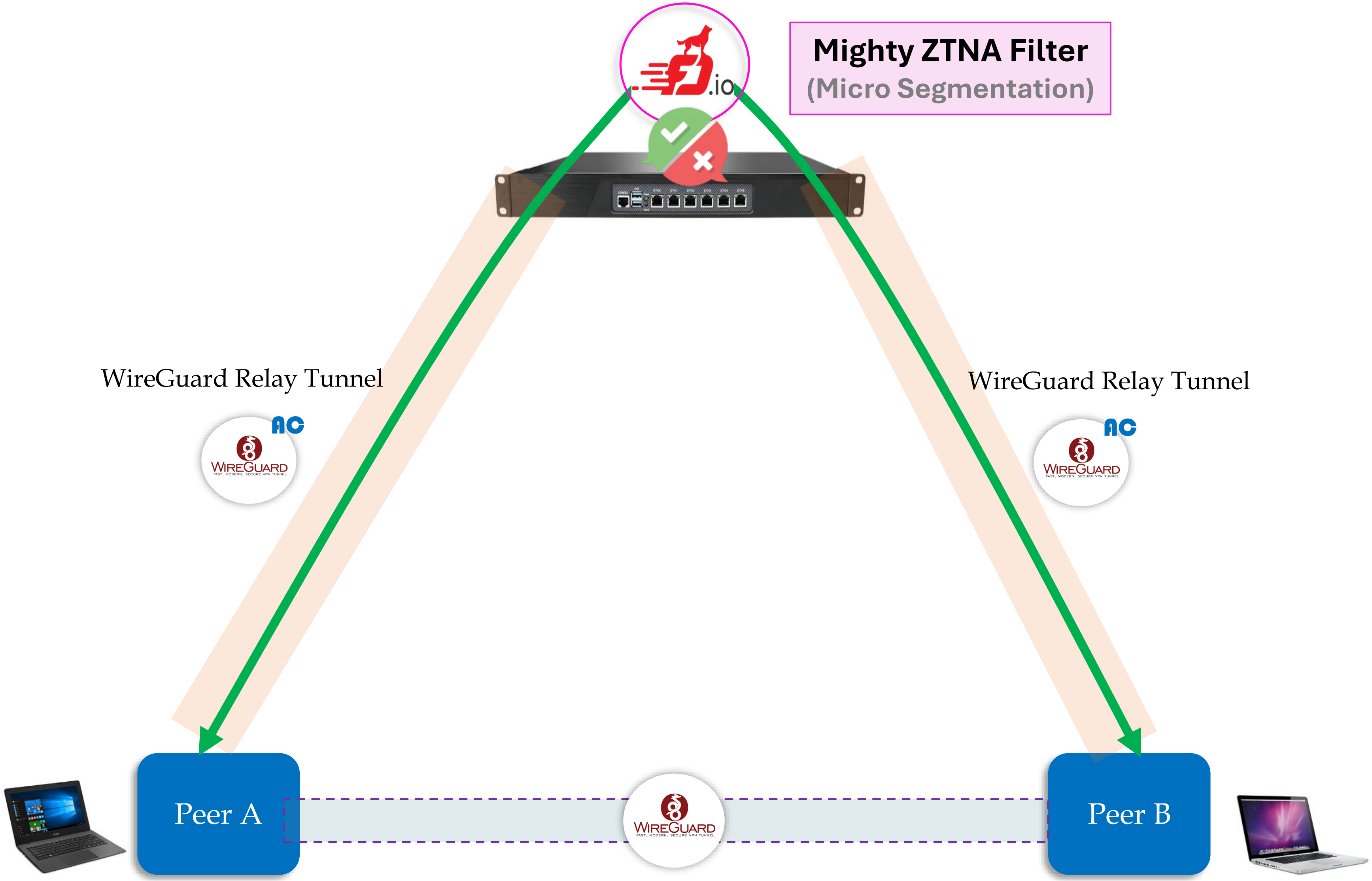
VPP ACL

DPDK

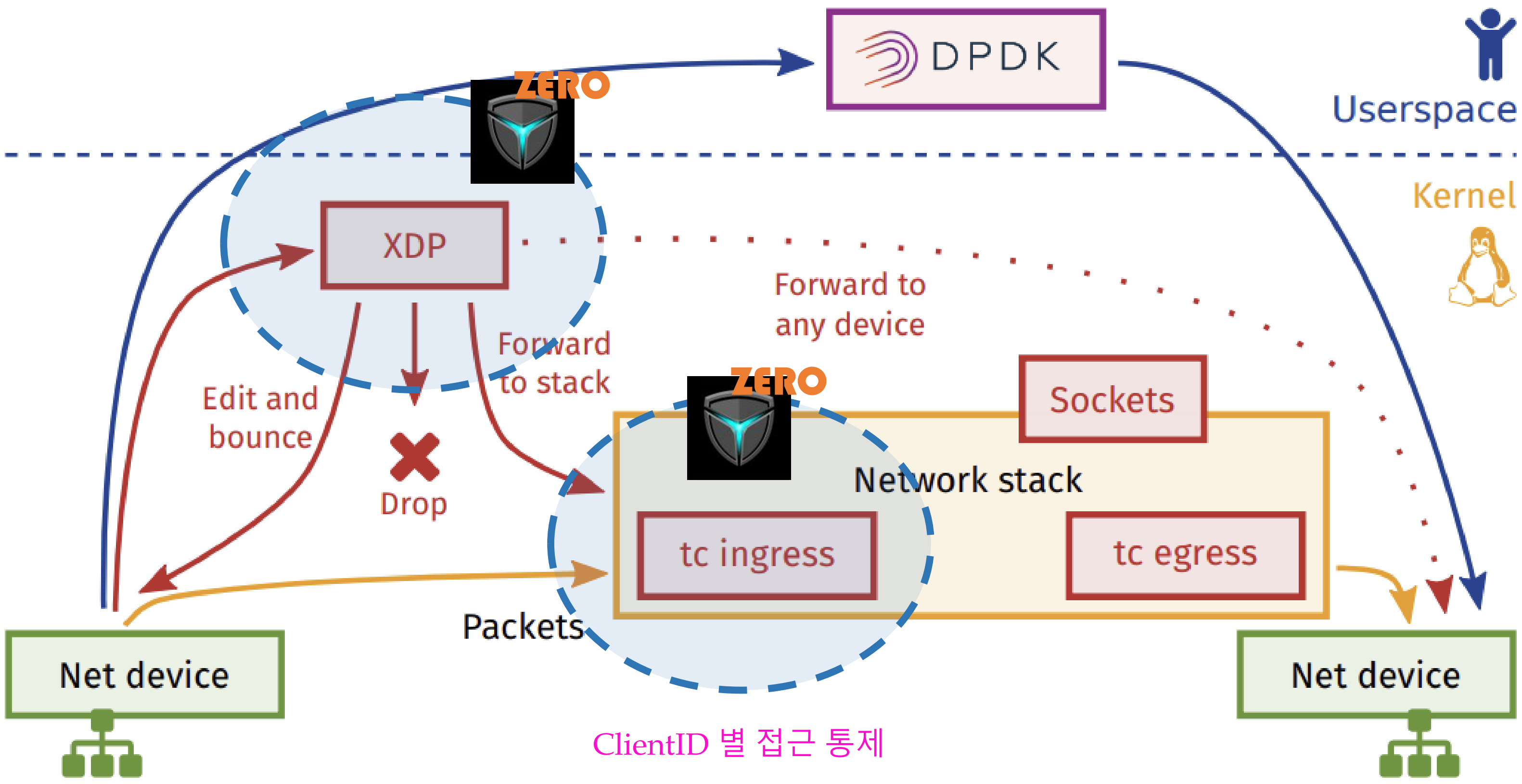
Mighty ZTNA Filter
(Micro Segmentation)

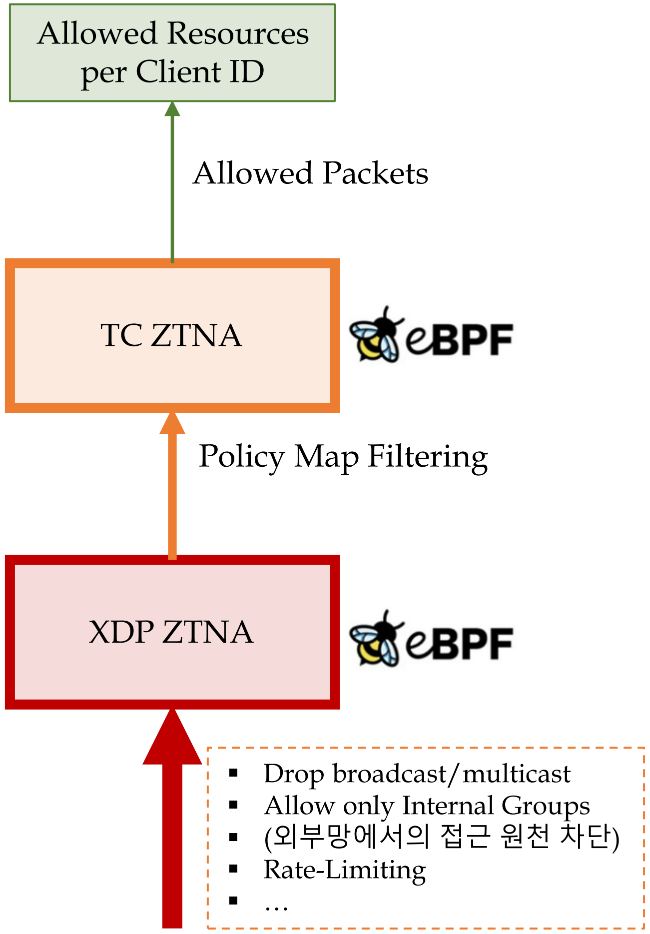
```
VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
          0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
          1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
          2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
          3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
          4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
          5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
          6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
          7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
          8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index: 1, 2
  applied outbound on sw_if_index:
  used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
          0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index:
  applied outbound on sw_if_index: 1, 2
  used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
          0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
          1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
          2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
          3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
          4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
          5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
          6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
          7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
          8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
          9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
         13: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
         14: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
         15: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         16: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         17: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         18: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
         19: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
         20: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         21: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         22: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         23: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
         24: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
         25: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5432
         26: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
         27: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 8080
  applied inbound on sw_if_index: 5
  applied outbound on sw_if_index:
  used in lookup context index: 5
acl-index 3 count 1 tag {mightyconnect-acl-egress-wg0}
          0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied outbound on sw_if_index: 5
```

4. MightyConnect Security Box(5) – VPP ZTNA(2)

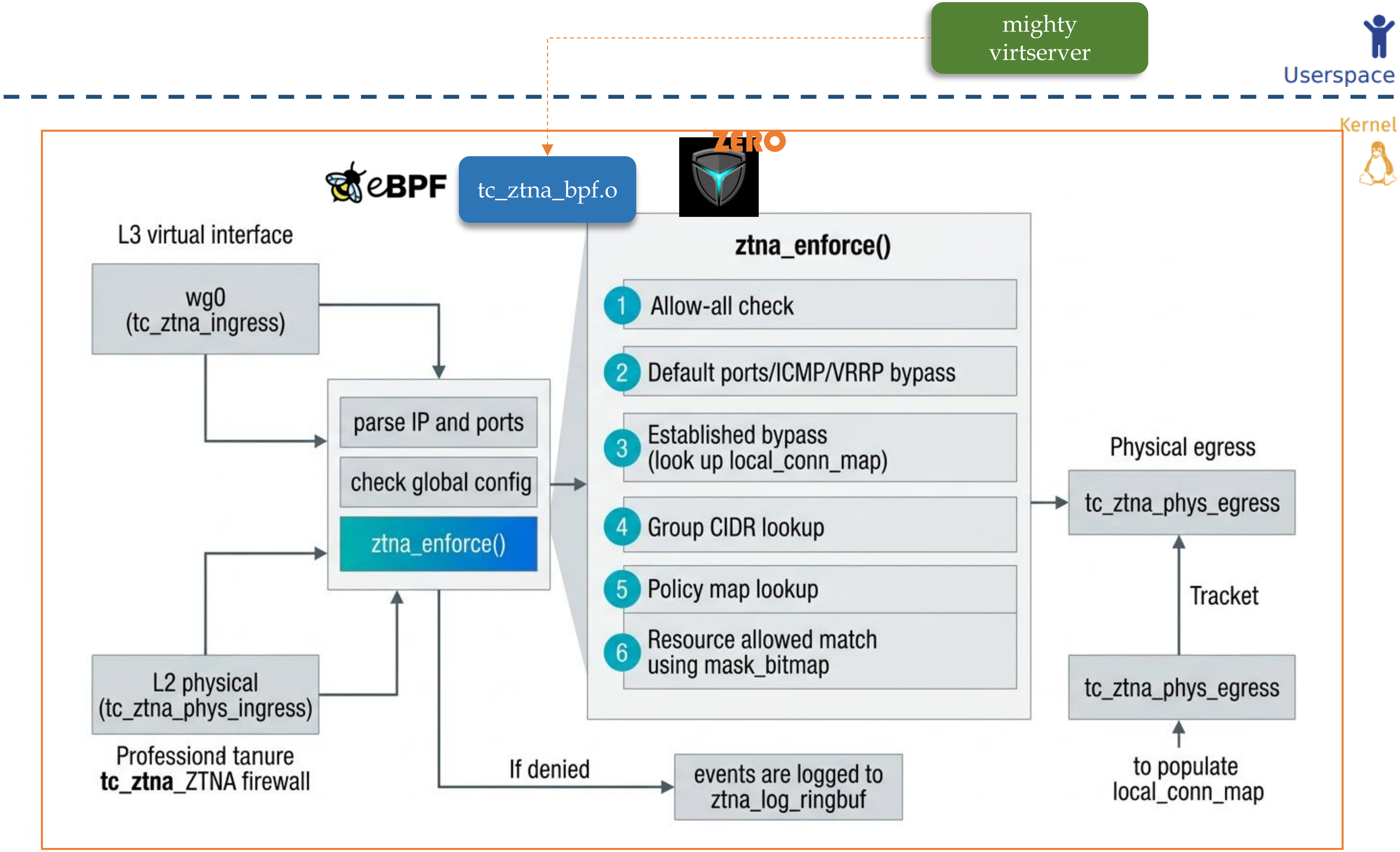


4. MightyConnect Security Box(6) – Kernel ZTNA(1)

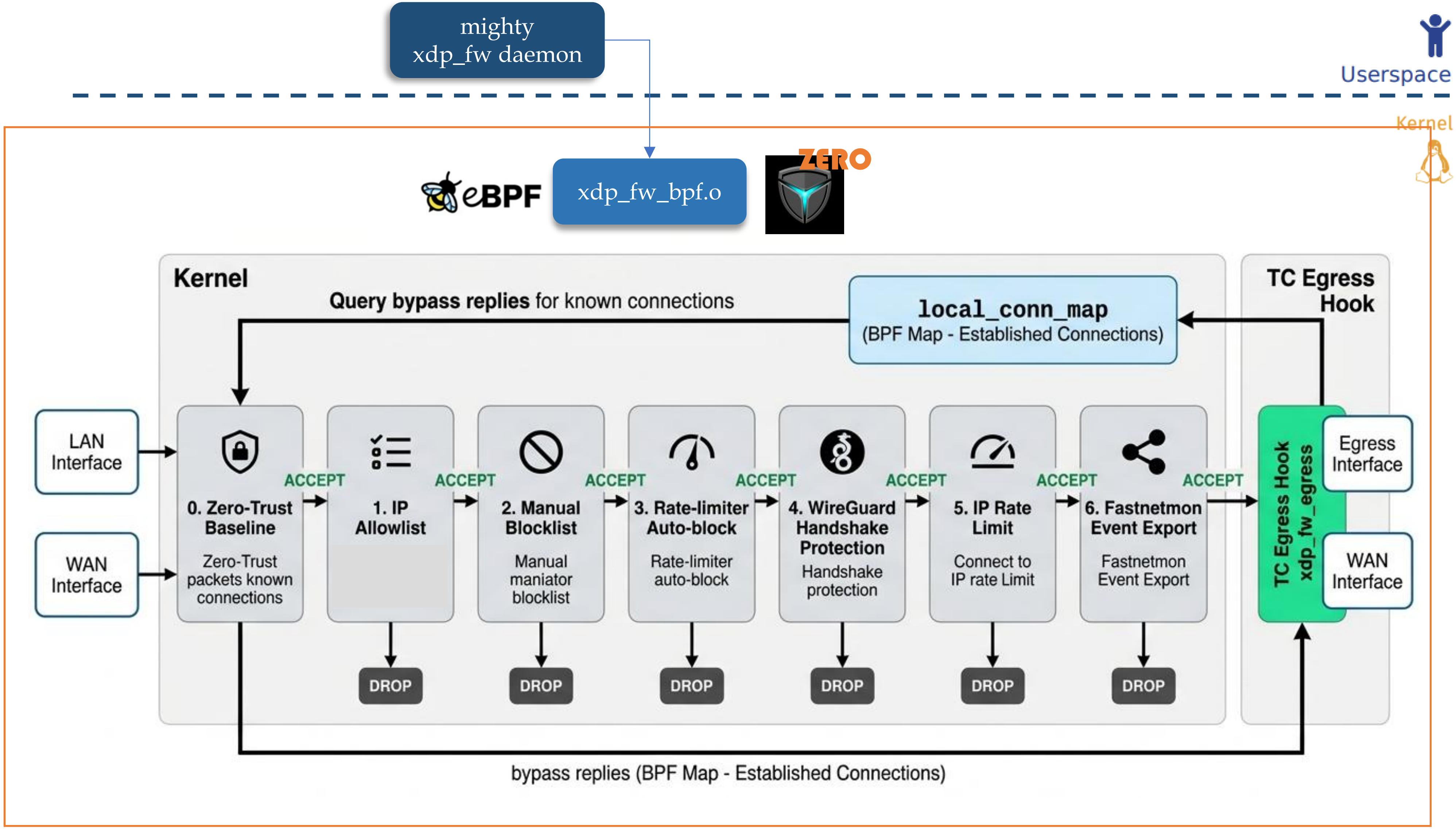




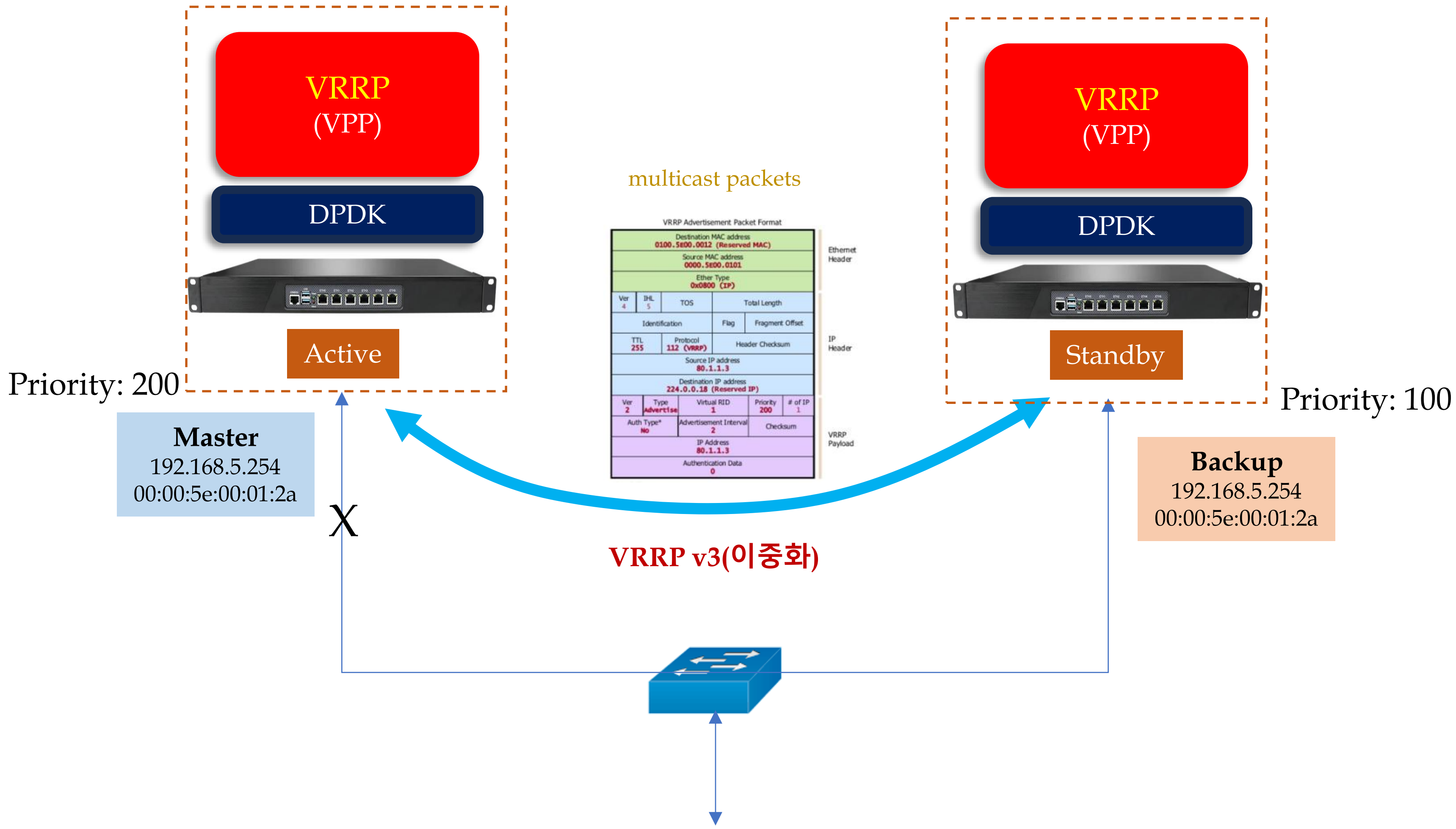
4. MightyConnect Security Box(6) – Kernel ZTNA(2)



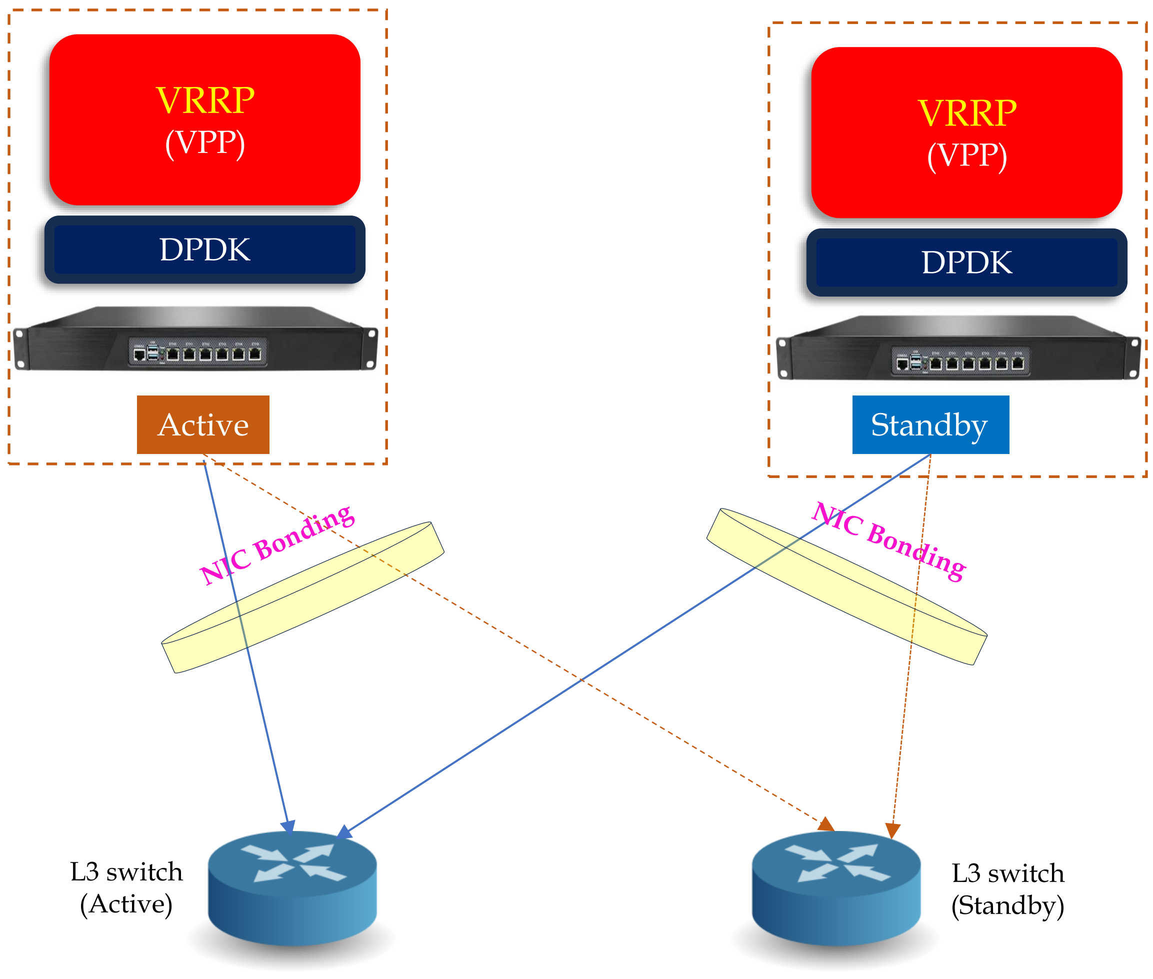
4. MightyConnect Security Box(6) – Kernel ZTNA(3)



4. MightyConnect Security Box(7) – Redundancy(1)



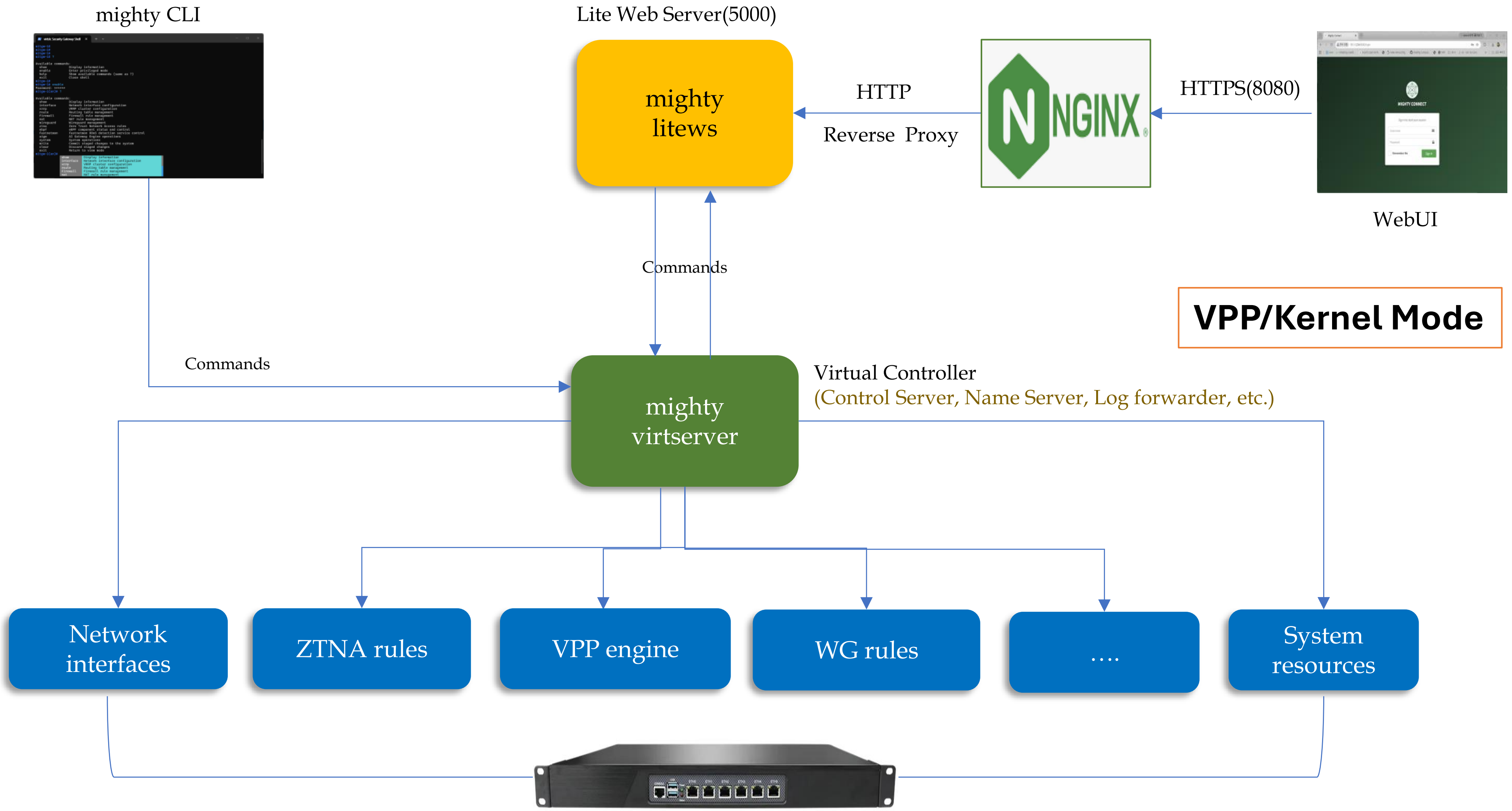
4. MightyConnect Security Box(7) – Redundancy(2)



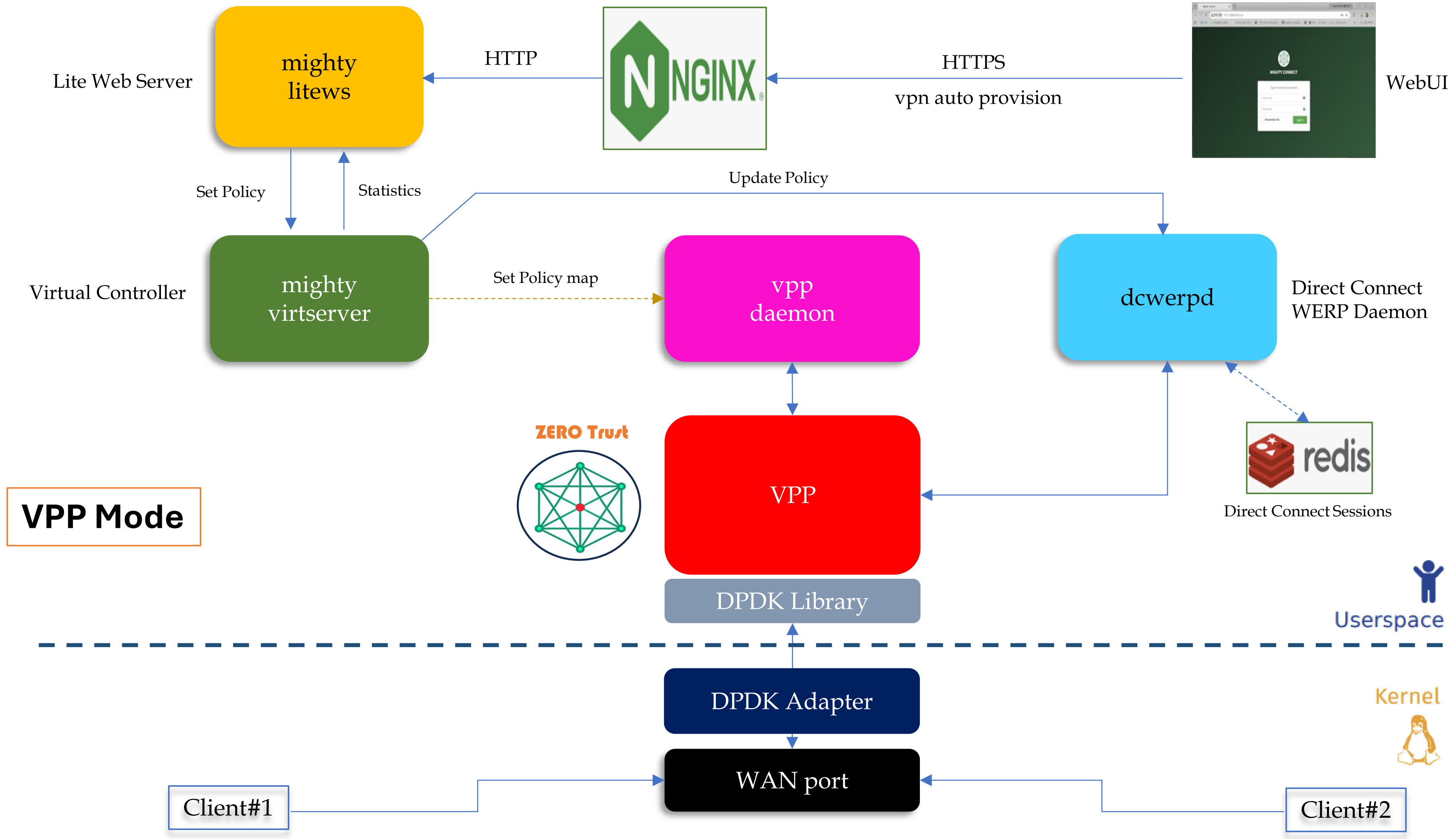
<NIC Bonding Mode>

- balance-rr
- active-backup
- balance-xor
- broadcast
- 802.3ad
- …

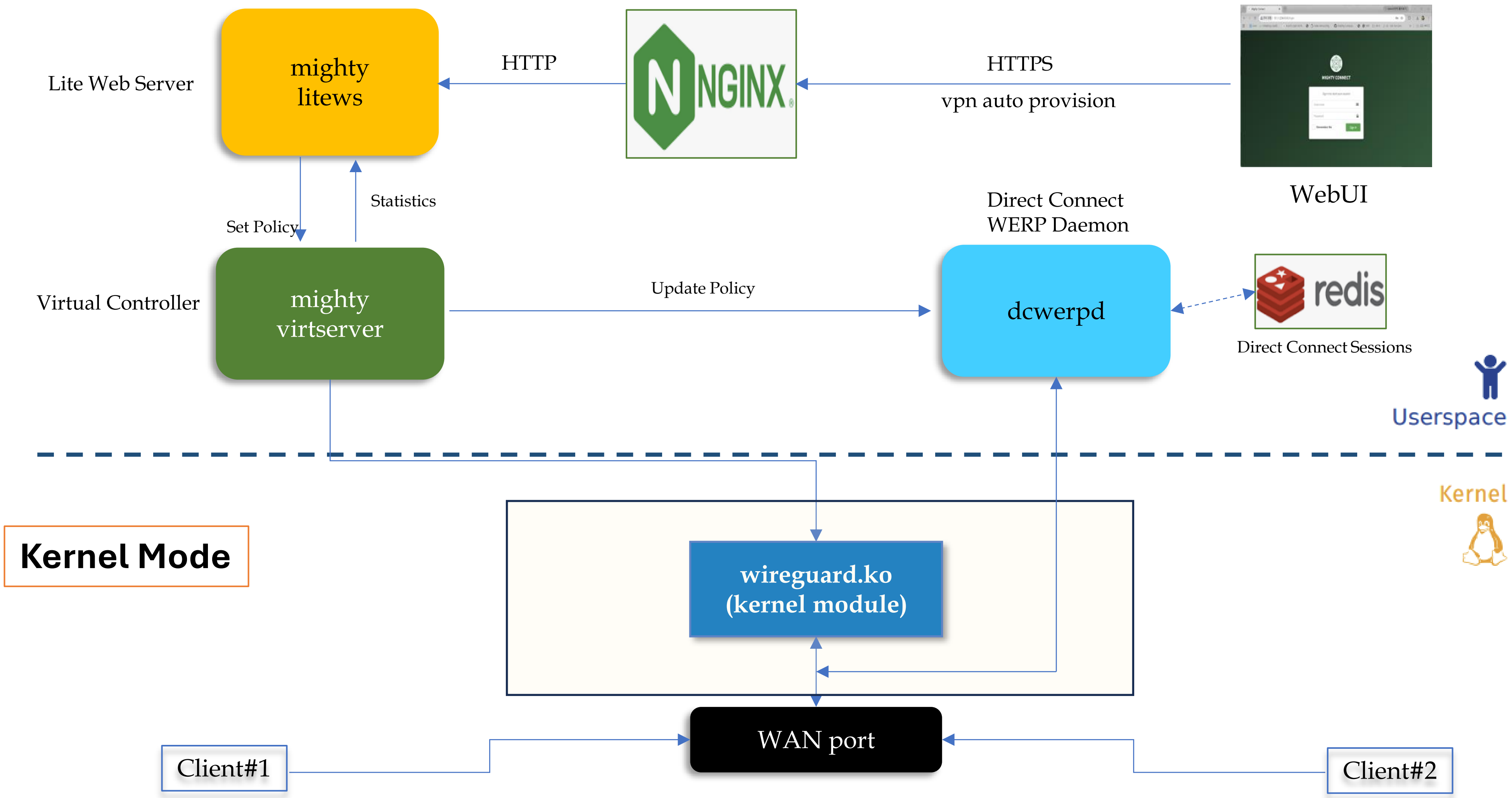
4. MightyConnect Security Box(8) – Mighty Engine(1)



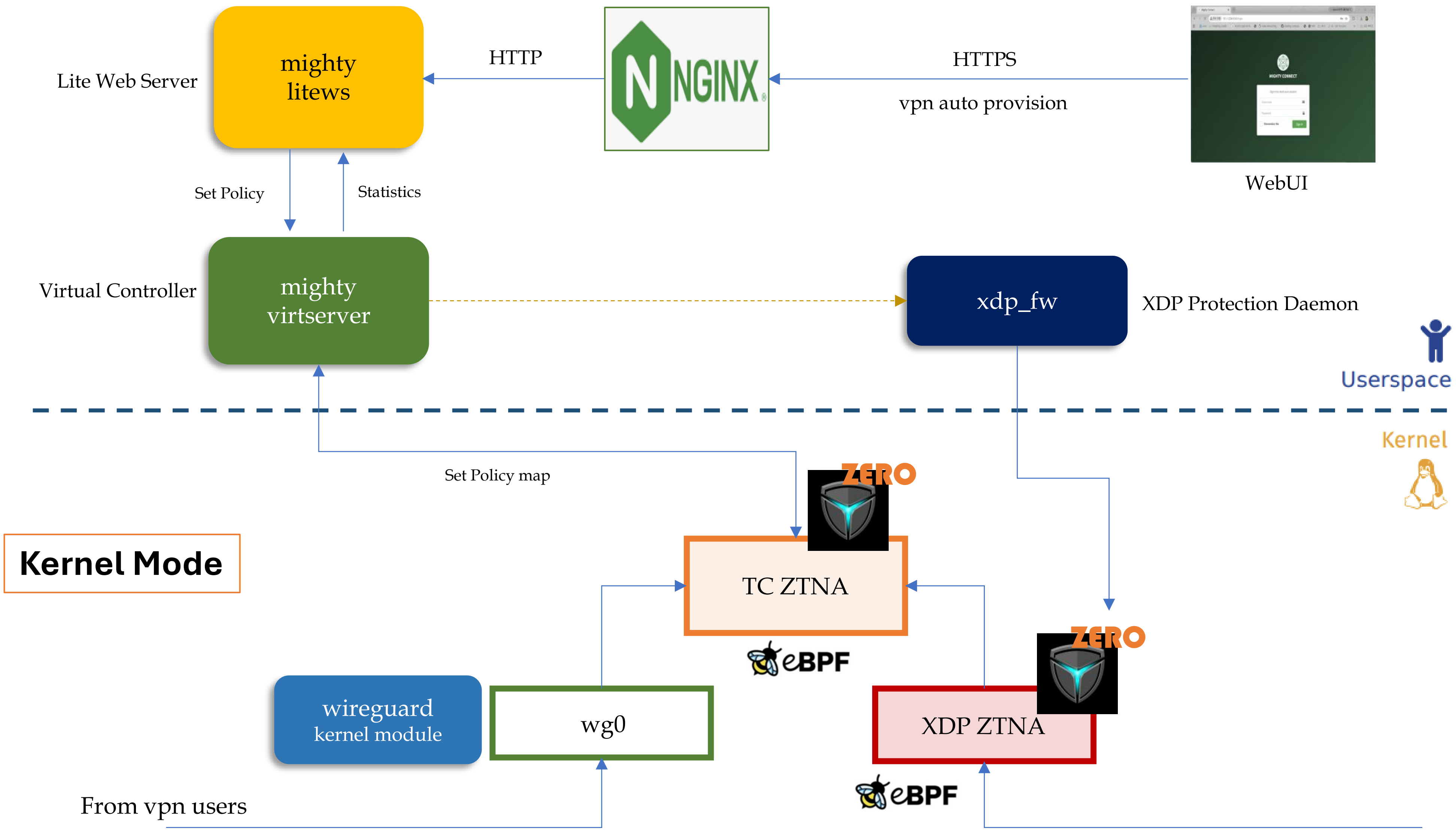
4. MightyConnect Security Box(8) – Mighty Engine(2)



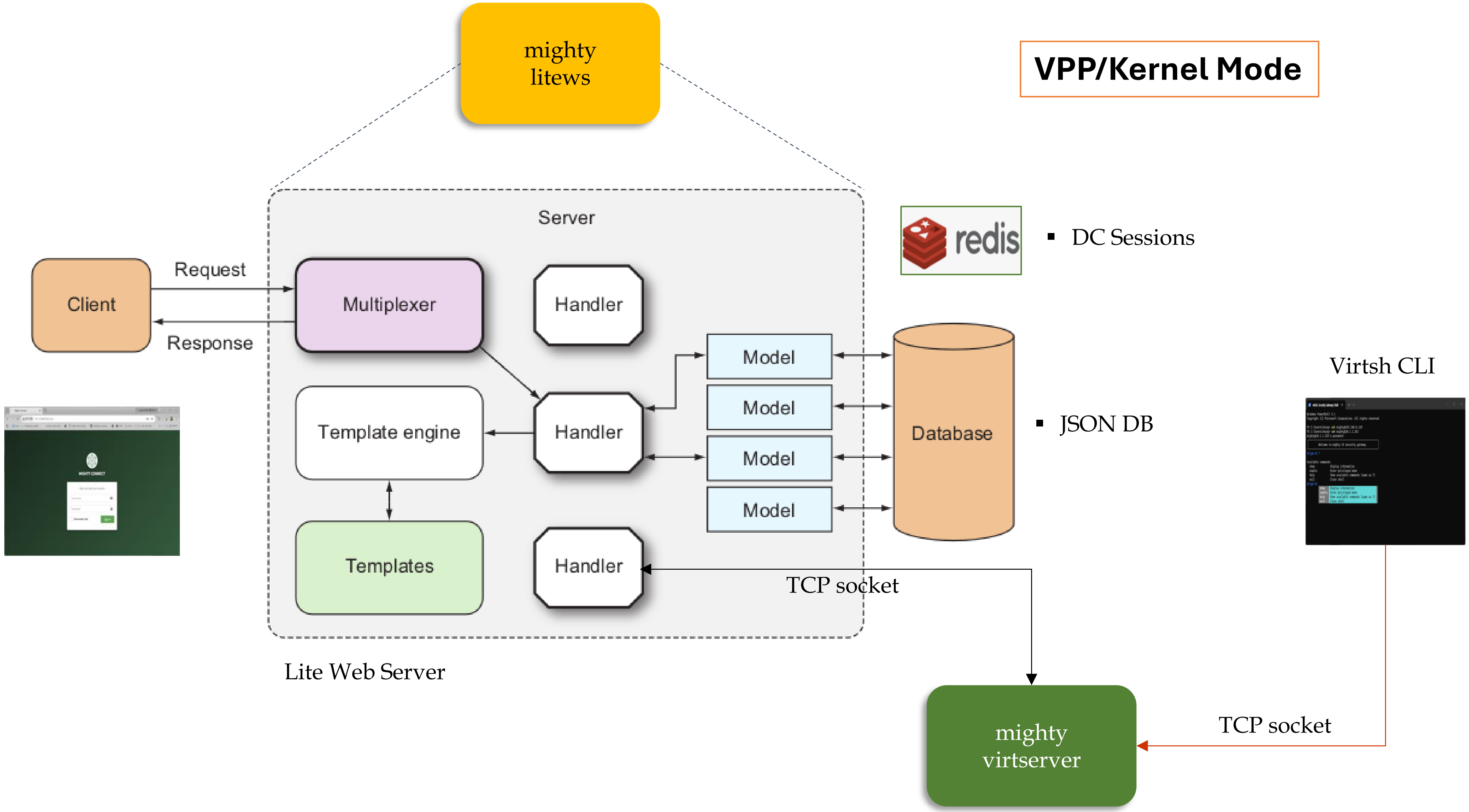
4. MightyConnect Security Box(8) – Mighty Engine(3-1)



4. MightyConnect Security Box(8) – Mighty Engine(3-2)

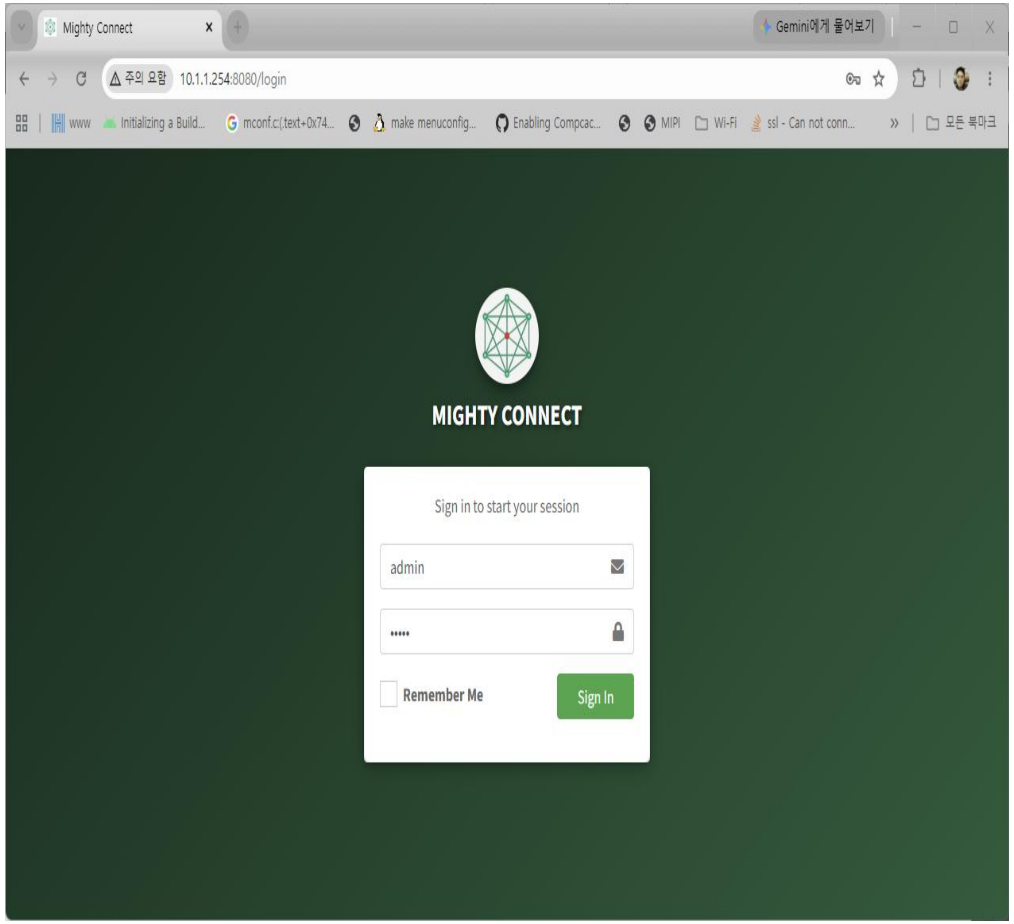


4. MightyConnect Security Box(8) – Mighty Engine(4)

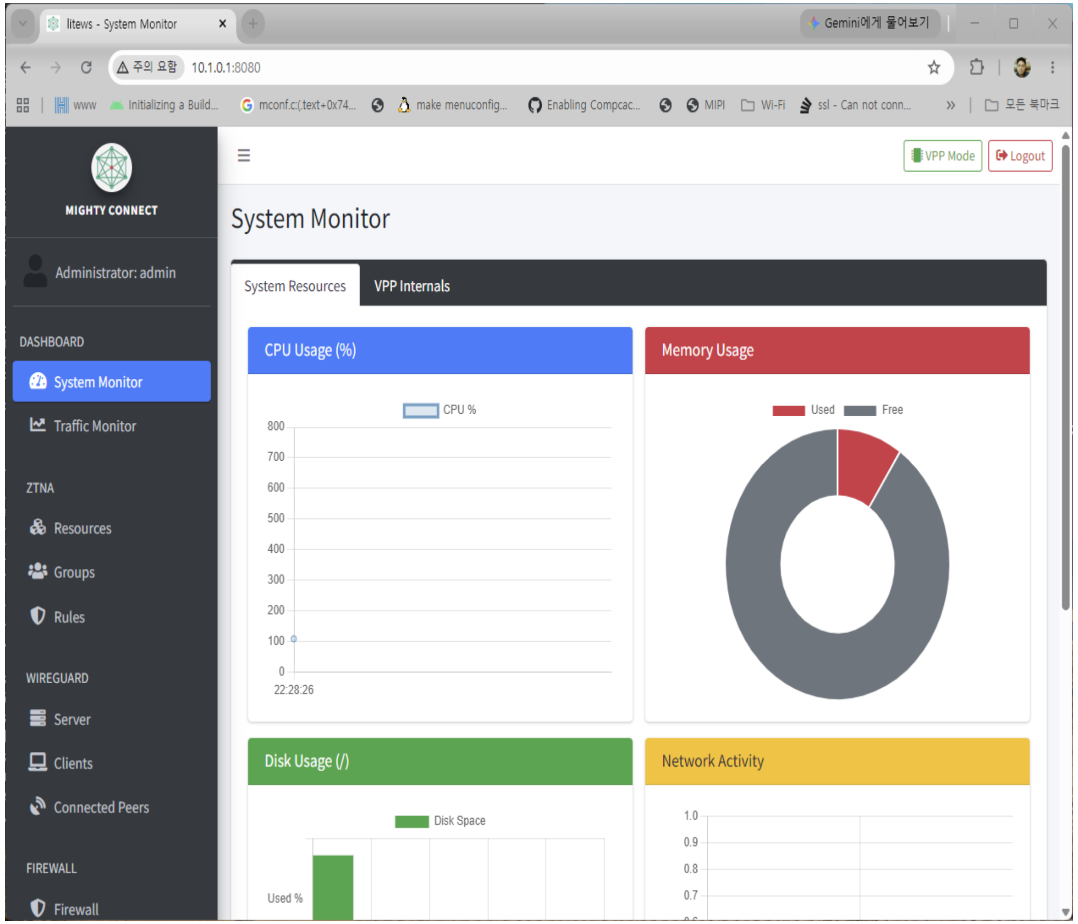


4. MightyConnect Security Box(9) – WebUI(1)

- WebUI Login Page

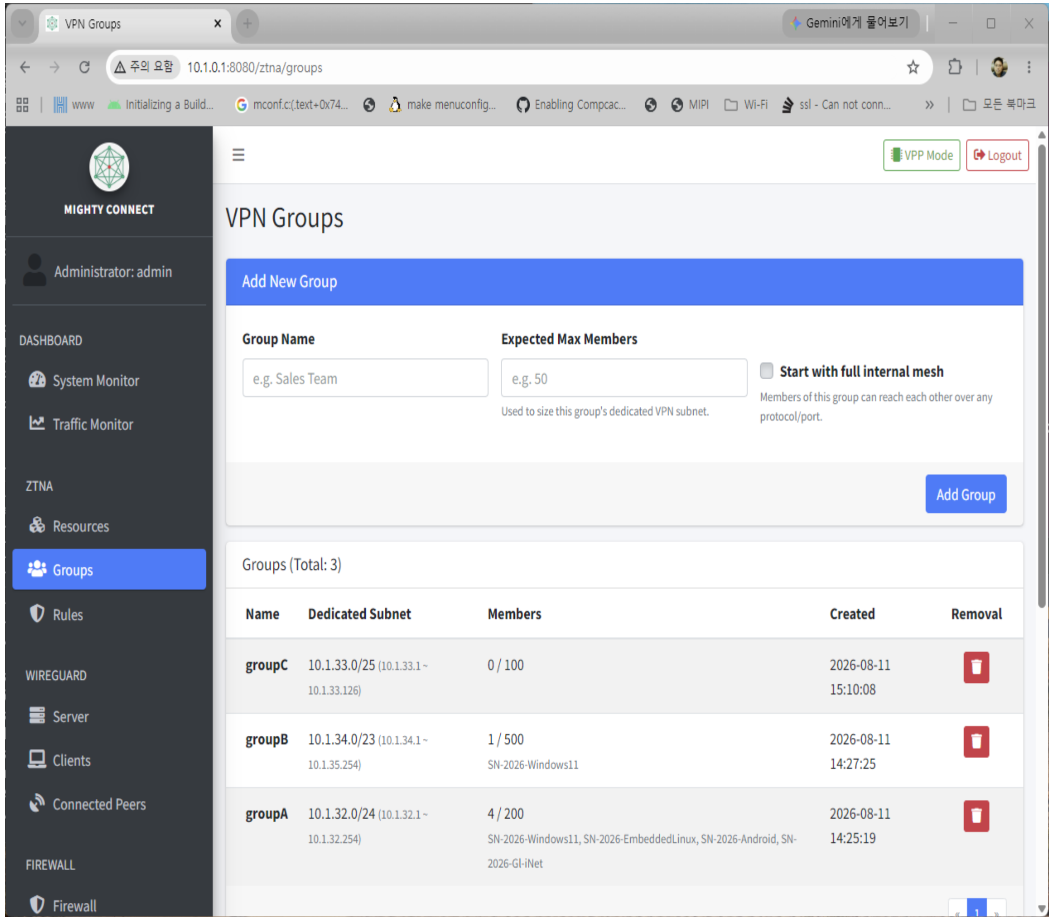


- Dashboard Page

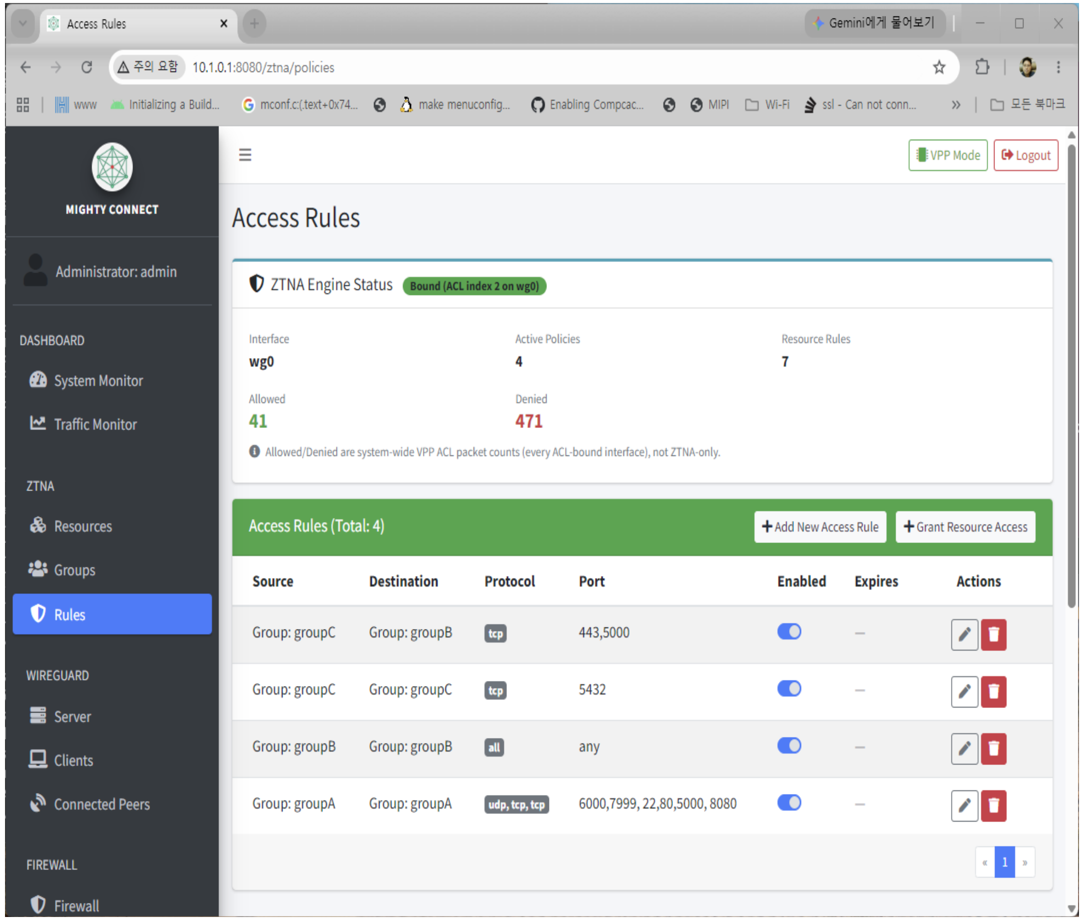


4. MightyConnect Security Box(9) – WebUI(2)

- ZTNA VPN Groups Page

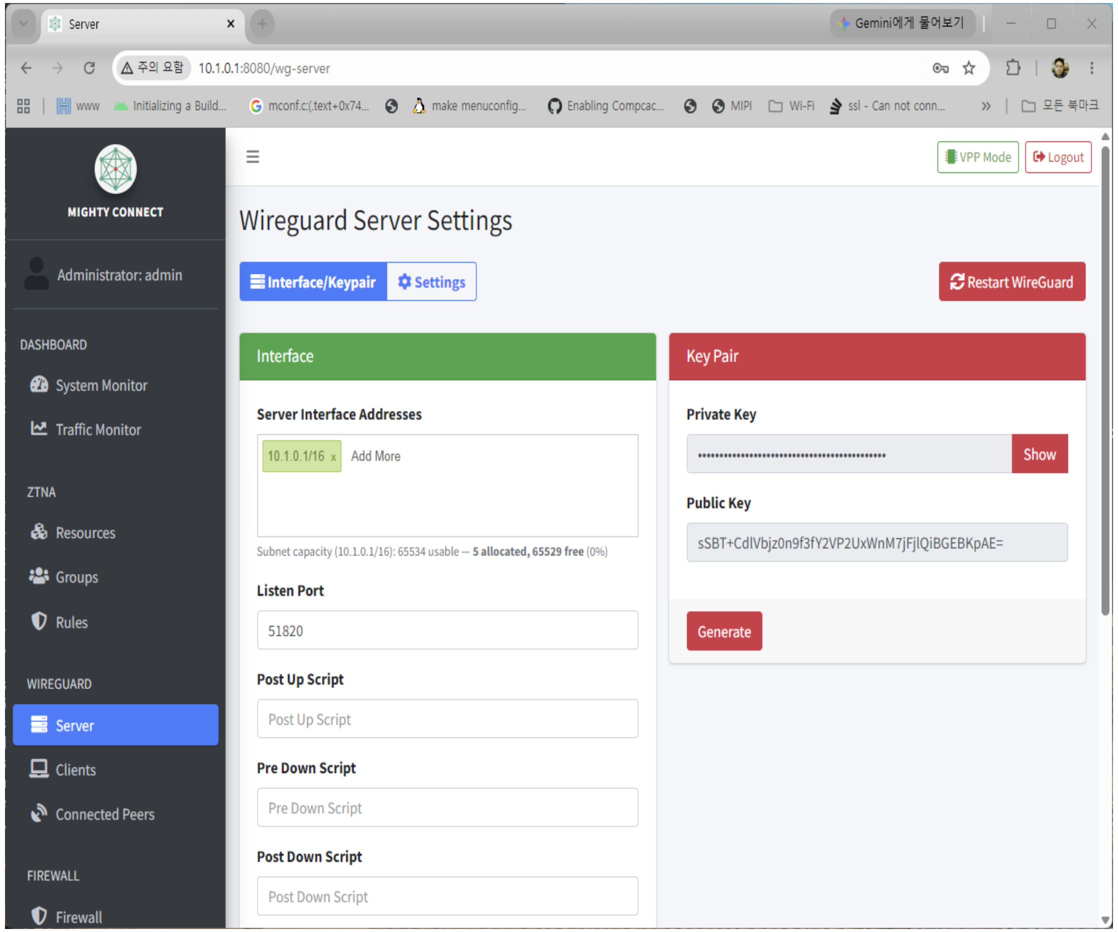


- ZTNA Access Rules Page

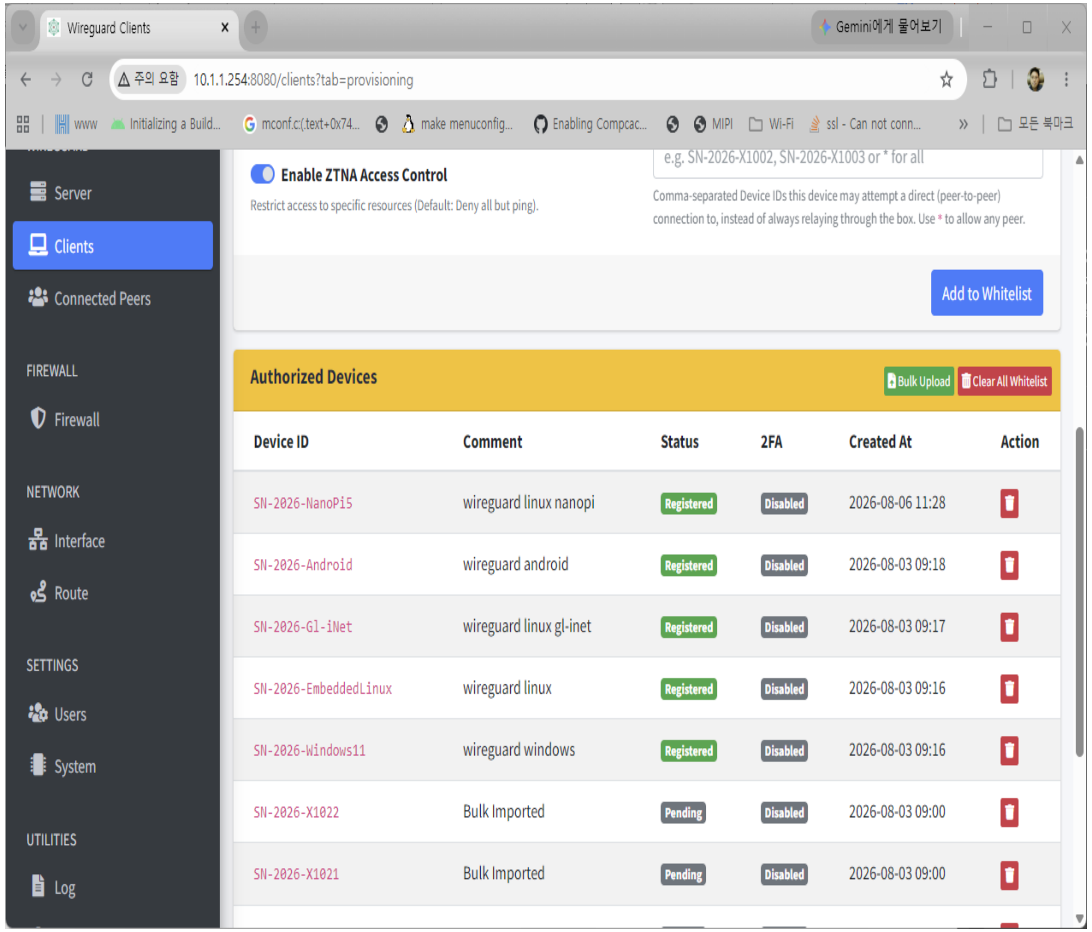


4. MightyConnect Security Box(9) – WebUI(3)

- WireGuard Server Page

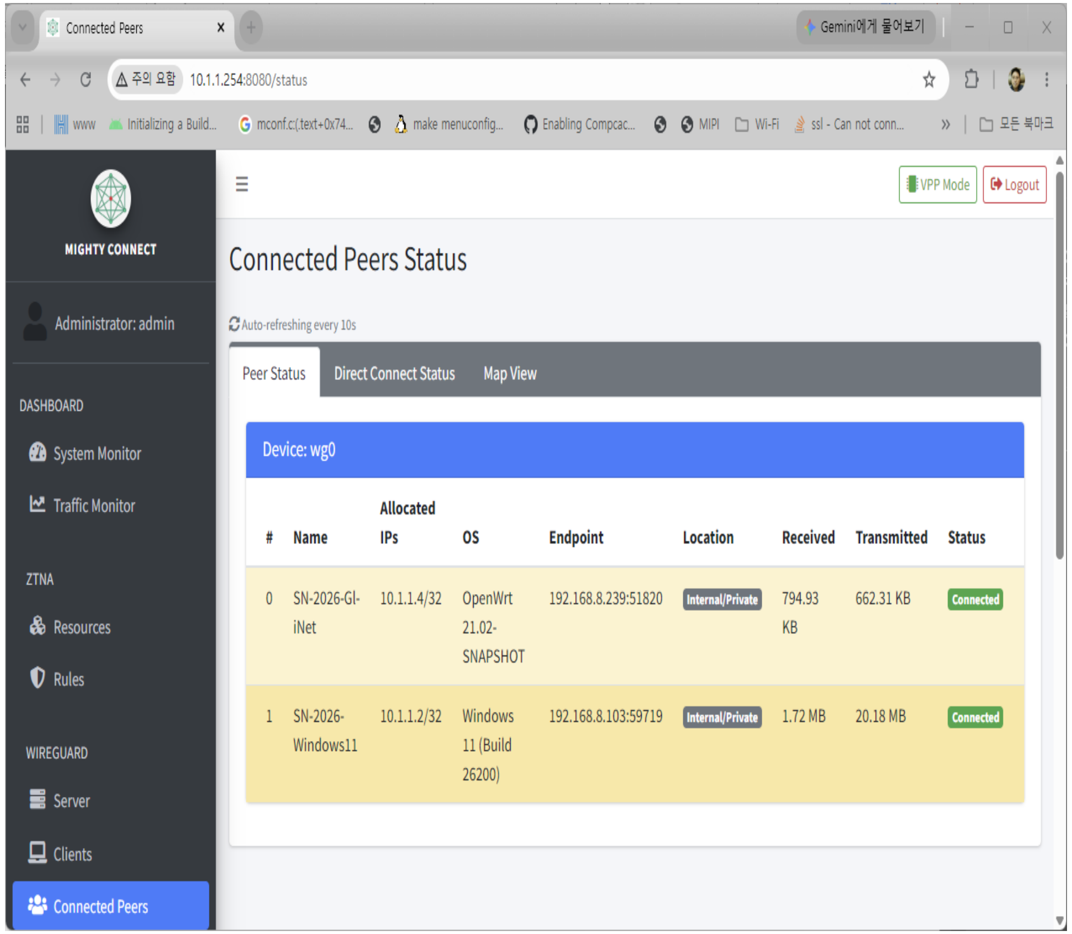


- WireGuard Client Auto Provision Page

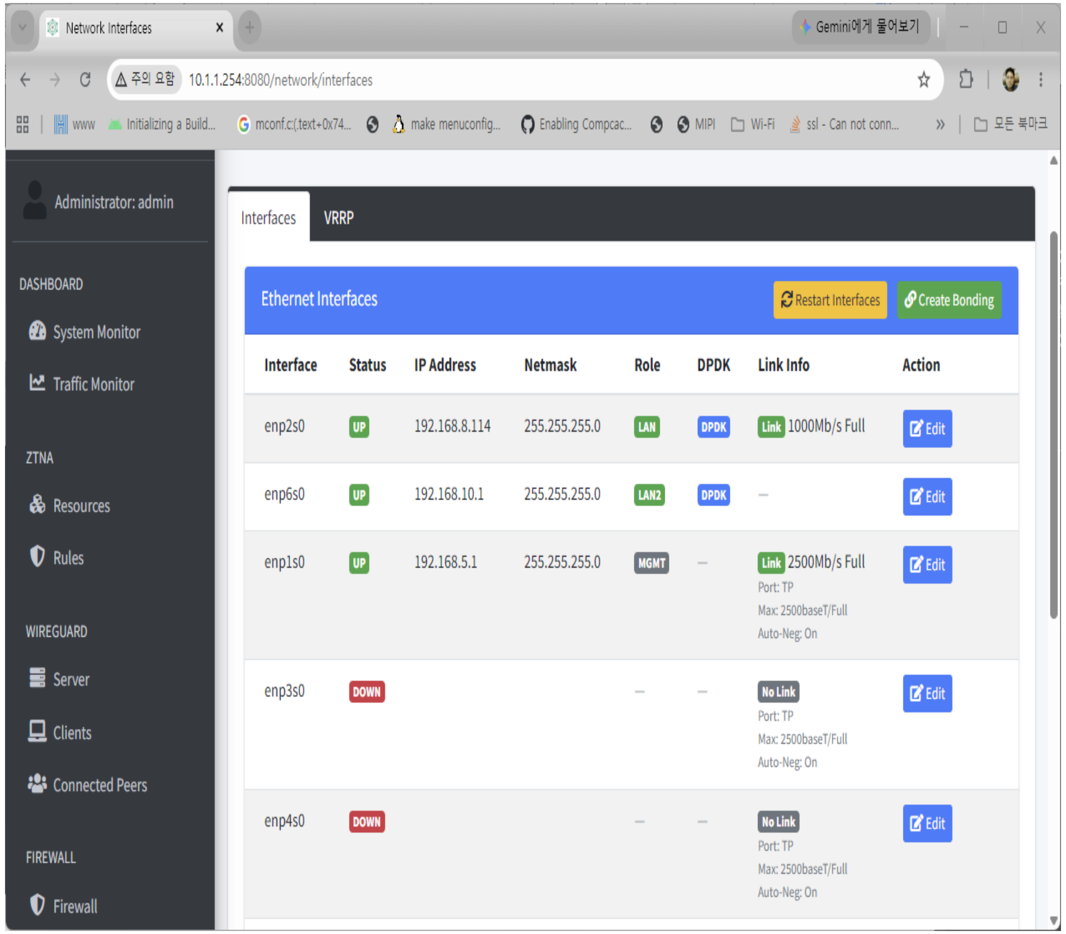


4. MightyConnect Security Box(9) – WebUI(4)

- WireGuard Connected Peers Page

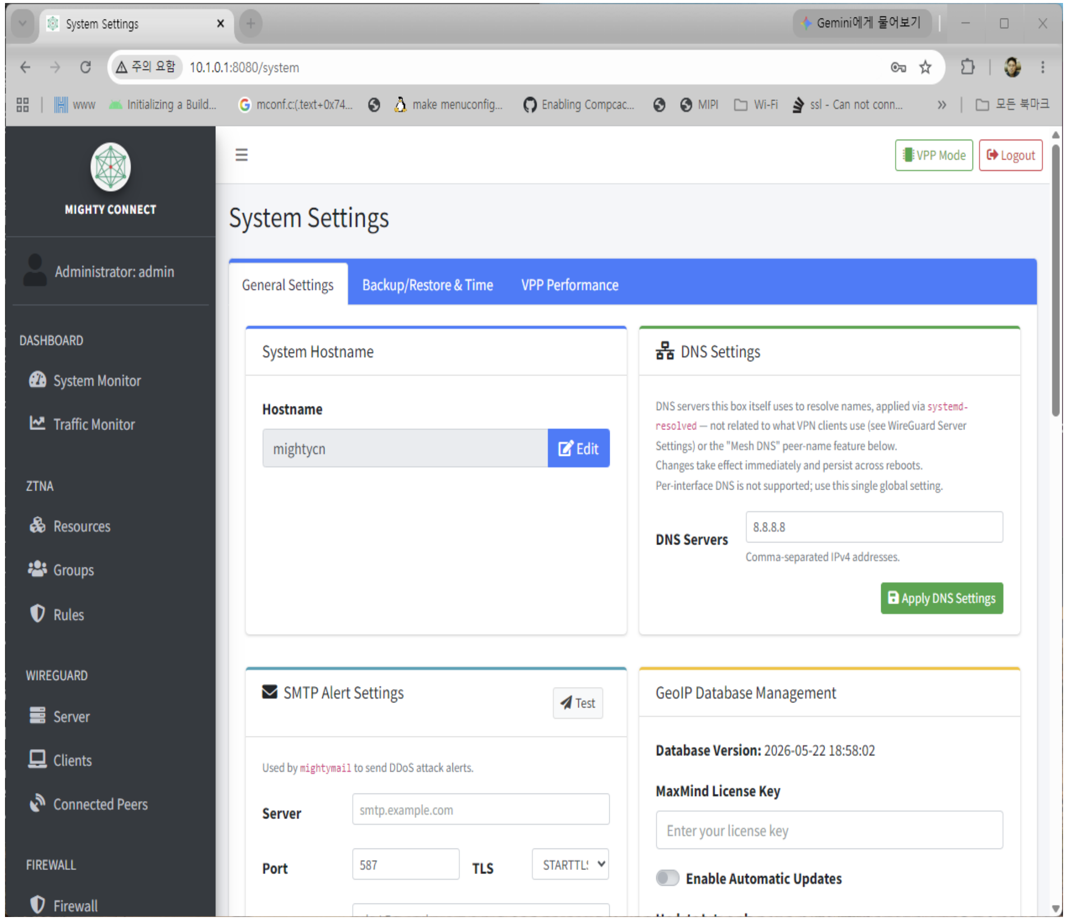


- Network Interfaces Page

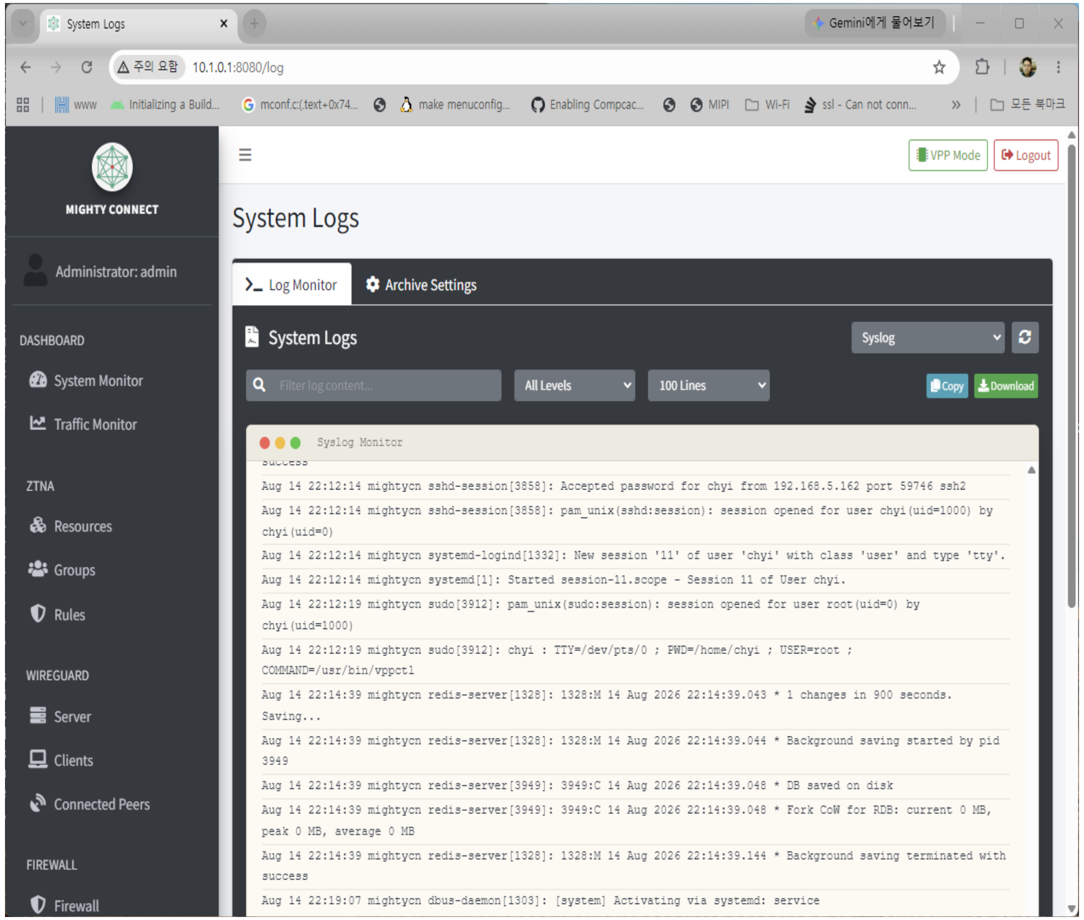


4. MightyConnect Security Box(9) – WebUI(5)

- System Settings Page



- Logs Page



4. MightyConnect Security Box(10) – CLI

```
PS C:\Users\chung> ssh mighty@10.1.1.254
mighty@10.1.1.254's password:
 ----------------------------------------------------
|         Welcome to mighty AI security gateway      |
 ----------------------------------------------------
mightycn# ?

Available commands:
  show           Display information
  enable         Enter privileged mode
  help           Show available commands (same as ?)
  exit           Close shell
mightycn# show interface
Interface  Status  IP Address                    DPDK
-----------------------------------------------------
enp2s0     UP      192.168.8.114/255.255.255.0   yes
enp6s0     UP      192.168.10.1/255.255.255.0    yes
tap4096    UP      -                             yes
tap4097    UP      -                             yes
tun4098    UP      -                             yes
wg0        UP      10.1.1.254/255.255.255.0      yes
enp1s0     UP      192.168.5.1/255.255.255.0     -
enp3s0     DOWN    -                             -
enp4s0     DOWN    -                             -
enp5s0     DOWN    192.168.20.1/255.255.255.0    -

VPP interfaces (show interface):
Name                Idx    State  MTU (L3/IP4/IP6/MPLS)     Counter          Count
enp2s0                          1      up          9000/0/0/0     rx packets                 83438
                                                                  rx bytes                31067596
                                                                  tx packets                 96924
                                                                  tx bytes                69994133
                                                                  drops                       1197
                                                                  punt                        2101
                                                                  ip4                        82166
                                                                  ip6                          286
                                                                  tx-error                       2
enp6s0                          2      up          9000/0/0/0     tx-error                       1
local0                          0     down          0/0/0/0       drops                          4
tap4096                         3      up          9000/0/0/0     rx packets                   547
                                                                  rx bytes                  163088
                                                                  tx packets                   862
                                                                  tx bytes                  191013
                                                                  drops                         16
                                                                  ip4                          459
                                                                  ip6                           17
tap4097                         4      up          9000/0/0/0
tun4098                         6      up          1420/0/0/0     rx packets                 41921
                                                                  rx bytes                40105173
                                                                  tx packets                 27211
                                                                  tx bytes                 2196120
                                                                  ip4                        41903
                                                                  ip6                           18
wg0                             5      up          1420/0/0/0     rx packets                 79471
```

```
mightycn# show ztna
Network ZTNA Rules (L4 Access Control):
Client Name (Label)    Client IP    Authorized Resources                                             Status
------------------------------------------------------------------------------------------------------------
SN-2026-Android        10.1.1.3/32  default-ping (icmp/10.1.1.0/24/any),                              Enabled
                                    ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443)
SN-2026-EmbeddedLinux  10.1.1.1/32  default-ping (icmp/10.1.1.0/24/any),                              Enabled
                                    ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                    http (tcp/10.1.1.0/24/80)
SN-2026-Gl-iNet        10.1.1.4/32  default-ping (icmp/10.1.1.0/24/any),                              Enabled
                                    ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                    http (tcp/10.1.1.0/24/80), litews (tcp/10.1.1.0/24/5000)
SN-2026-NanoPi5        10.1.1.5/32  default-ping (icmp/10.1.1.0/24/any),                              Enabled
                                    ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                    http (tcp/10.1.1.0/24/80)
SN-2026-Windows11      10.1.1.2/32  default-ping (icmp/10.1.1.0/24/any),                              Enabled
                                    ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                    http (tcp/10.1.1.0/24/80), postgresql (tcp/10.1.1.0/24/5432),
                                    litews (tcp/10.1.1.0/24/5000),
                                    boxlitews (tcp/10.1.1.254/32/8080)

Always Allowed (all clients, bypasses policy_map entirely): DNS tcp/53, DNS udp/53, ICMP

VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
          0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
          1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
          2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
          3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
          4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
          5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
          6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
          7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
          8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index: 1, 2
  applied outbound on sw_if_index:
  used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
          0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index:
  applied outbound on sw_if_index: 1, 2
  used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
          0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
          1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
          2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
          3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
          4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
          5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
          6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
          7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
          8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
          9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
--More--
```

4. MightyConnect Security Box(11) – H/W Spec(1)

고급(상급) 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-5000 / 2U Rackmount
CPU	Intel® Xeon® Scalable Processor Gold 5218 2CPU(32 cores, 64 threads), 2.3GHz
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/4TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(2)

중급 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-2500 / 2U Rackmount
CPU	Intel® Xeon® D-2779 1 CPU(16 core, 32 threads)
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(3)

하급 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-1000 / 1U Rackmount
CPU	Intel® Xeon® D-2146NT 1 CPU(8 core, 16 threads)
RAM	DDR4 32GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 1 GbE RJ45 2 ports
Power Input	200W Low-noise AC-DC power supply

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

5. MightyConnect Devices

MightyConnect Devices

5. MightyConnect Devices

MightyConnect™ Box

MightyConnect™ Mini

MightyConnect™ Windows

MightyConnect™ Ubuntu

MightyConnect™ macOS/iOS

MightyConnect™ Android

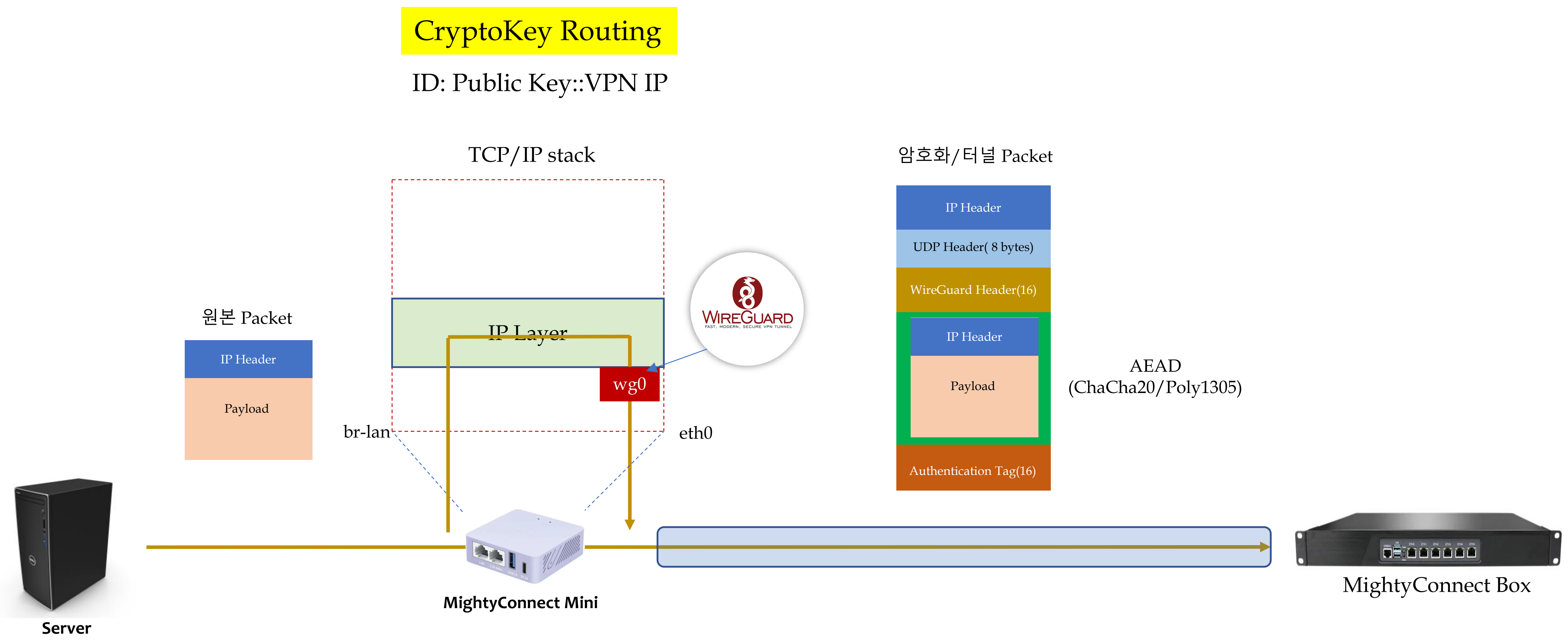
5.1 MightyConnect Mini

MightyConnect Mini Security Gateway

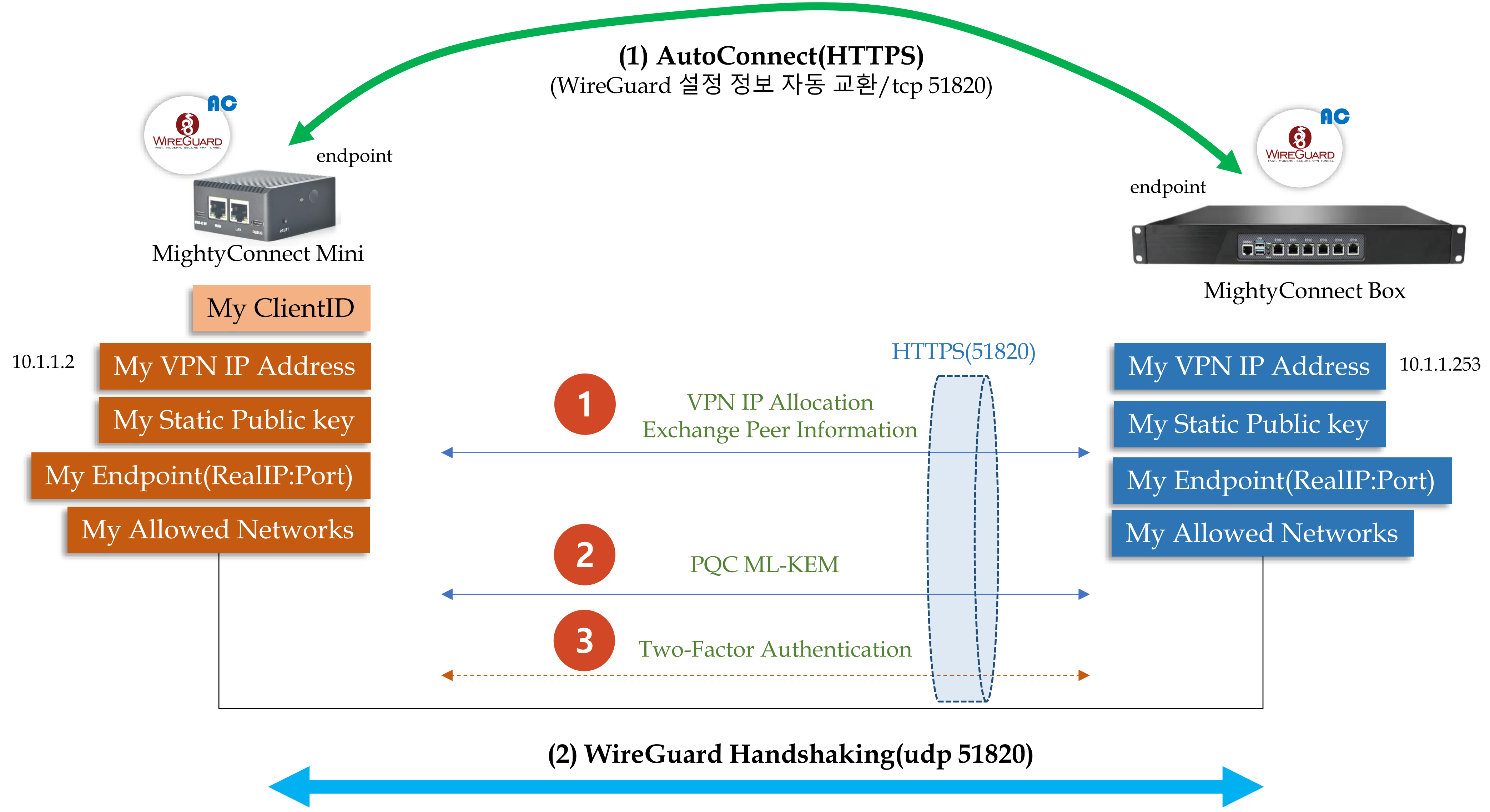
GL-iNet Series
https://www.gl-inet.com/

NanoPi Series
https://www.friendlyelec.com/

5.1 MightyConnect Mini(1) – Overview

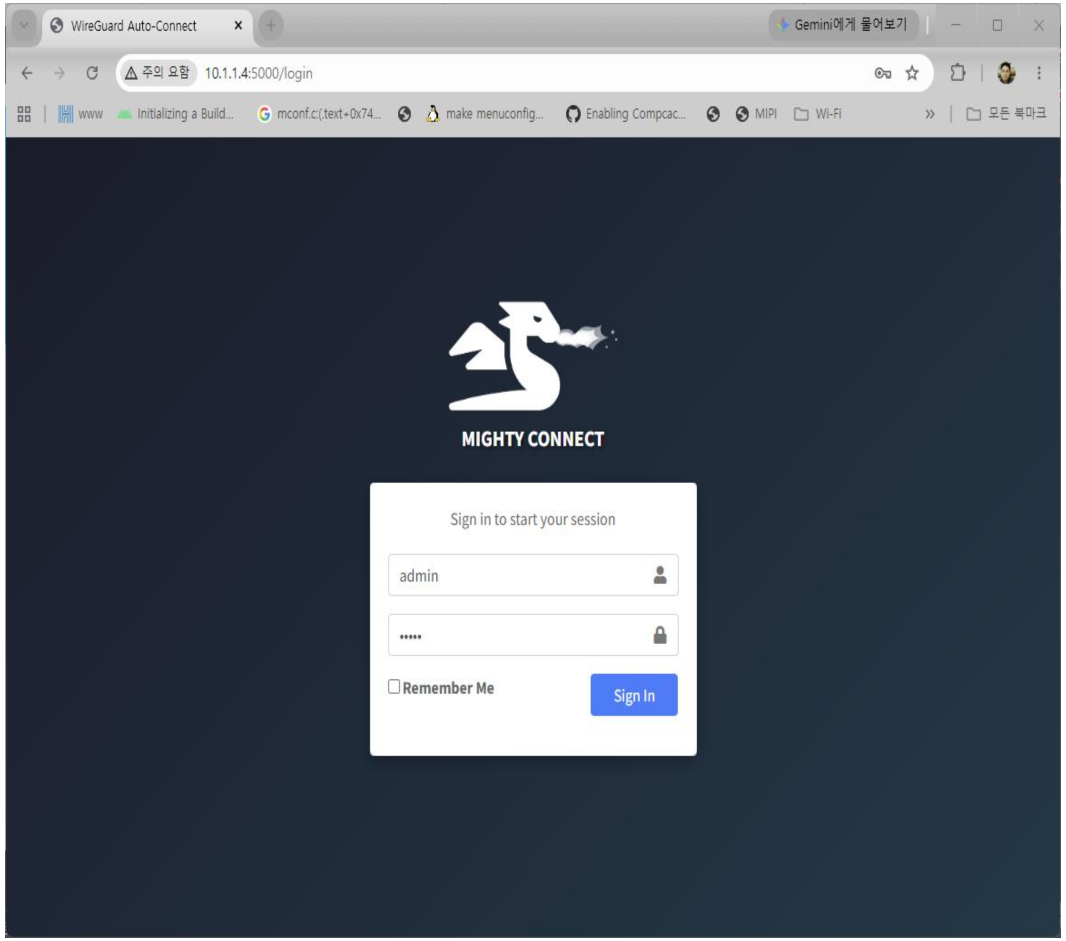


5.1 MightyConnect Mini(2) – Auto Connection

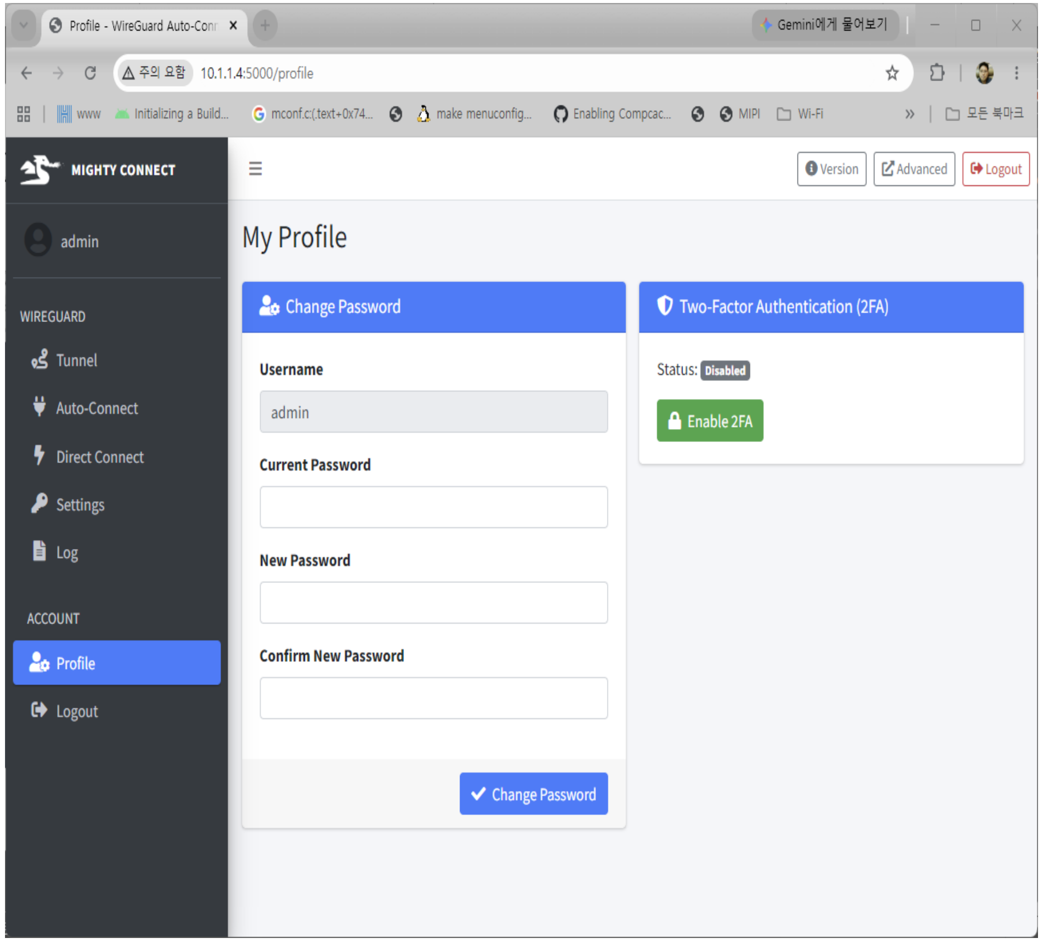


5.1 MightyConnect Mini(3) – WebUI(1)

- Login Page

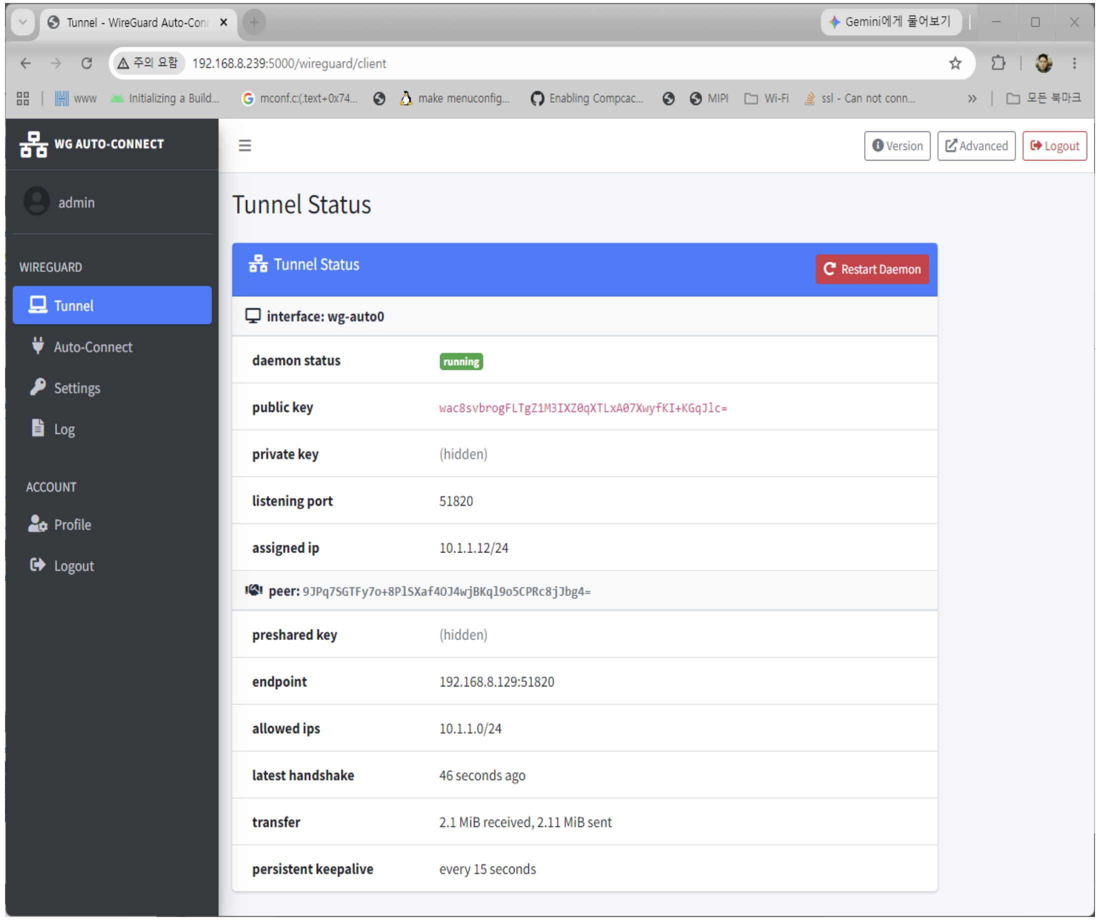


- Admin User Page

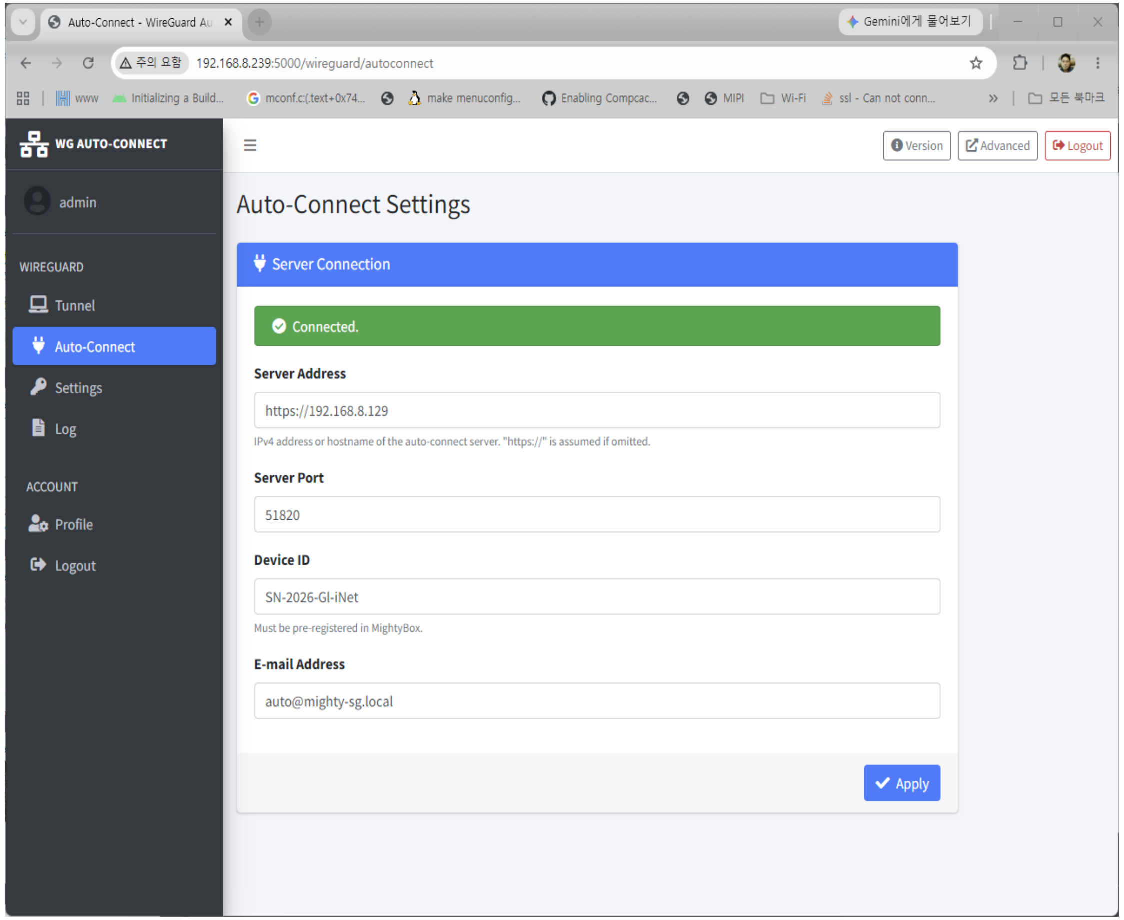


5.1 MightyConnect Mini(3) – WebUI(2)

- Tunnel Status Page

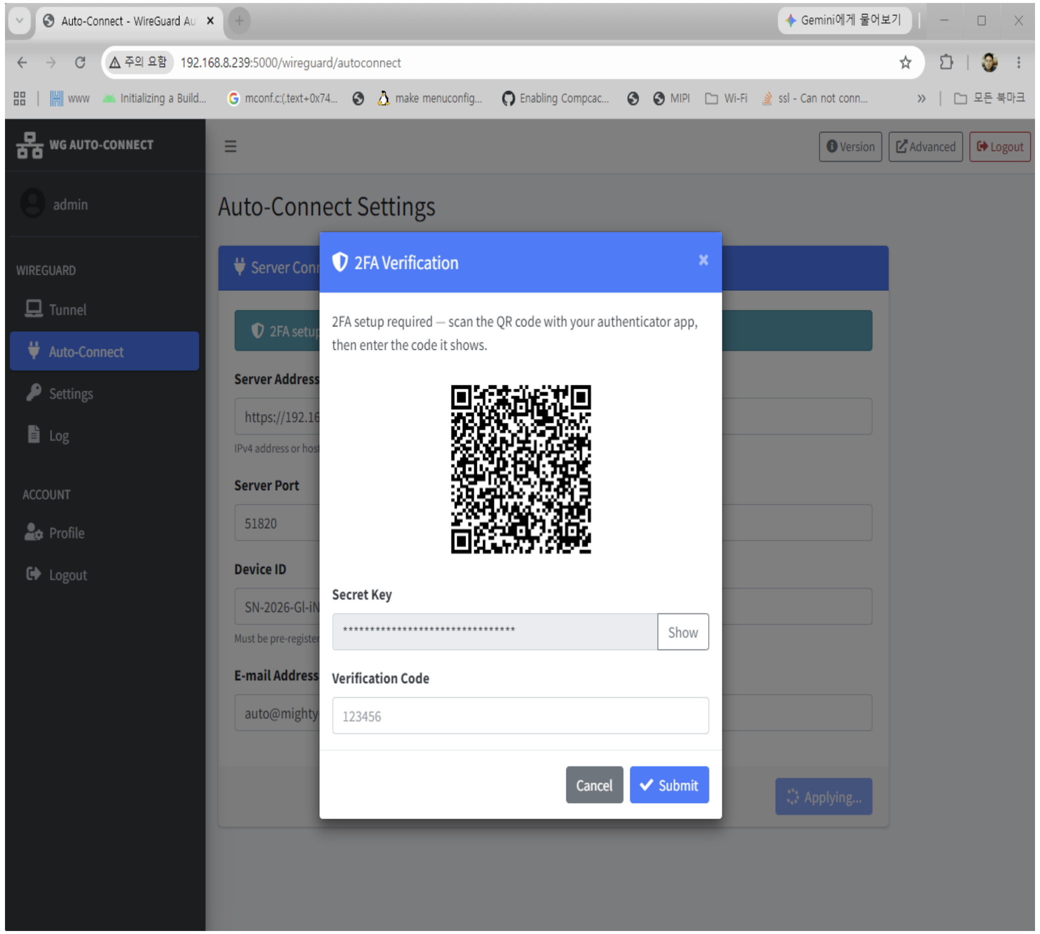


- Auto Connection Page

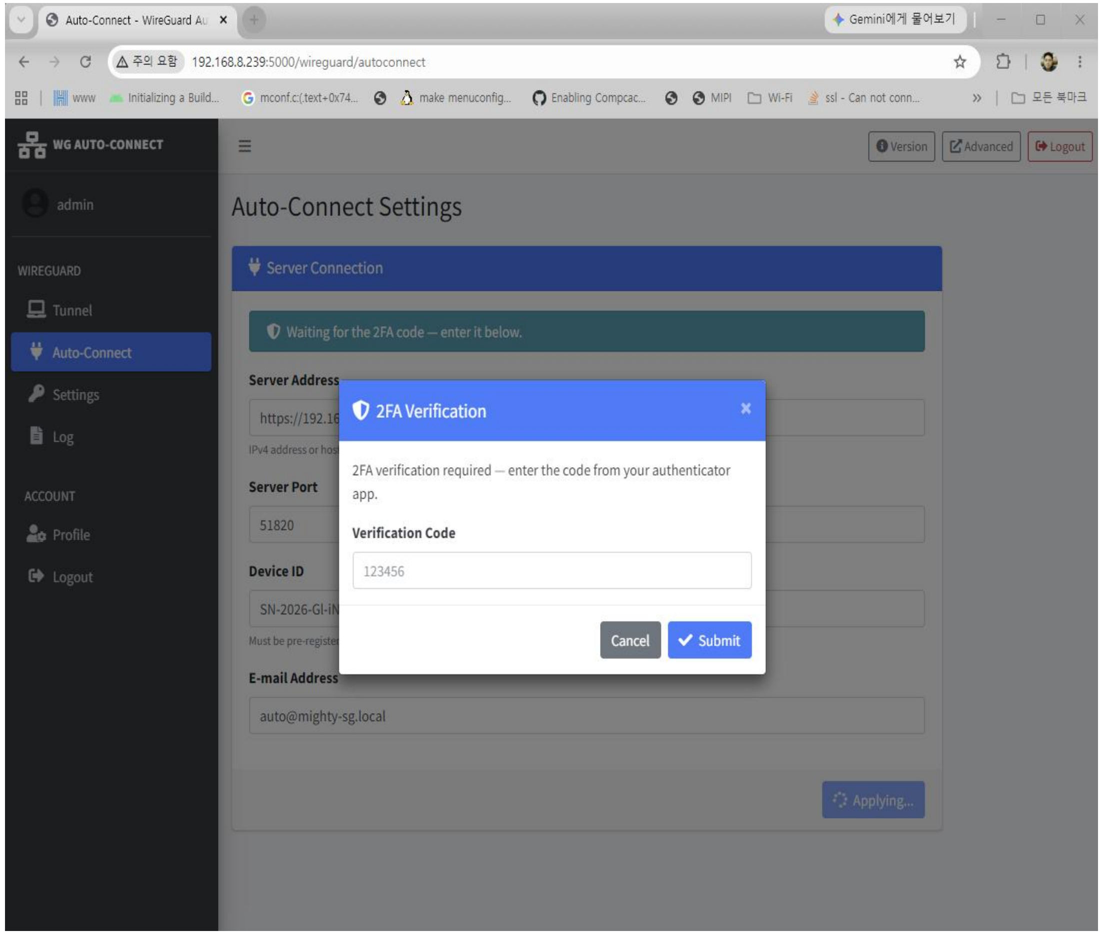


5.1 MightyConnect Mini(3) – WebUI(3)

- Auto Connection 2FA Step1 Page

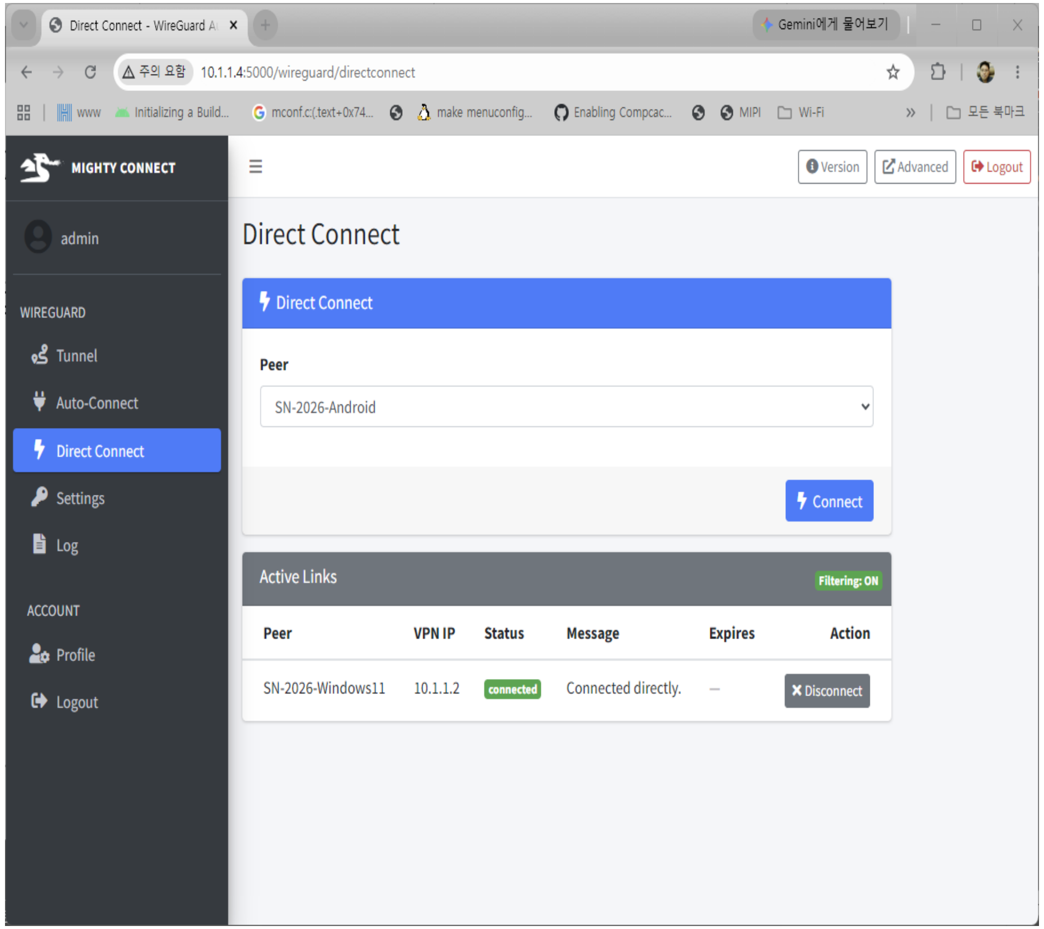


- Auto Connection 2FA Step2 Page

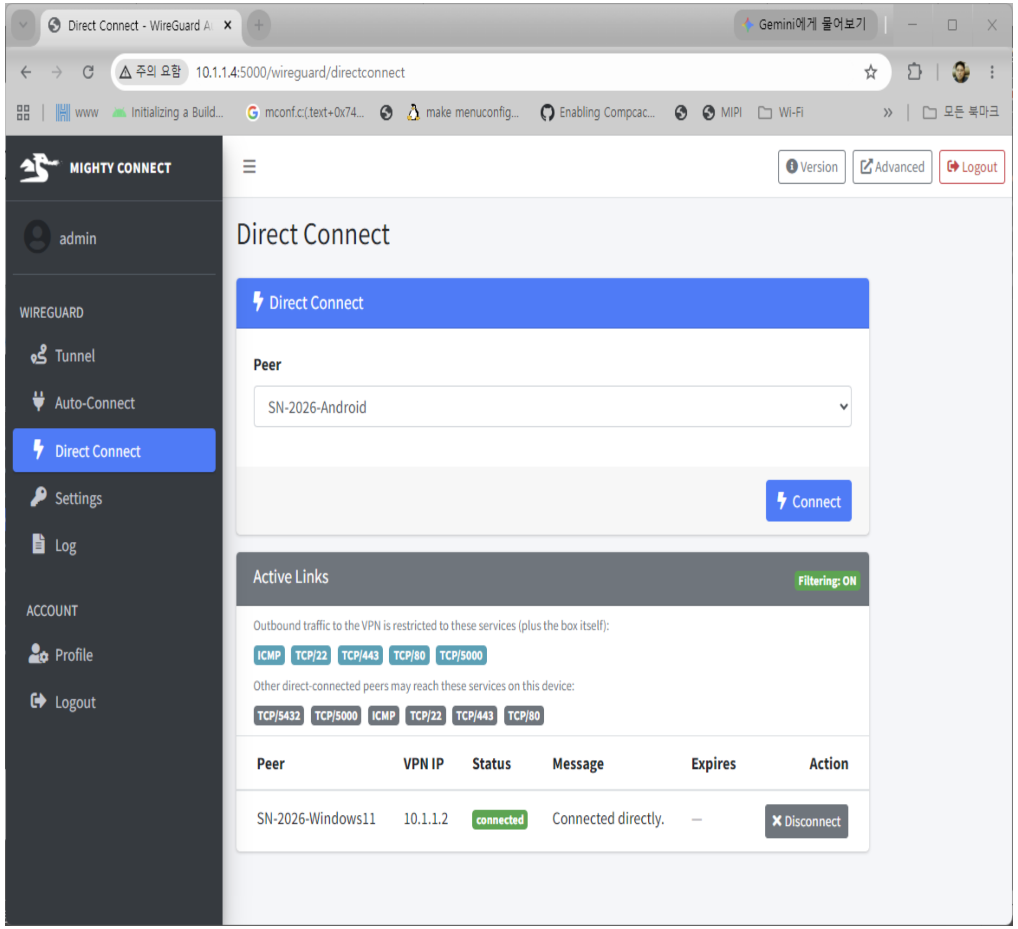


5.1 MightyConnect Mini(3) – WebUI(4)

- Direct Connect Page

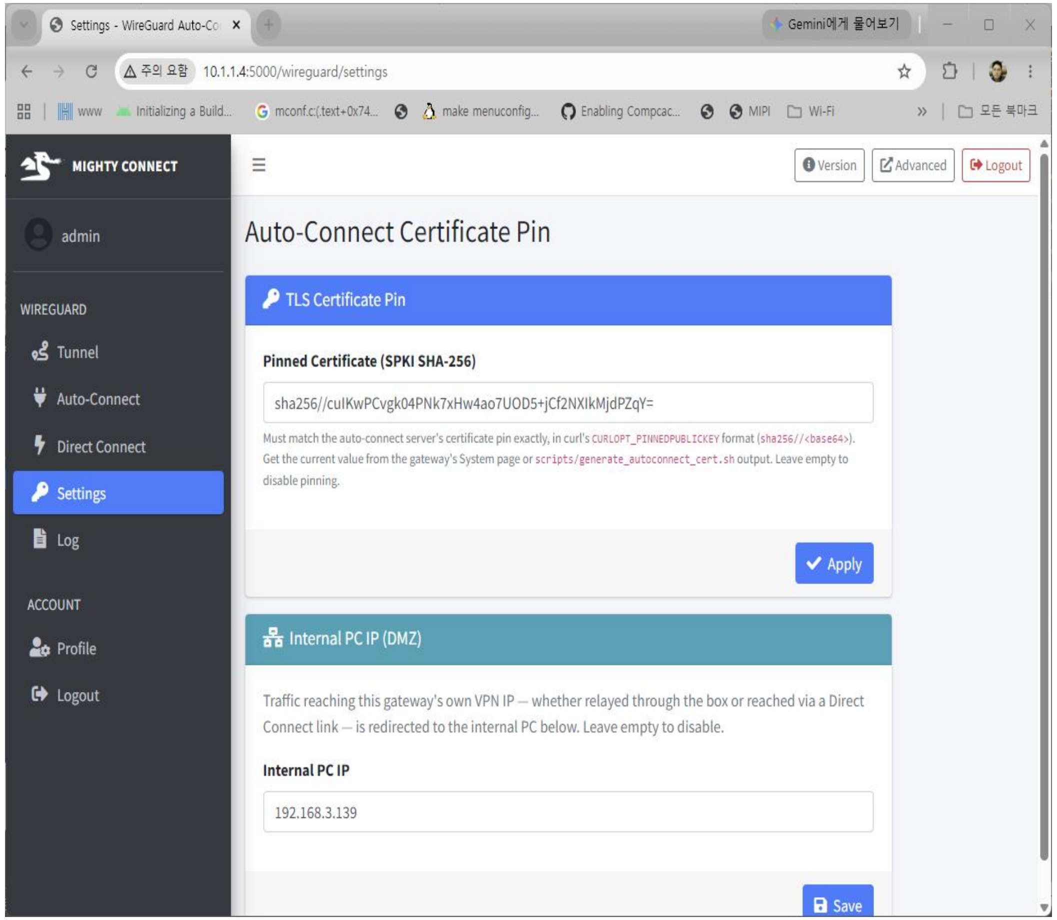


- Direct Connect Filter Rules

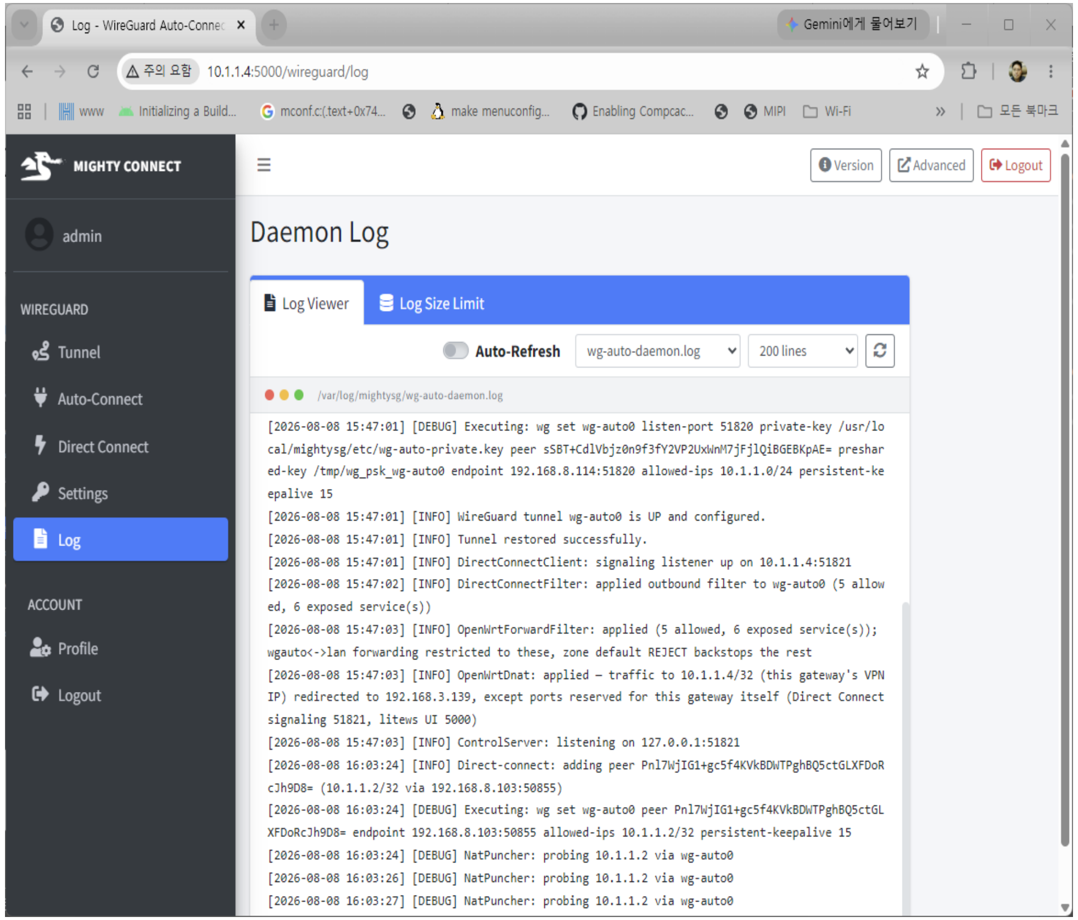


5.1 MightyConnect Mini(3) – WebUI(5)

- Settings Page



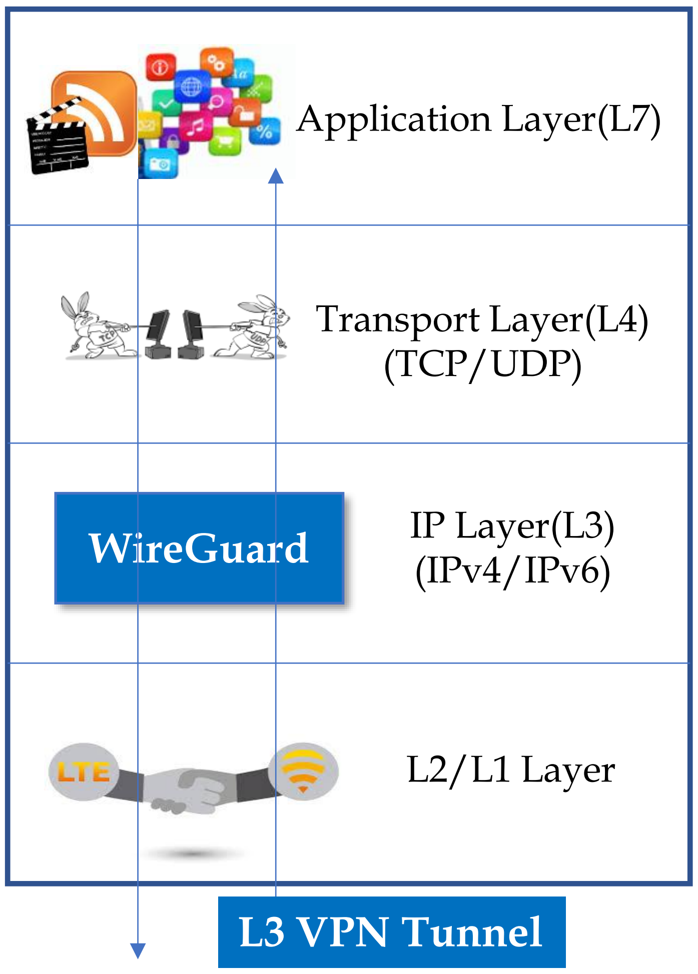
- Log Page

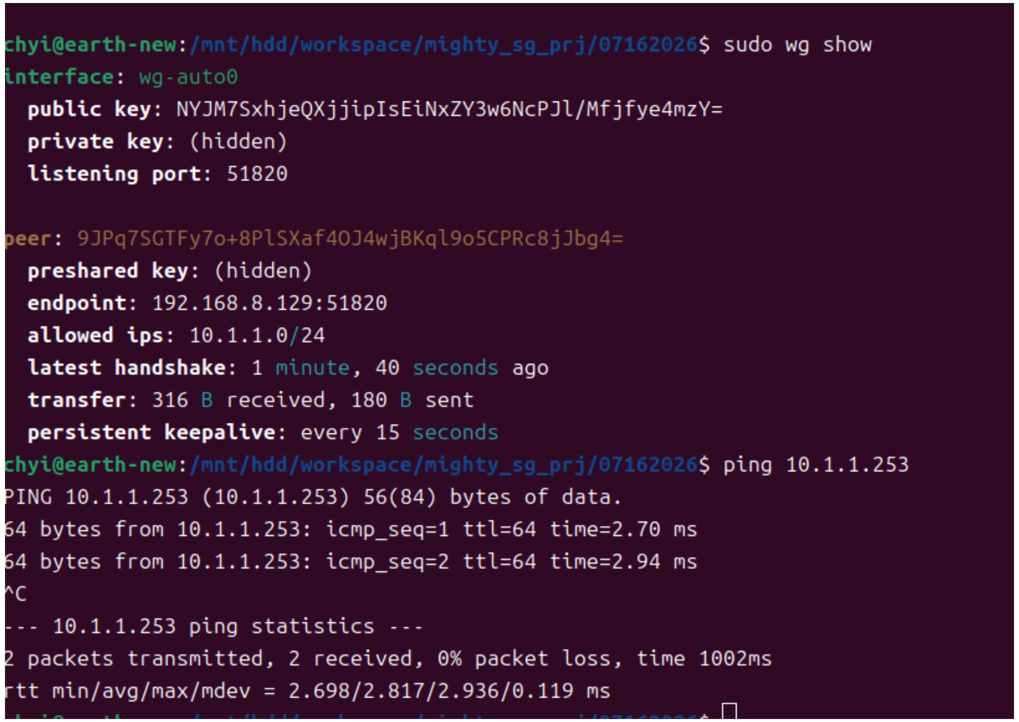


5.2 MightyConnect Ubuntu

5.2 MightyConnect Ubuntu(1) – WireGuard

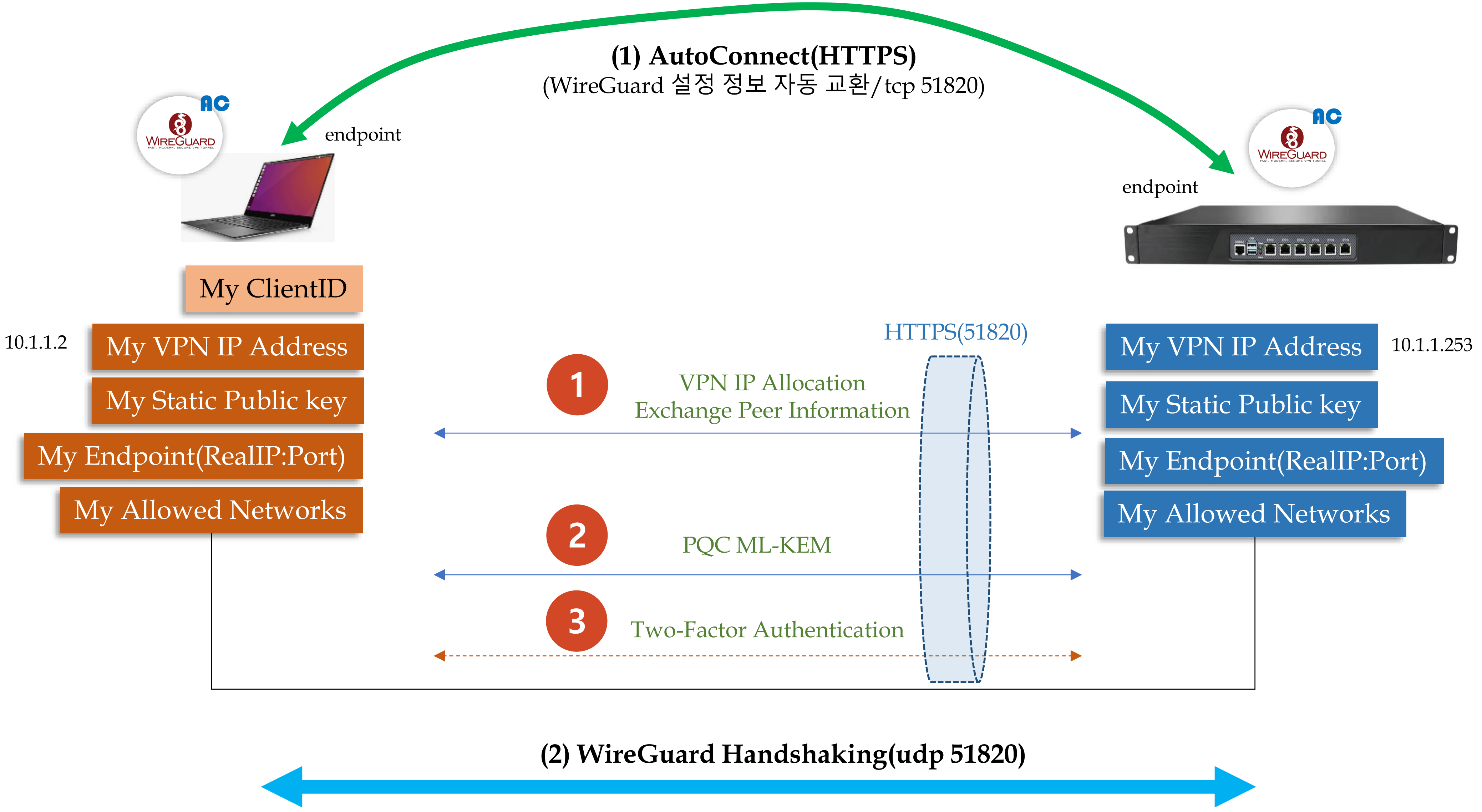
Cryptokey Routing





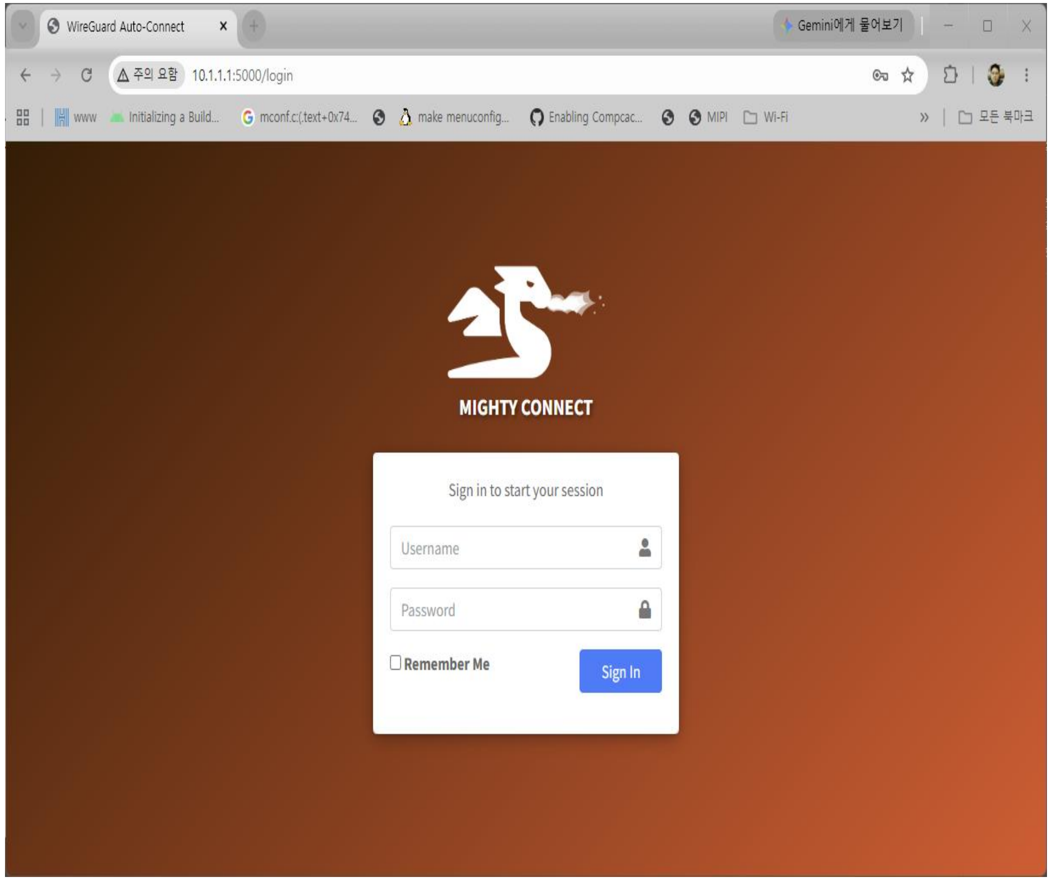
```
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ sudo wg show
interface: wg-auto0
  public key: NYJM7SxhjeQXjjipIsEiNxZY3w6NcPJl/Mfjfye4mzY=
  private key: (hidden)
  listening port: 51820

peer: 9JPq7SGTFy7o+8PlSXaf4OJ4wjBKql9o5CPRc8jJbg4=
  preshared key: (hidden)
  endpoint: 192.168.8.129:51820
  allowed ips: 10.1.1.0/24
  latest handshake: 1 minute, 40 seconds ago
  transfer: 316 B received, 180 B sent
  persistent keepalive: every 15 seconds
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ ping 10.1.1.253
PING 10.1.1.253 (10.1.1.253) 56(84) bytes of data.
64 bytes from 10.1.1.253: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 10.1.1.253: icmp_seq=2 ttl=64 time=2.94 ms
^C
--- 10.1.1.253 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.698/2.817/2.936/0.119 ms
```

5.2 MightyConnect Ubuntu(2) – Auto Connection

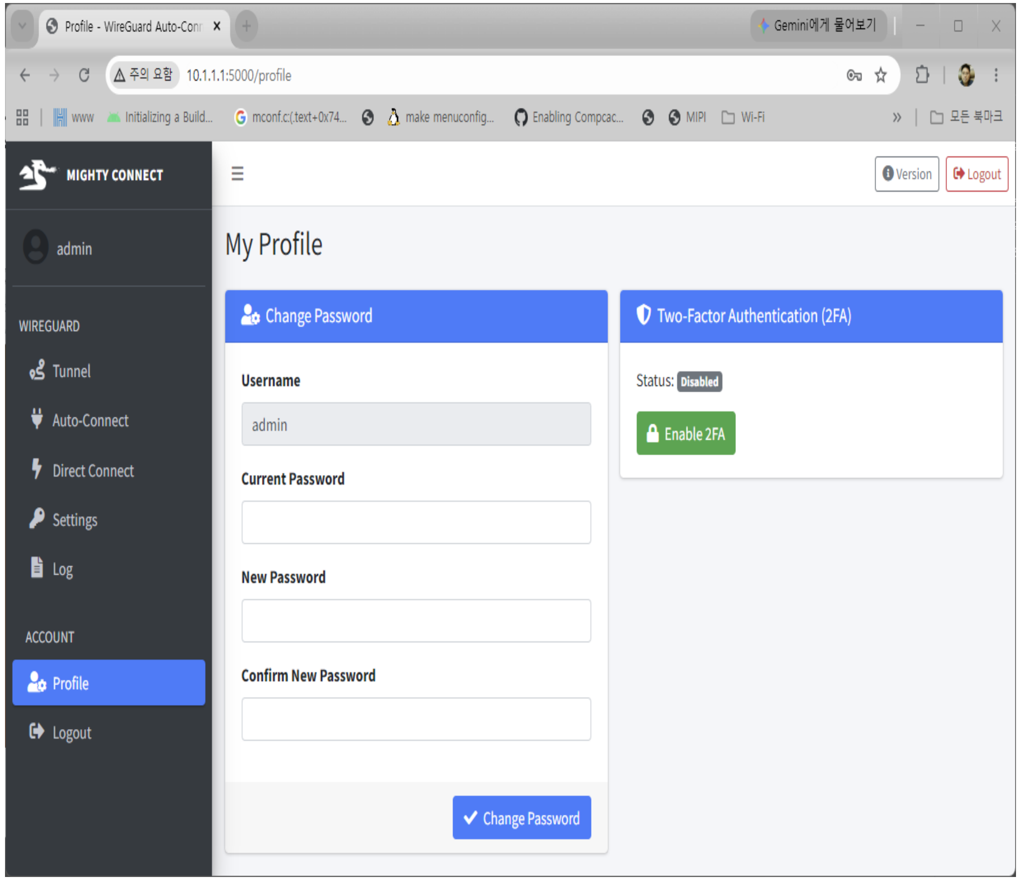


5.2 MightyConnect Ubuntu(3) – WebUI(1)

- Login Page

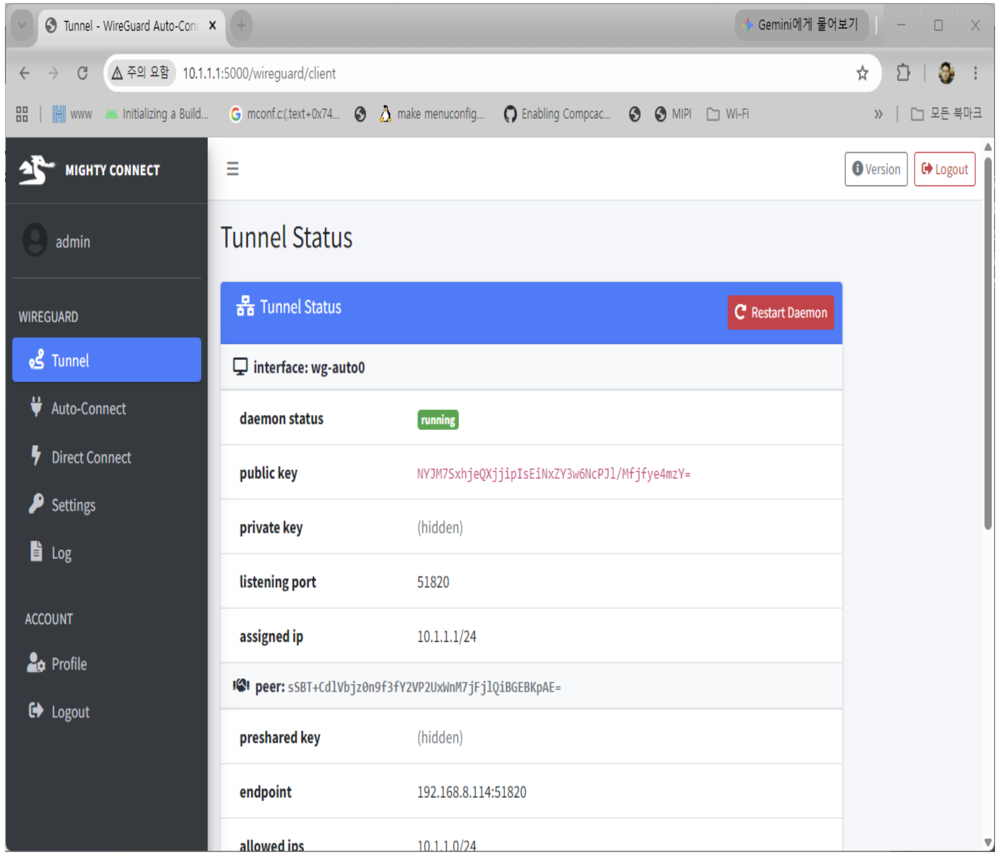


- Admin User Page

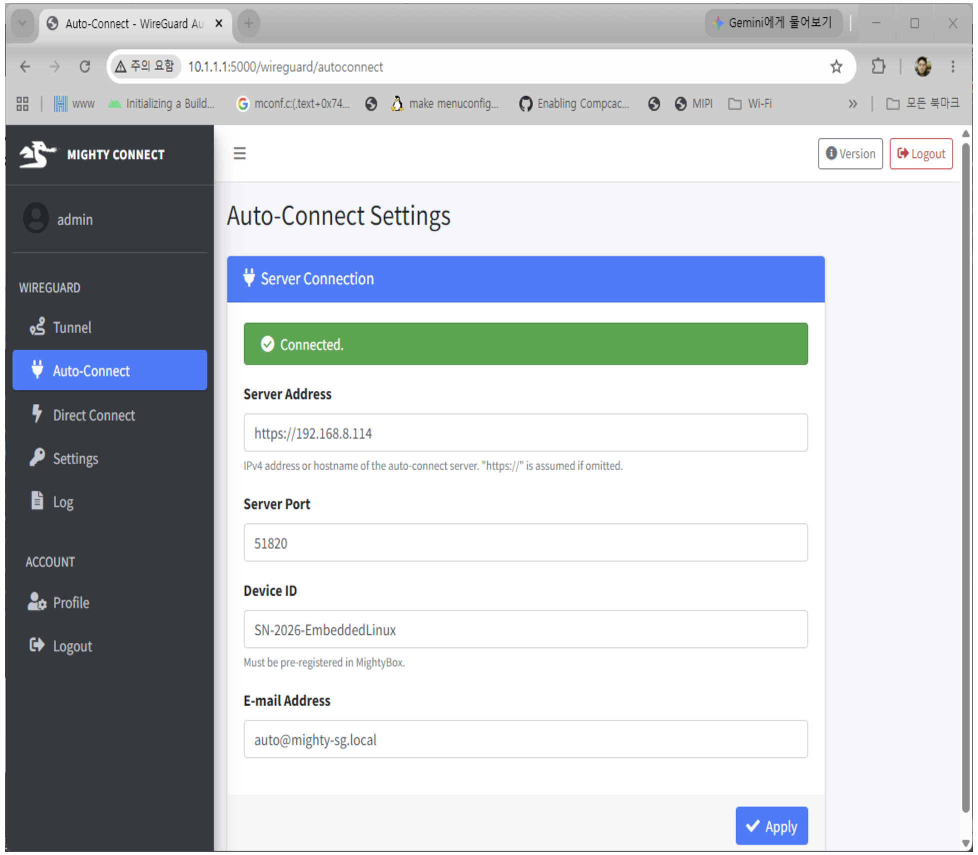


5.2 MightyConnect Ubuntu(3) – WebUI(2)

- Tunnel Status Page

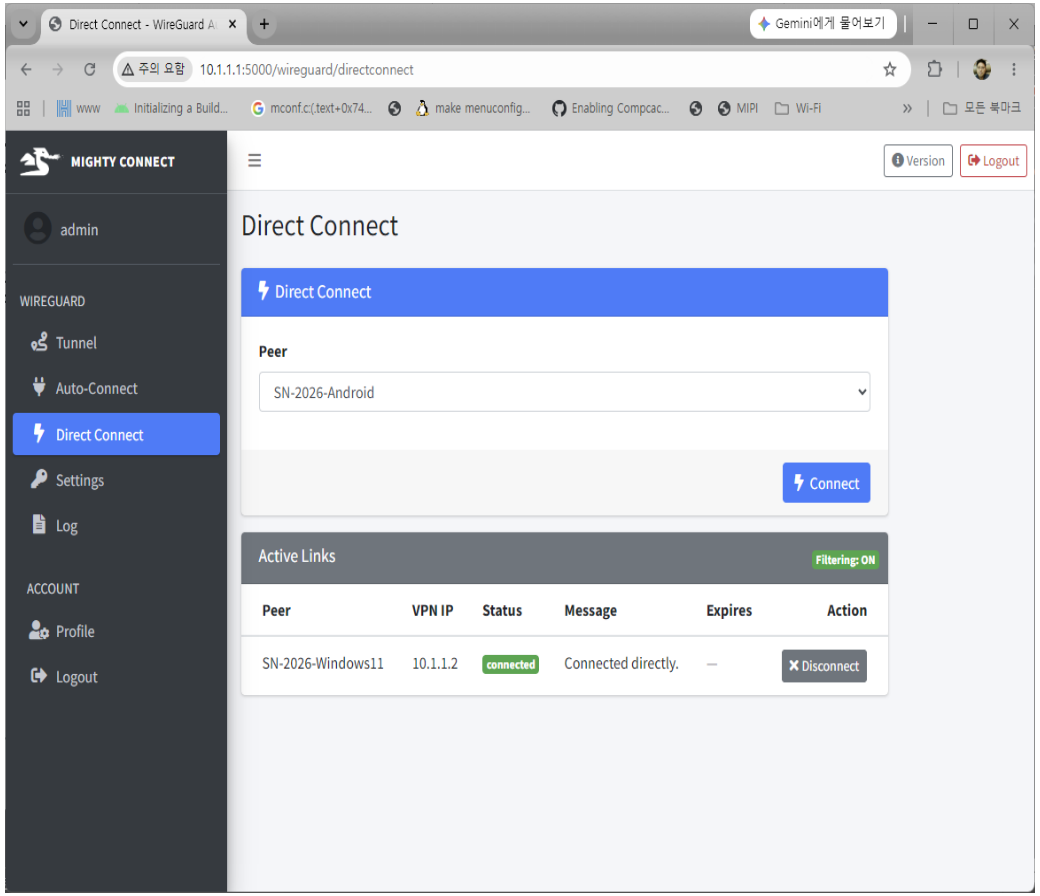


- Auto Connection Page

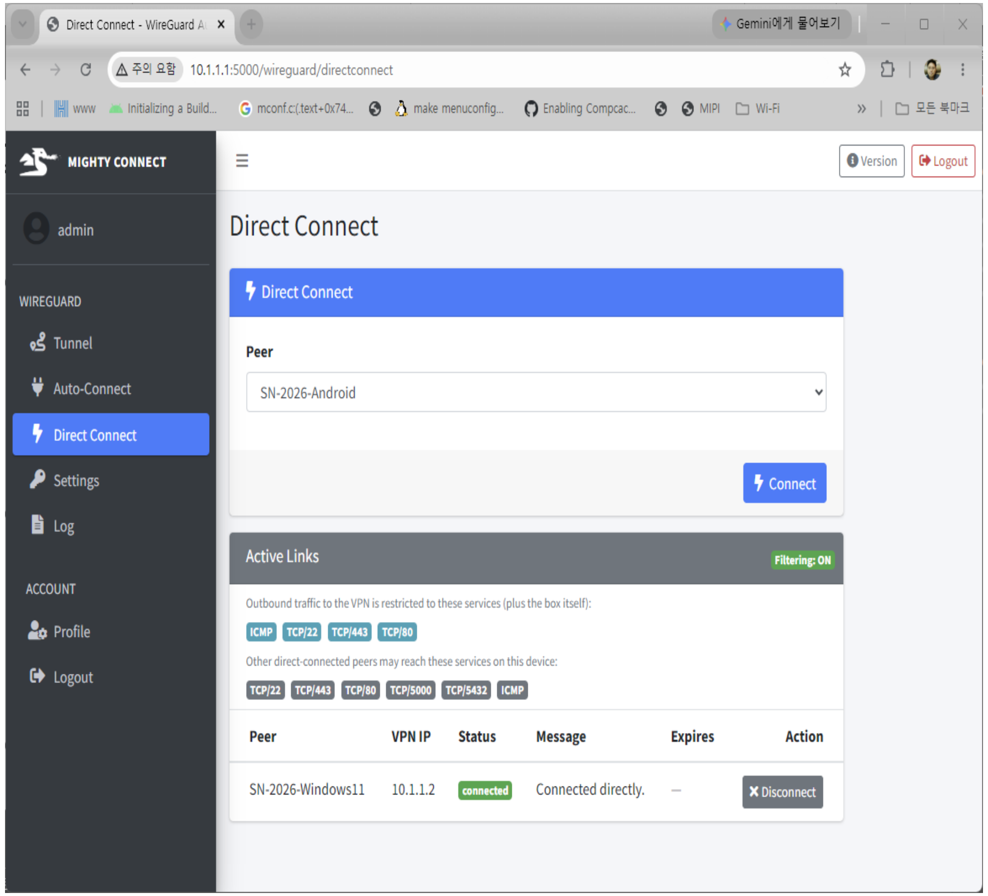


5.2 MightyConnect Ubuntu(3) – WebUI(3)

- Direct Connect Page



- Direct Connect Filtering

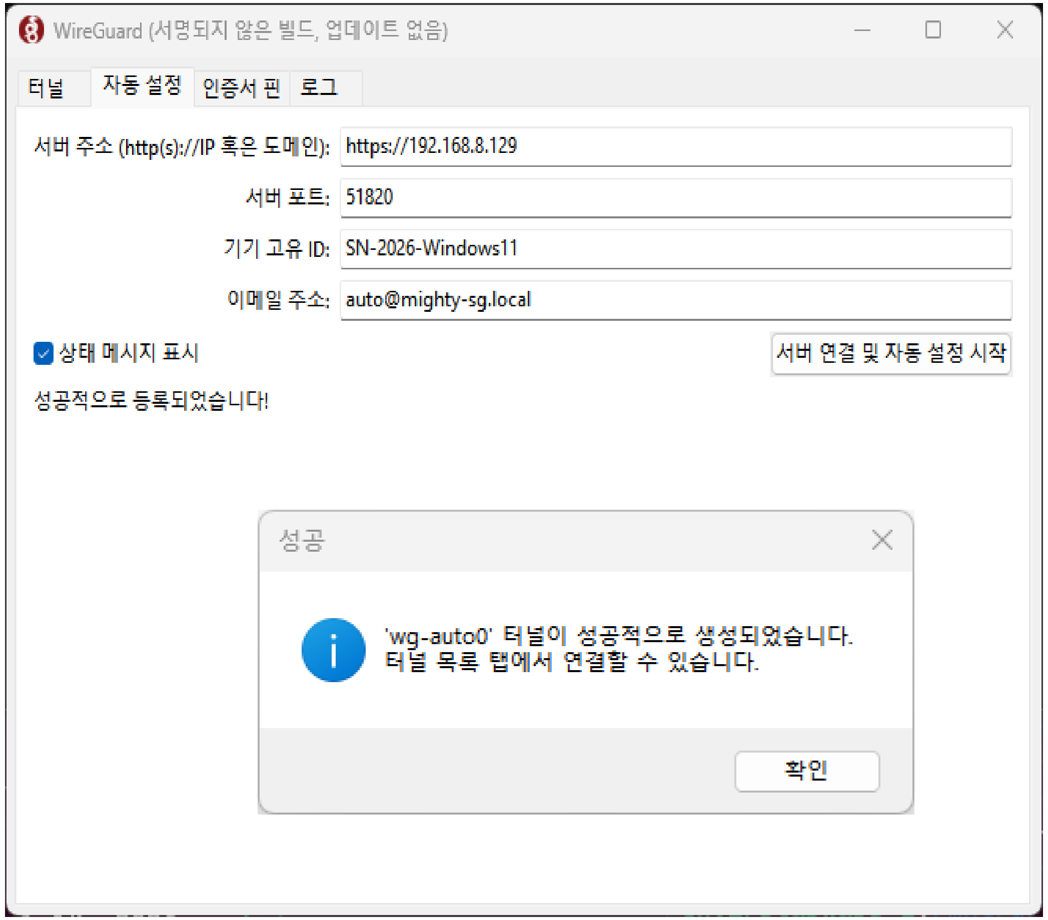


5.3 MightyConnect Windows

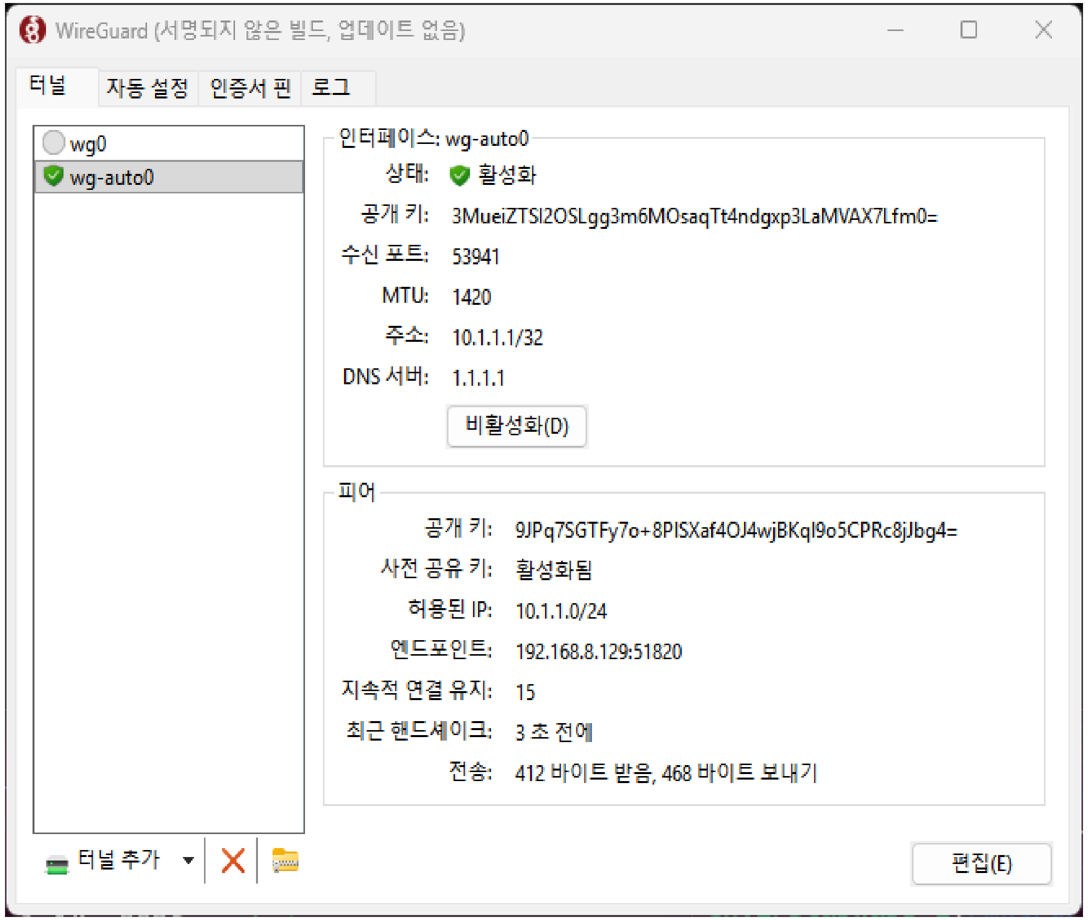
MightyConnect Windows Agent

5.3 MightyConnect Windows(1) – User Interface(1)

- Auto Connection Page

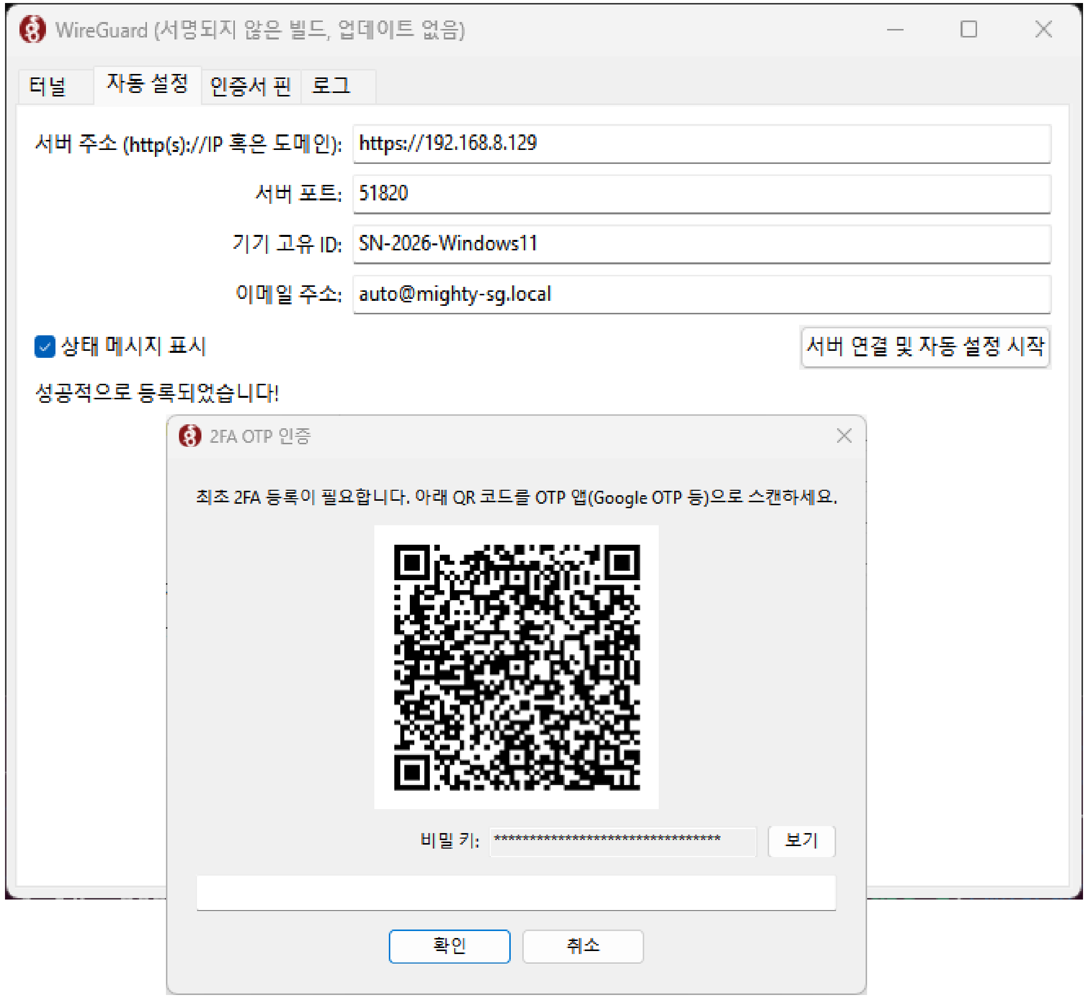


- wg-auto0 interface 생성 및 터널 활성화 Page

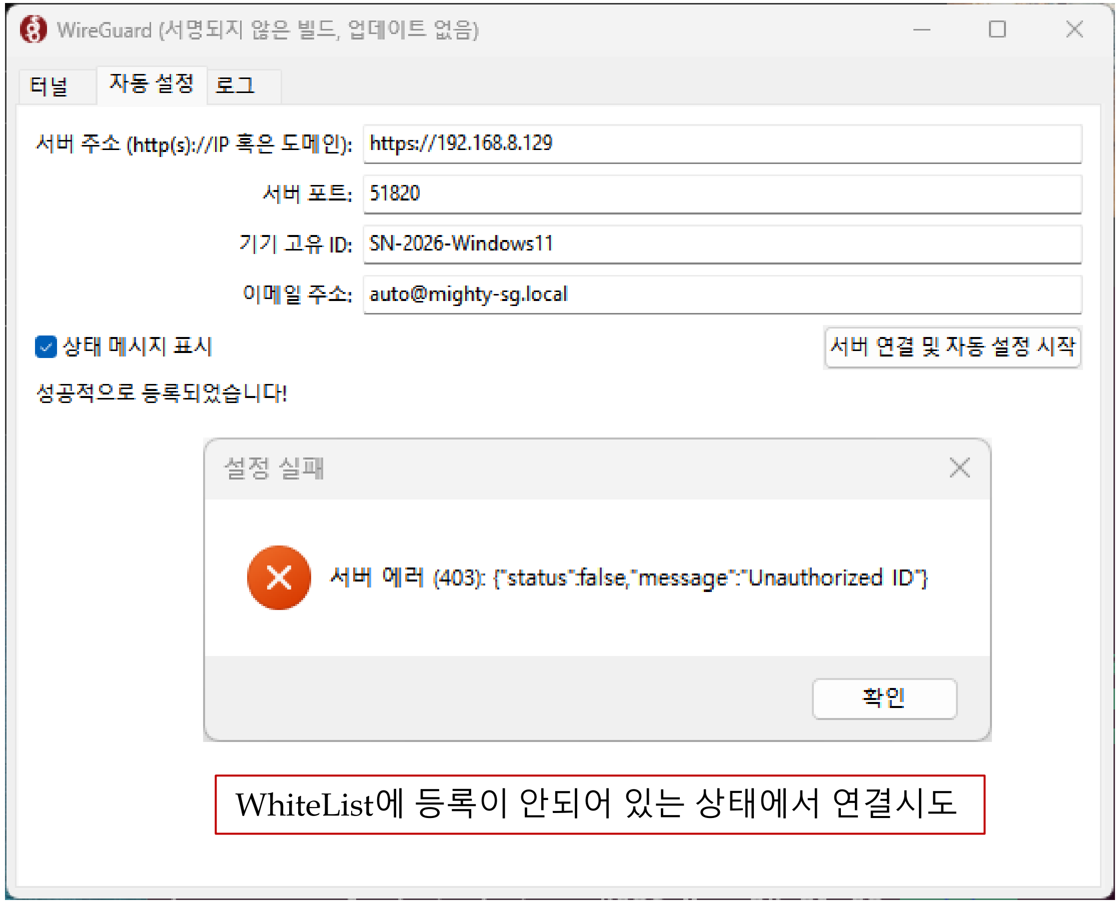


5.3 MightyConnect Windows(1) – User Interface(2)

- Auto Connection – 2FA 인증 Page

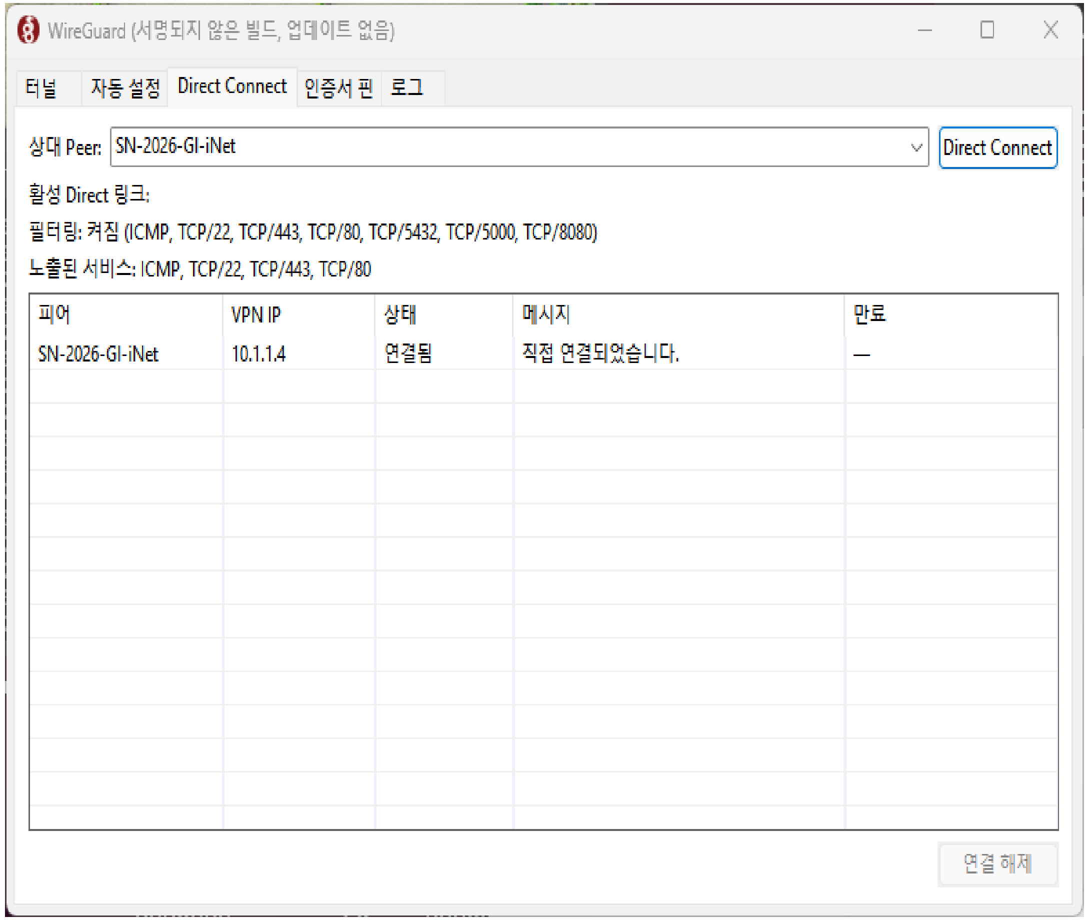


- Auto Connection – 인증 실패 예시

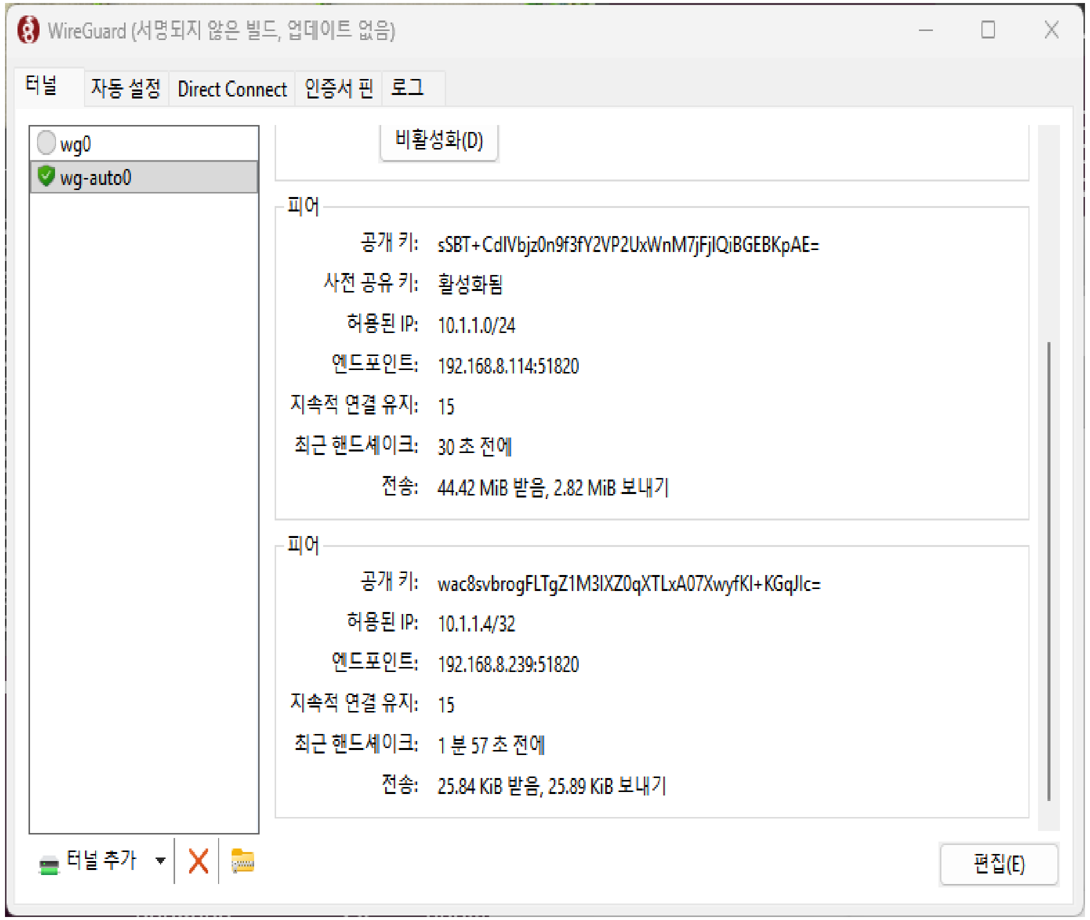


5.3 MightyConnect Windows(1) – User Interface(3)

- Direct Connect Page

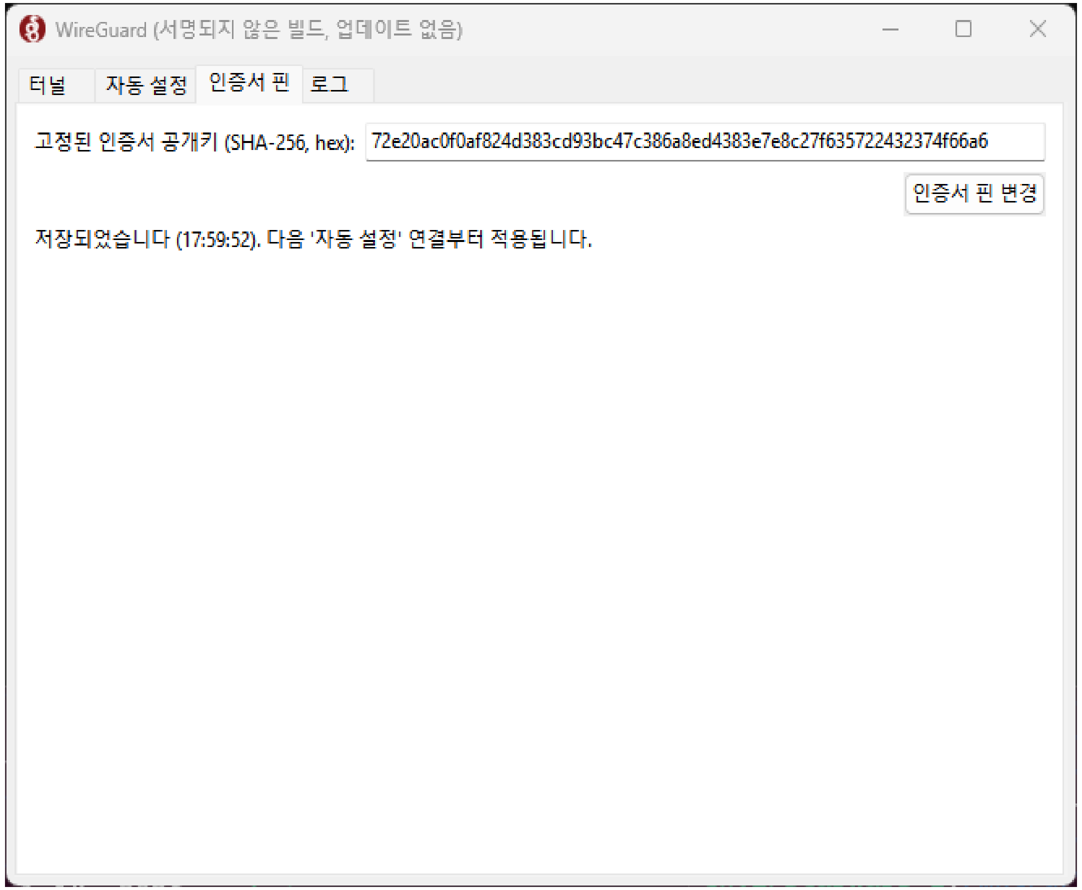


- Direct Connect Peer 확인

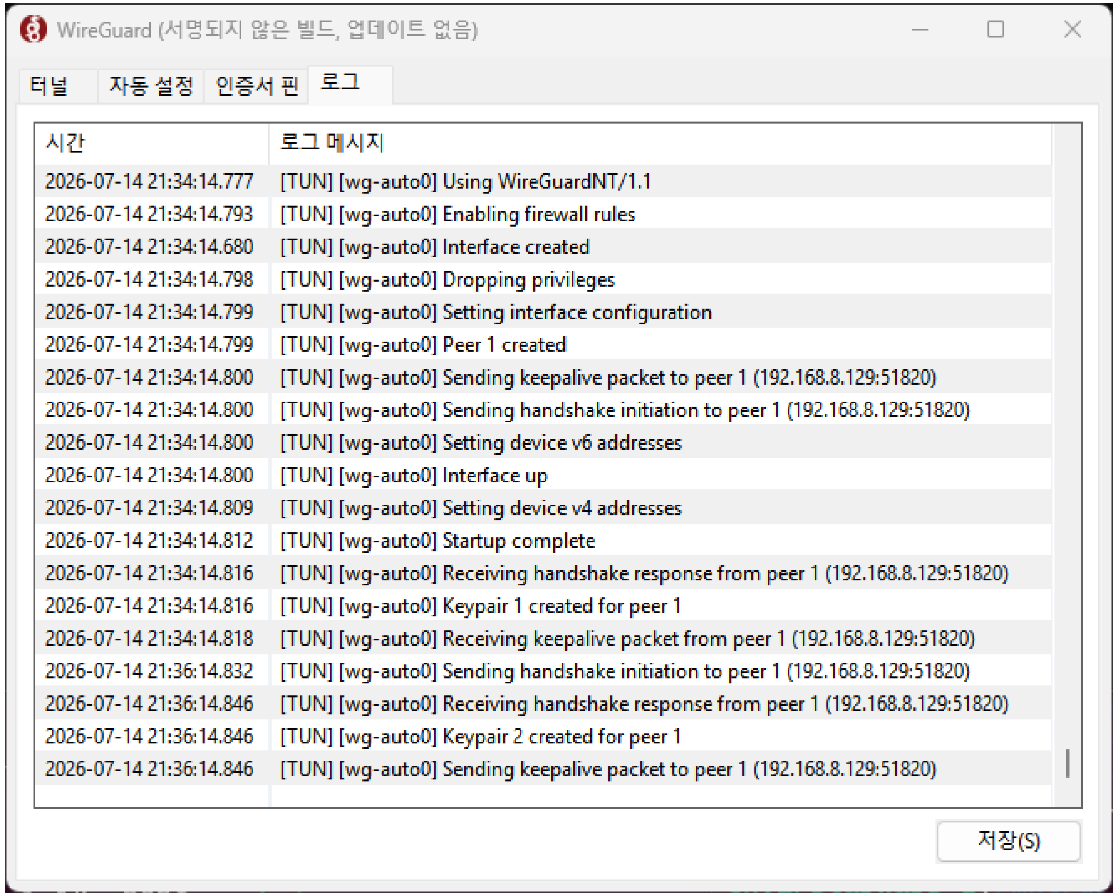


5.3 MightyConnect Windows(1) – User Interface(4)

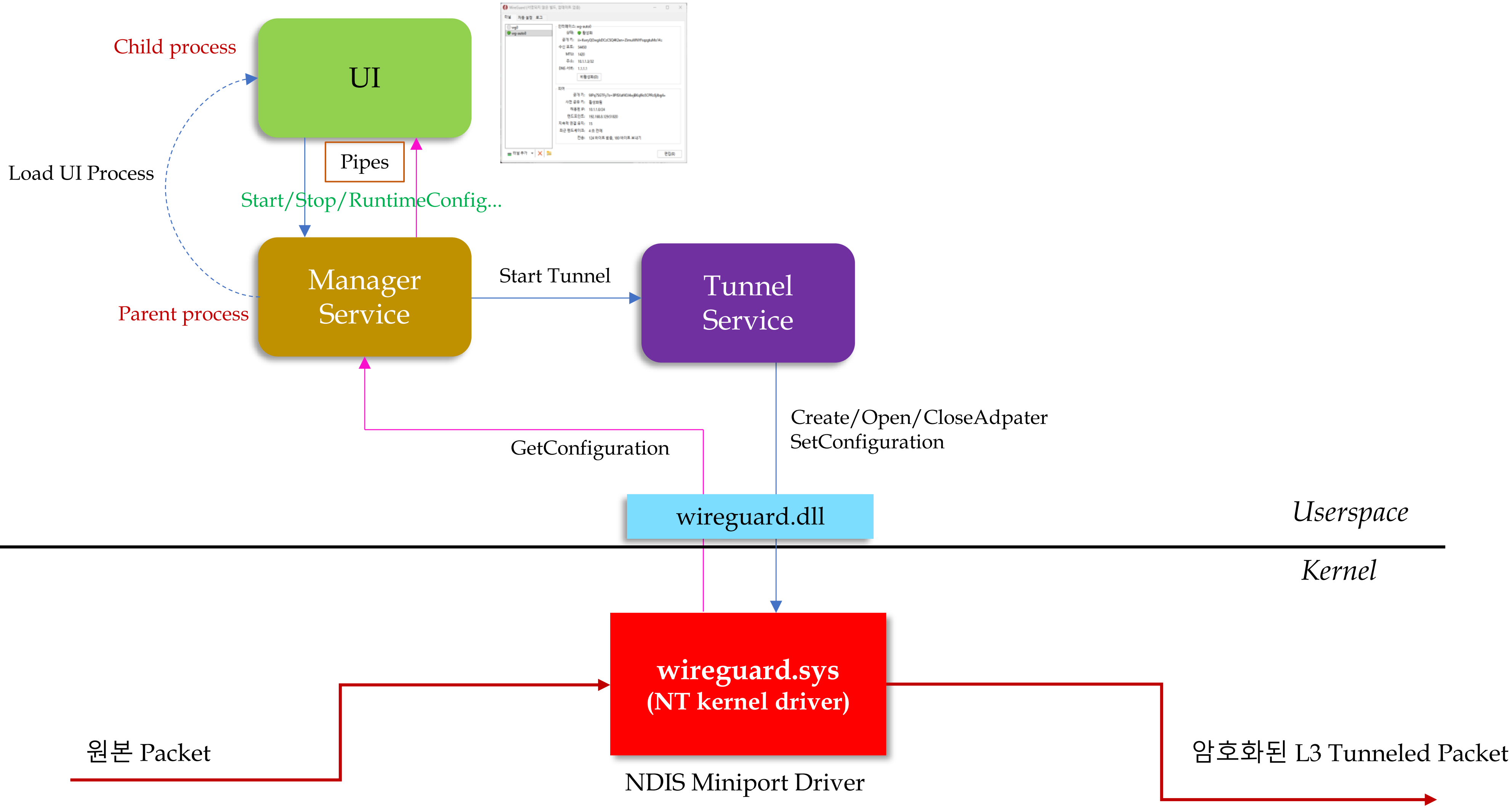
- X.509 인증서 Pin 값 변경 Page



- Log Page



5.3 MightyConnect Windows(2) – Architecture

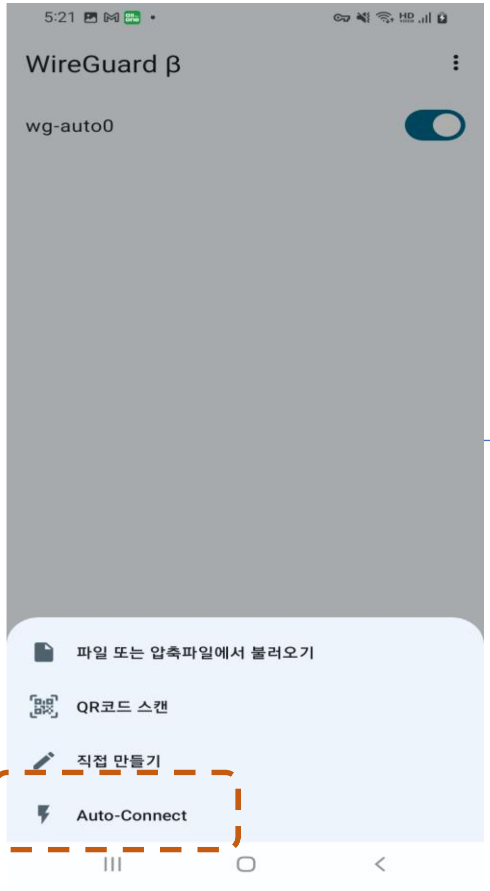


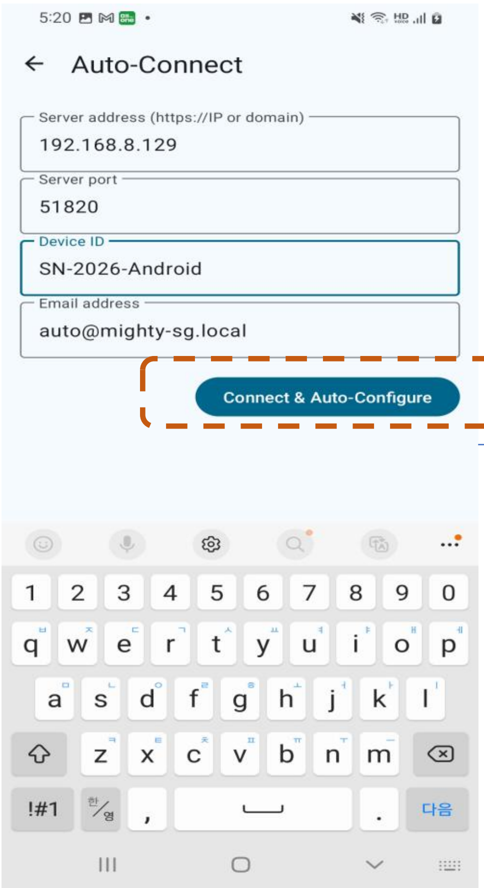
5.4 MightyConnect Android

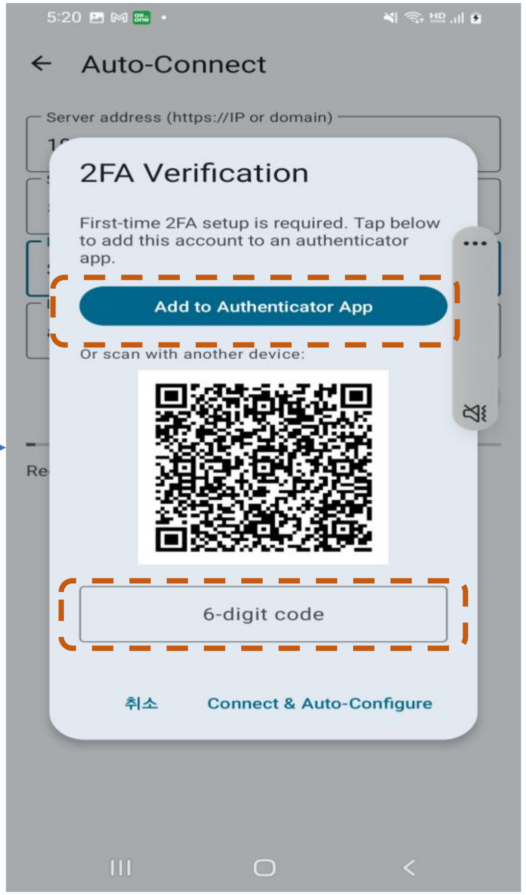
Android App

5.4 MightyConnect Android(1) – Auto Connect(1)

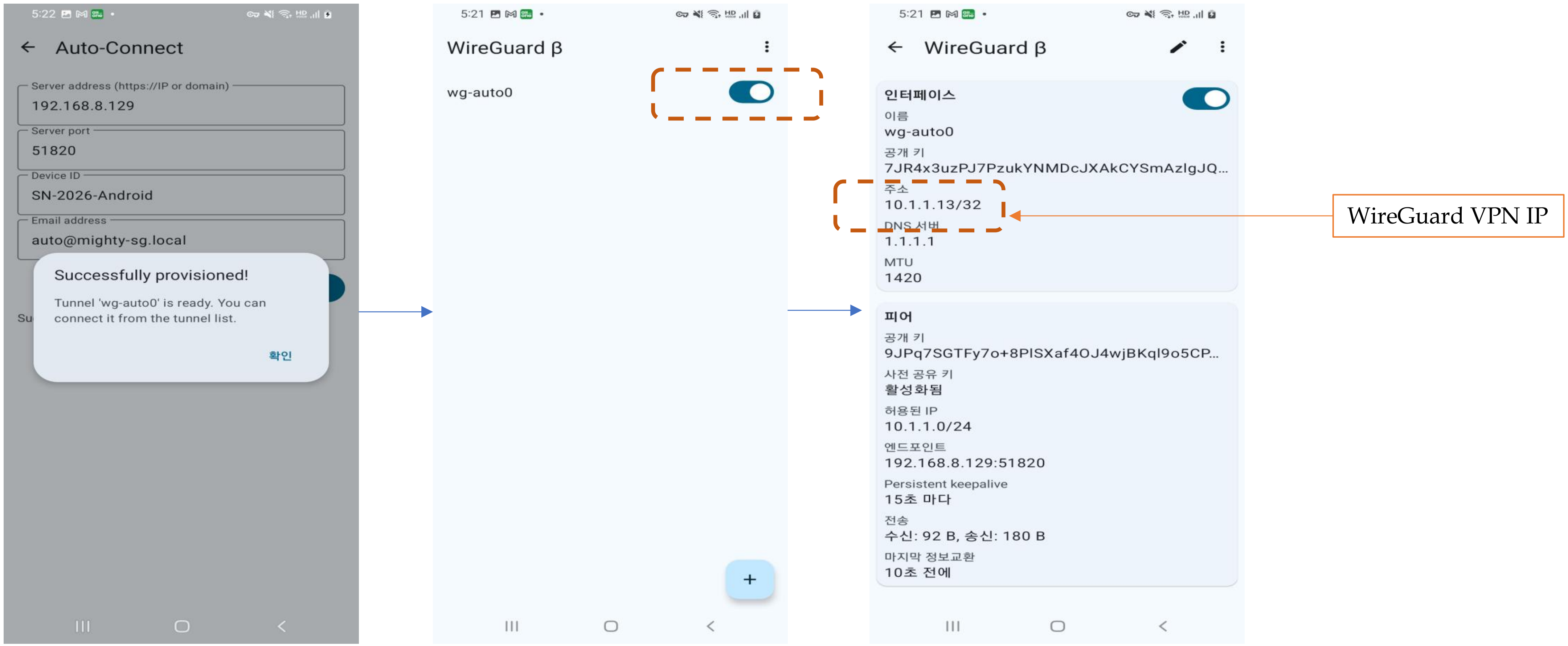
터널 추가 버튼



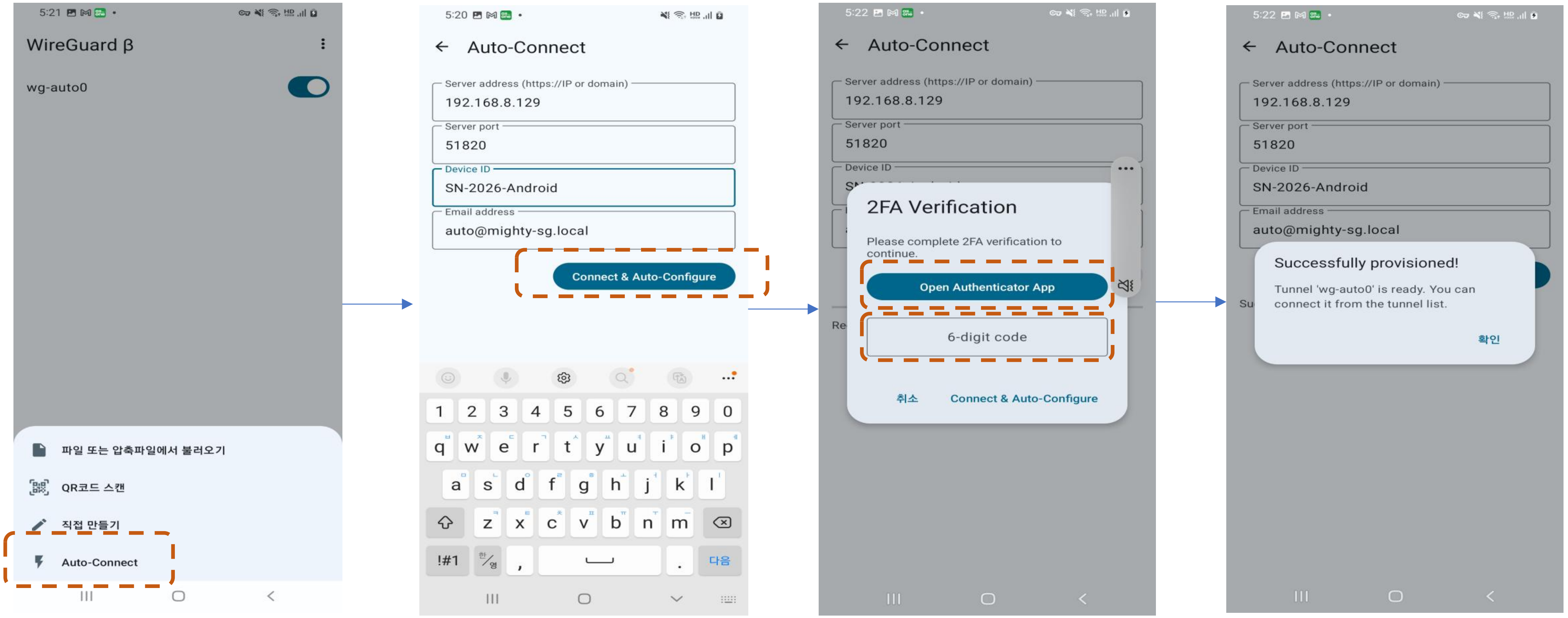




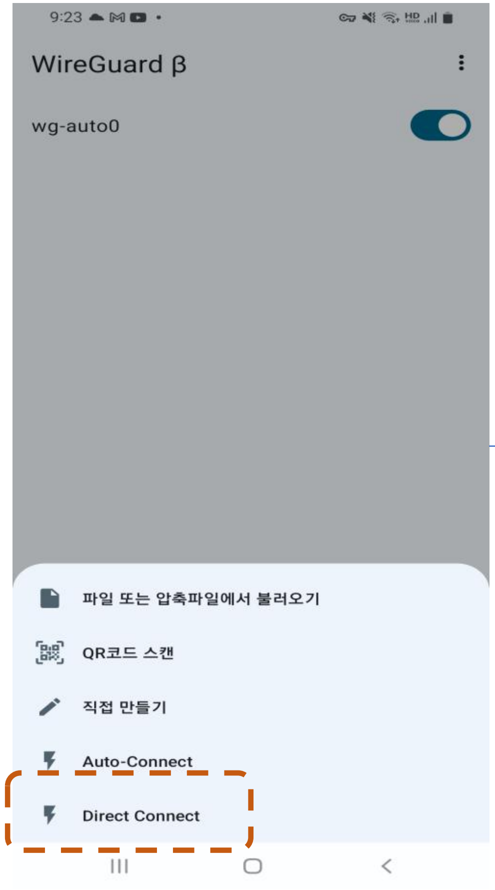
5.4 MightyConnect Android(1) – Auto Connect(2)

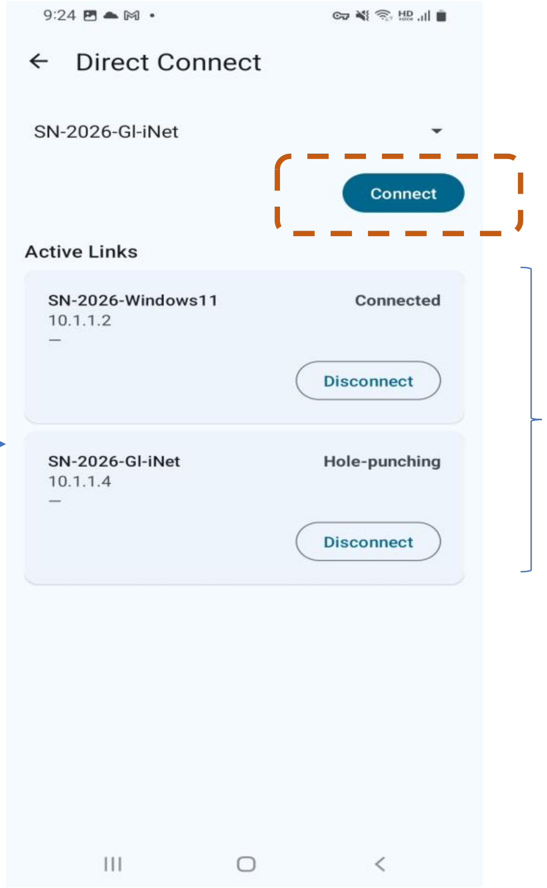


5.4 MightyConnect Android(1) – Auto Connect(3)



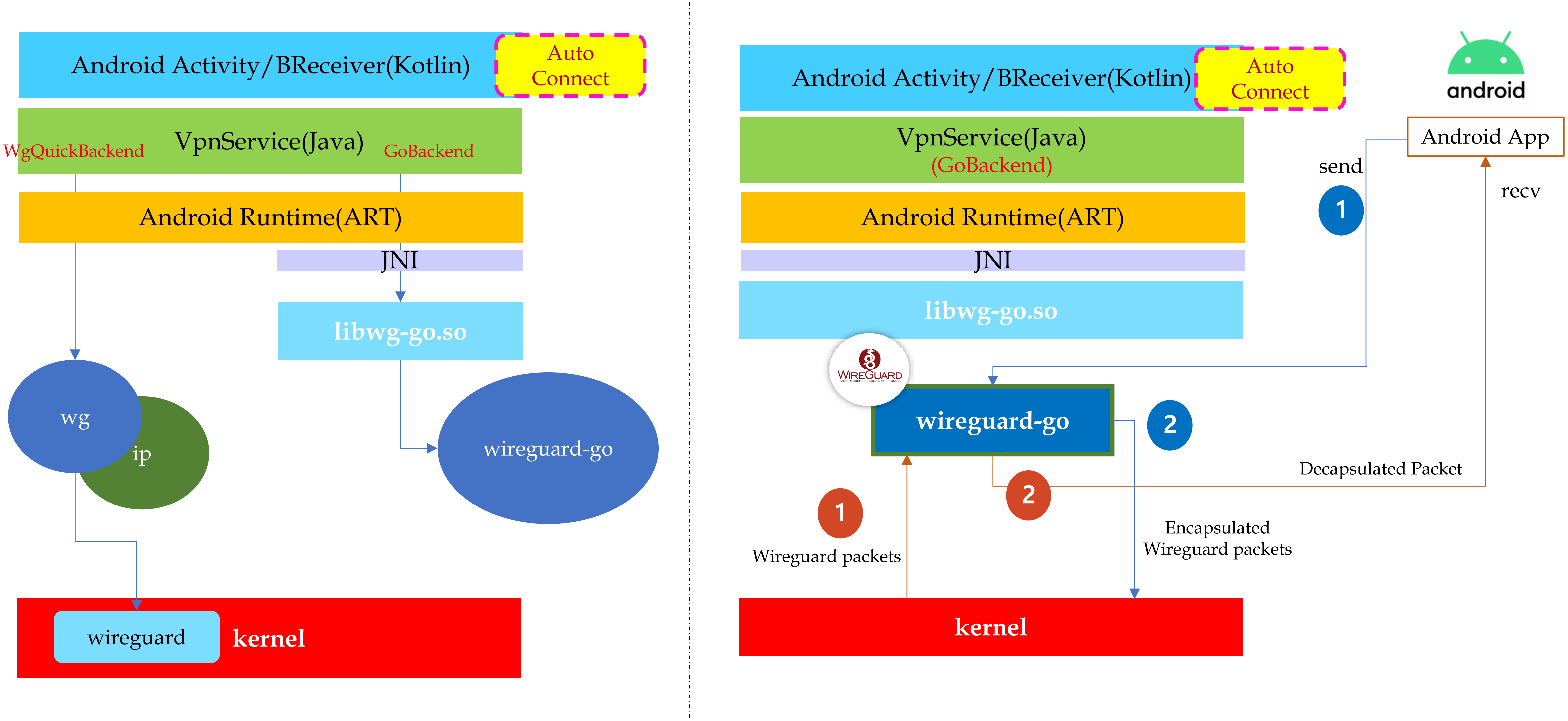
5.4 MightyConnect Android(2) – Direct Connect





P2P Sessions

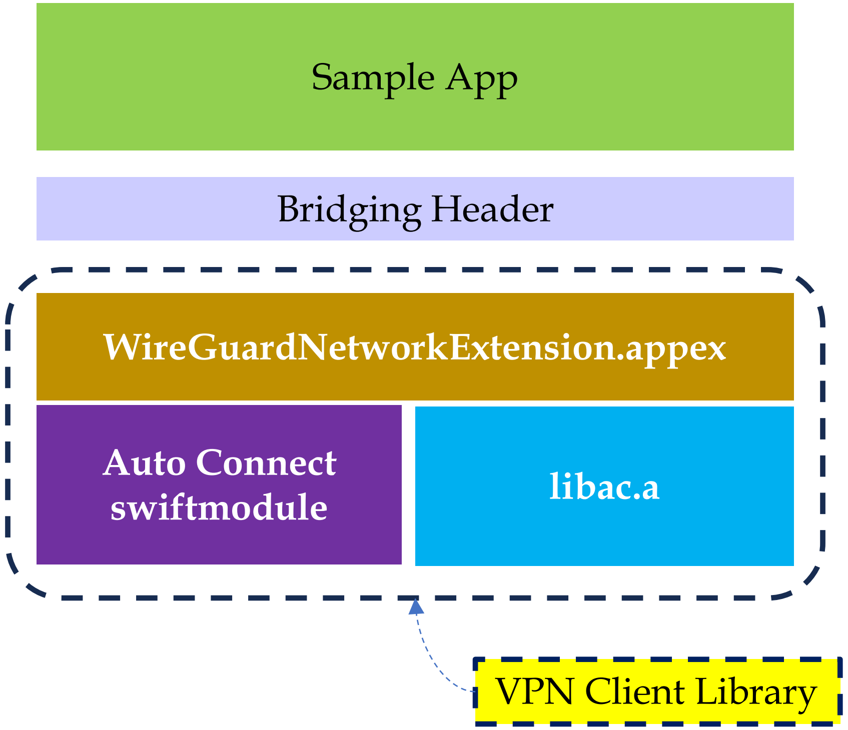
5.4 MightyConnect Android(3) – Architecture

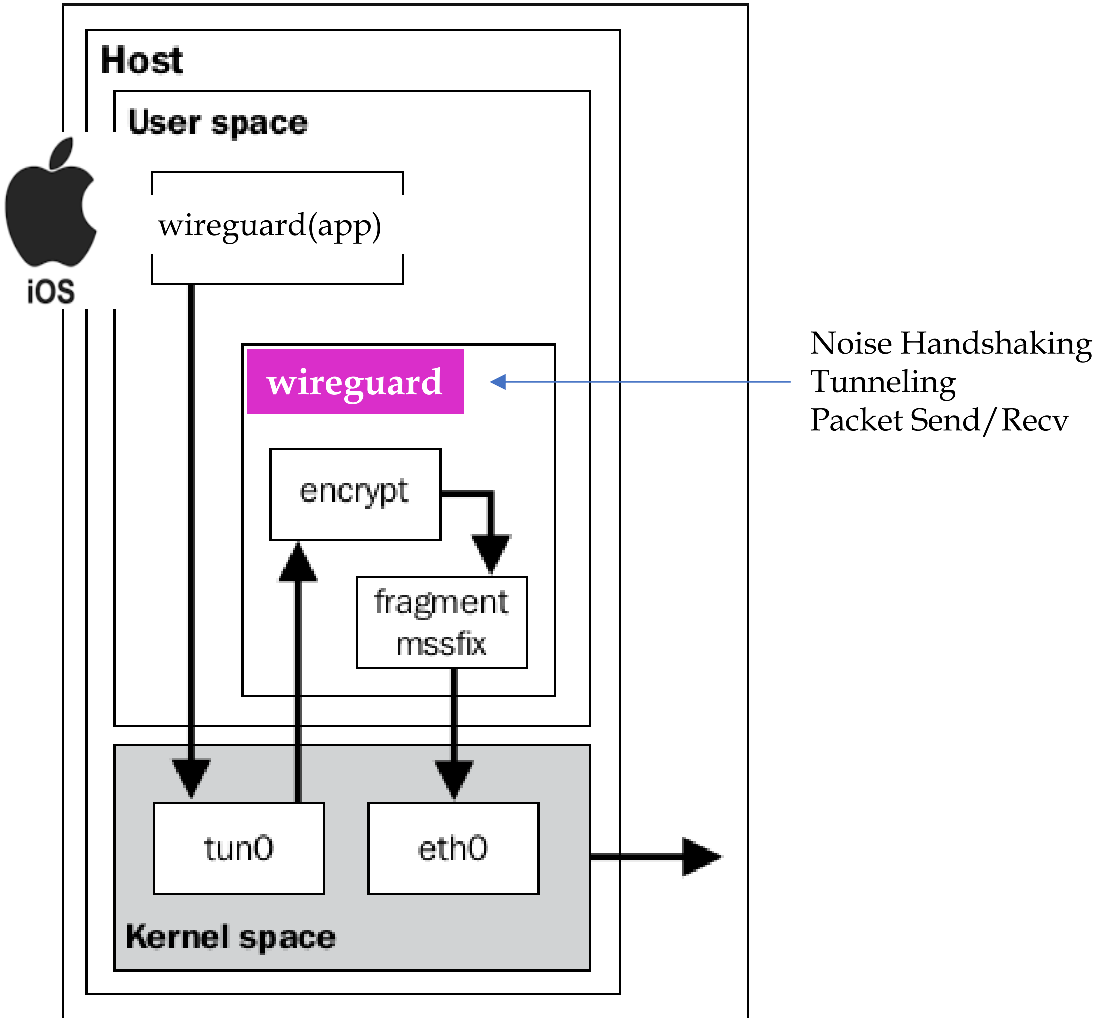


5.5 MightyConnect macOS/iOS

MightyConnect macOS/iOS Agent

5.5 MightyConnect macOS/iOS – Architecture





6. References

6. References

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- [13] macOS & iOS is a registered trademark of Apple.
- [14] Ubuntu is a registered trademark of Canonical Ltd.
- [15] And Google~

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